

AS/NZS 1596:2014



Australian/New Zealand Standard

The storage and handling of LP Gas

Superseding AS/NZS 1596:2008

AS/NZS 1596:2014



This joint Australian/New Zealand standard was prepared by joint Technical Committee ME-015, Storage and Handling—Liquefied Petroleum Gases. It was approved on behalf of the Council of Standards Australia on 14 April 2014 and on behalf of the Council of Standards New Zealand on 27 March 2014.

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Australian/New Zealand Standard

The storage and handling of LP Gas

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PREFACE

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee ME-015, Storage and Handling—Liquefied Petroleum Gases, to supersede AS/NZS 1596:2008, *The storage and handling of LP Gas*.

This edition is the result of a comprehensive review of the 2008 edition, in order to update procedures in light of industry best practice, and correct and update references.

A new informative Appendix, 'Impact protection', for assessing the requirements for the protection of tanks and cylinders from impact, has been introduced.

A new informative Appendix, 'Transport of portable LP Gas cylinders for trade use', covering the transport of cylinders for trade use, has been included. This has drawn heavily on safety notes and bulletins produced by various government bodies following an increase in safety incidents in this area. Addressing additional safety concerns regarding the hazardous nature of working in underground pits, a number of changes have been made in this area.

Appendix K has been reviewed to bring the emergency planning aspects of the Standard in line with current emergency services terminology and practice.

A significant number of minor changes have been made to align the Standard with New Zealand regulatory practice and it is envisaged that the Standard may be called up in legislation in New Zealand in the future.

This Standard has been revised in terms of safety and technical matters. The regulatory authorities in each state and territory determine how and when Standards are to be applied. The method of application may be full, part, with or without modifications or not at all. This application is not directed by the Standard. The relevant regulatory authority should be consulted regarding the application of this Standard.

The terms 'normative' and 'informative' have been used in this Standard to define the application of the appendix to which they apply. A 'normative' appendix is an integral part of a Standard, whereas an 'informative' appendix is only for information and guidance.

Statements expressed in mandatory terms in notes to tables are deemed to be requirements of this Standard.

Where Australian and New Zealand references are separated by the word 'or', a slash (/) or parentheses, Australian references apply in Australia, and New Zealand references apply in New Zealand. Joint Standards apply in both Australia and New Zealand.

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STANDARDS AUSTRALIA/STANDARDS NEW ZEALAND

Australian/New Zealand Standard

The storage and handling of LP Gas

SECTION 1 SCOPE AND GENERAL

1.1 SCOPE AND OBJECTIVE

This Standard specifies requirements for the location, design, construction, commissioning and operation of installations for the storage and handling of LP Gas, and includes the management of emergencies.

The objective of this Standard is to provide designers, planners, operators and regulators with technical and procedural requirements for installations for the safe storage and handling of LP Gas.

1.2 APPLICATION**1.2.1 Application of Standard**

This Standard shall be read as defining the minimum requirements of acceptability.

This Standard applies to installations commenced after its publication date. Unless specifically indicated in this Standard, it shall not apply to existing installations.

NOTE: 'Commenced', for the purpose of this Clause, means the point at which site or equipment construction has begun.

The relevant authority having jurisdiction may determine the extent of application of this Standard.

This Standard does not apply to the following situations:

- (a) Refrigerated storage of LP Gas.
- (b) Underground storage of LP Gas in mined caverns or geological formations.
- (c) Plant or equipment in which LP Gas is processed or produced, or vessels that form an integral part of that processing equipment (but does include post manufacturing storage).
- (d) Industrial gas-consuming equipment (see AS 3814 or AS/NZS 5601.1).
- (e) Automotive installations, as addressed in AS/NZS 1425 and AS 4983.
- (f) Refrigerating systems, as addressed in the AS/NZS 1677 series.
- (g) Land transport equipment and operations, as covered—
 - (i) in Australia, by the ADG Code (but excluding references to this Standard that appear in the ADG Code) and AS 2809, Parts 1 and 3; or
 - (ii) in New Zealand, a road tank wagon shall be constructed in accordance with the *Hazardous Substances (Tank Wagons and Transportable Containers) Regulations 2004*. The *Code of Practice HSNO COP 29, Flammable Gases and Oxygen Tank Wagons*—A code of practice for the design and construction of vehicles for bulk transport of flammable gases and oxygen by road is a means of compliance. Land transport shall be consistent with the *Land Transport Rule: Dangerous Goods 2005*, as amended.

- (h) Air transport equipment and operations.
- (i) LP Gas systems for caravans, catering vehicles, marine craft or aircraft (see AS/NZS 5601.1), or as covered in the relevant parts of the AS 2809 series.
- (j) Refrigeration systems for vehicles.

1.2.2 Relationship to other Standards

Figure 1.1(A) and (B) illustrate the application of this Standard and its relationship with AS/NZS 5601.1 for gas storage containers connected in use.

Where only LP Gas is stored, this Standard applies. For the storage and handling of other gases in cylinders, or where LP Gas is stored with other gases in cylinders, AS 4332 should be consulted.

1.2.3 Application of sections

Sections 3, 11, 12 and 13 set out general requirements that apply to the storage and handling of LP Gas in quantities greater than minor storage.

Section 2 sets out conditions and requirements for minor storage and usage of LP Gas in cylinders.

Section 4 sets out specific requirements for the storage and usage of LP Gas in cylinders, in quantities that are greater than, or do not comply with, minor storage.

Sections 5, 6 and 7 set out requirements for the storage and handling of LP Gas in tanks, in various locations.

Sections 8 and 9 set out specific requirements for cylinder filling operations in Australia and New Zealand respectively.

Section 10 sets out specific requirements for automotive filling installations, including service stations.

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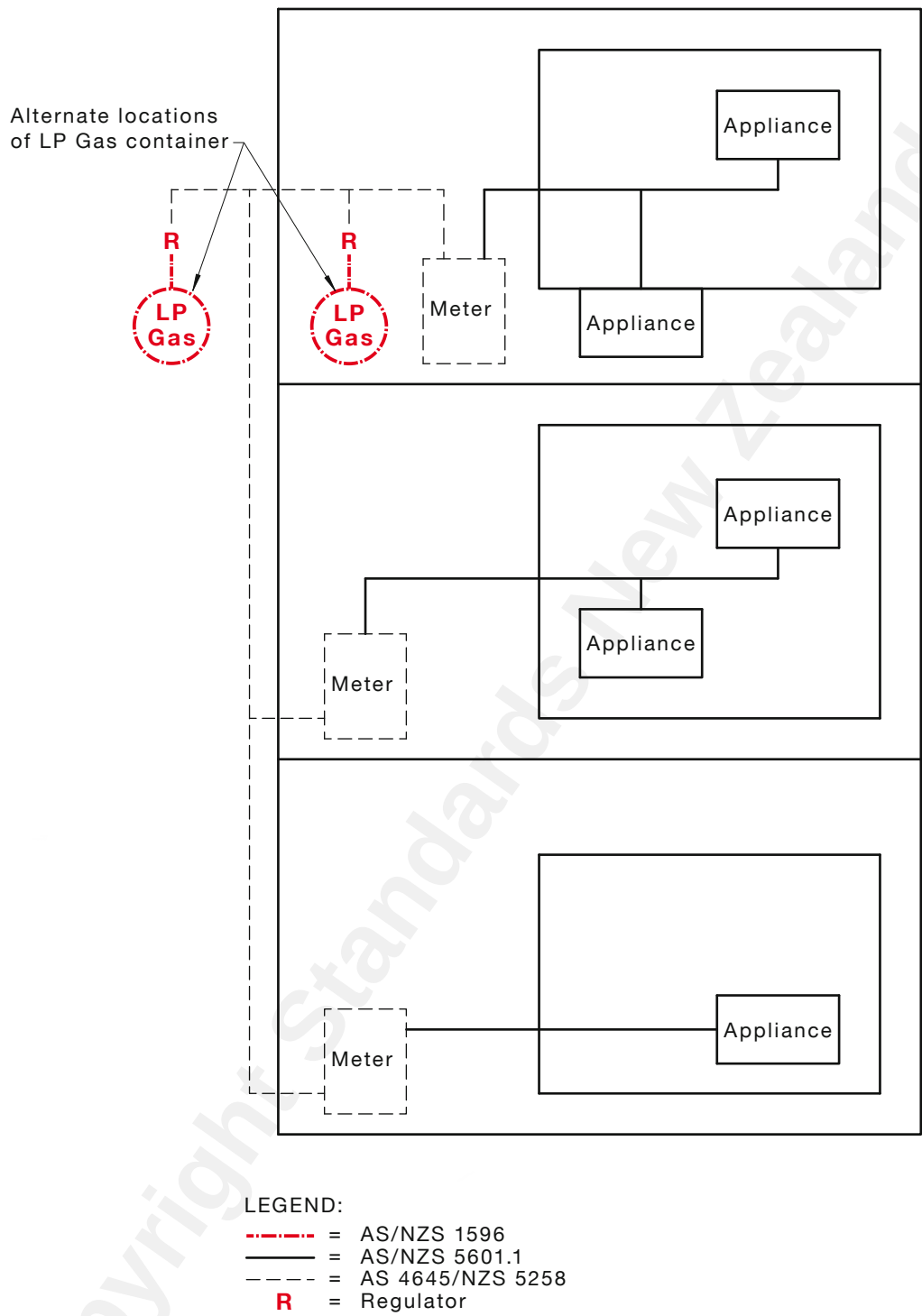
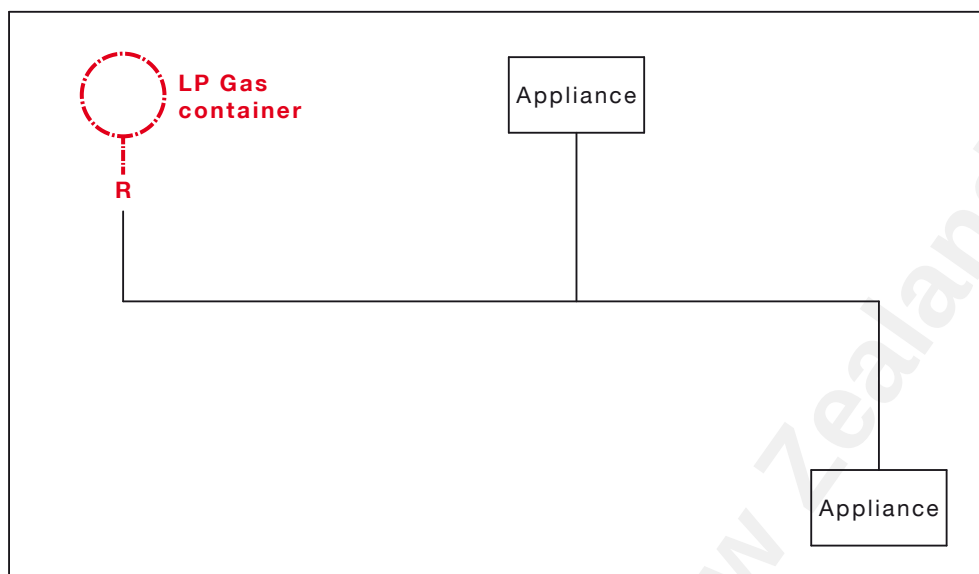


FIGURE 1.1(A) LP GAS CONTAINER SUPPLYING SEVERAL SITES BY RETICULATION FROM A SINGLE LP GAS SUPPLY



LEGEND:

- = AS/NZS 1596
- = AS/NZS 5601.1
- R** = Regulator

FIGURE 1.1(B) LP GAS CONTAINER SUPPLYING SEVERAL APPLIANCES ON A SINGLE SITE

1.3 REFERENCED DOCUMENTS

The documents referred to in this Standard are listed in Appendix A.

NOTE: For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

1.4 DEFINITIONS

For the purpose of this Standard, the definitions below apply.

1.4.1 ADG Code

The *Australian Code for the Transport of Dangerous Goods by Road and Rail* (commonly known as ADG Code) published by the National Transport Commission.

1.4.2 Aerial wiring

A power line placed above the ground and in the open air, but not including bus bars, telephone lines, or any direct current conductors used as traction trolley wires.

1.4.3 Authority

The authority having statutory (legal) control of a particular matter.

1.4.4 Automotive filling installation

A place where automotive LP Gas fuel tanks mounted on vehicles are intended to be filled from dispensers. [See also fleet refuelling installation (Clause 1.4.23) and service station (Clause 1.4.53).]

1.4.5 Boundary

The boundary of the whole of the site under the same occupancy as that on which the installation is included.

1.4.6 Capacity (of a tank or cylinder)

The total volume of the space enclosed within the cylinder or tank, expressed in litres (L) or kilolitres (kL) respectively.

NOTE: This is often referred to as 'water capacity'.

1.4.7 Combustible

A solid substance capable of undergoing combustion when tested in accordance with AS 1530.1.

1.4.8 Combustible liquid

Any liquid, other than a flammable liquid, that has a flash point, and has a fire point less than its boiling point.

NOTE: AS 1940 provides test methods of combustible liquid.

1.4.9 Compound

An area bounded by natural ground contours or by a bund, and intended to retain spillage or leakage.

NOTE: A pit or tank may be used to provide the same function.

1.4.10 Container

Anything that contains or can contain LP Gas. It does not include a vehicle or a freight container.

1.4.11 Cylinder

A container that falls within the scope of AS 2030.1 or, in New Zealand, an approved container as relevant to LP Gas.

1.4.12 Dangerous goods

(Australia) Substances and articles that—

- (a) satisfy the UN tests and criteria for determining whether they are dangerous goods; or
- (b) are listed in the ADG Code or are determined to be dangerous goods by the regulatory authority.

(New Zealand) Substances and articles that have one or more hazard Classes assigned to them in accordance with the HSNO Act, including packaging and empty containers that have not been cleaned after containing hazardous substances.

For the purposes of this Standard (for use in New Zealand) the term 'dangerous good' is synonymous with 'hazardous substance' and both are used interchangeably throughout the document.

In New Zealand, 'dangerous goods' is a term used by the *Land Transport Rule: Dangerous Goods 2005*, which applies during transport. For the storage and handling of substances, the HSNO Act applies, and the term 'hazardous substances' is used.

NOTE: UN tests and criteria are specified in the *UN Manual of Tests and Criteria and the Recommendations on the Transport of Dangerous Goods—Model Regulations*.

1.4.13 Deadman's handle

A device that is activated by the person operating the system, to start the flow of fuel, and that requires constant actuation by the operator to maintain the flow of fuel. The flow of fuel is stopped if the device is released.

1.4.14 Decanting

A procedure in which the liquid phase of LP Gas is transferred from one container to another by utilizing the difference in pressure between the two containers.

1.4.15 Direct connection

A filling connection mounted directly at the tank and designed to spray-fill into the tank vapour space.

1.4.16 Emergency shut-down system

A control system that facilitates safe LP Gas shut-down in an emergency.

1.4.17 Excess-flow valve

A valve that is normally open, but closes automatically when a predetermined flow rate in a particular direction has been exceeded.

1.4.18 Filling connection

A loading connection that allows the transfer of LP Gas from a tanker to a tank, and vice versa.

1.4.19 Fire resistance level (FRL)/Fire resistance rating (FRR)

A measure of the fire resistance of a material or structure as determined in accordance with AS 1530.4 or NZS/ISO 834. It consists of three numerals representing, in order, the period of resistance for—

- (a) structural adequacy;
- (b) integrity; and
- (c) insulation.

For example, 60/60/30 means 60 min of resistance for each of Items (a) and (b), and 30 min for Item (c).

NOTE: In New Zealand the term 'fire resistance rating' (FRR) is used. It has the same meaning as FRL.

1.4.20 Firewall

A wall or other barrier constructed and placed with the object of preventing the spread of fire or the radiation of heat from any one place to some other place.

1.4.21 Fixed liquid level gauge or fixed ullage gauge

A gauge that indicates the maximum standard liquid level in the container.

1.4.22 Flammable liquid

For Australia, a flammable liquid as defined in the ADG Code.

For New Zealand, a flammable liquid as defined in New Zealand Hazardous Substances (Classification) Regulations.

1.4.23 Fleet refuelling installation

A place where only industrial or commercial vehicles are refuelled, and where the retail selling of LP Gas to the general public does not occur.

1.4.24 Fusible link

A safety device consisting of a material having a suitably low melting point, and which is intended to yield or melt at a predetermined temperature.

1.4.25 Gas distributor

A person, or persons, or company which receives LP Gas in bulk for distribution to the general public, or to agents who in turn sell the LP Gas to the general public.

1.4.26 Gas free

For flammable gases, the concentration of that gas or vapour in the atmosphere of the container being less than 5% of the lower explosive limit (LEL) for the substance when sampled at ambient temperature.

1.4.27 Hazardous area

An area in which an explosive gas atmosphere is or may be expected to be present, in quantities such as to require special precautions for the construction, installation and use of equipment.

NOTES:

- 1 The classification of hazardous areas due to flammable gases is specified in AS/NZS 60079.10.1.
- 2 (New Zealand) The HSNO Act uses the term 'hazardous atmosphere zones'; however, for the purposes of this Standard the term 'hazardous area' and corresponding definition should be used.

1.4.28 Hazardous substance (New Zealand)

See definition for 'dangerous goods'.

In New Zealand, for the purposes of this Standard the term 'dangerous good' is synonymous with 'hazardous substance' and both are used interchangeably throughout the document.

NOTE: In New Zealand, the term 'hazardous substance' is used in the following Act and regulations:

- (a) The HSNO Act together with the regulations.
- (b) Group Standards and Transfer Notices made under that Act.

1.4.29 Heat radiation barrier

A shield interposed between a source of radiant heat and an object, in order to reduce the amount of radiant heat received by the object to an acceptable level.

1.4.30 Hot work

Any condition that is likely to produce a source of ignition or be an ignition source.

NOTE: Examples of hot work include the following:

- (a) Welding of flame cutting equipment.
- (b) Matches, naked lights, cigarettes.
- (c) Grinding, drilling or other activities that could create heat or spark.
- (d) Electrical equipment not suitable for hazardous areas.
- (e) Spark ignition engines or compression ignition engines not suitable for hazardous areas.
- (f) Hand tools that could create a spark.

1.4.31 Ignition source

A source of energy sufficient to ignite an explosive atmosphere.

NOTE: Examples of ignition sources include naked flames, exposed incandescent material, electrical welding arcs, mechanical or static sparks, and electrical, electronic or mechanical equipment not suitable for use in hazardous locations.

1.4.32 In situ filling

A cylinder filling procedure, where a cylinder installed on a customer's premises is filled from a tanker.

1.4.33 Installation

All of the facilities on a site that are used for storing or handling LP Gas.

1.4.34 Internal component

In relation to a fitting or component, one that is constructed so that its significant working parts are within the container shell, and damage to exposed portions will not prevent effective safe seating or closure.

1.4.35 Internal safety control valve (ISC)

A quick-closing internal valve incorporating an internal excess-flow valve.

1.4.36 Liquefied petroleum gas (LP Gas)

A hydrocarbon fluid composed predominantly of any of the following hydrocarbons, or mixtures of all or any of them: propane (C_3H_8), propylene (C_3H_6), butane (C_4H_{10}) or butylenes (C_4H_8). Unless specifically stated otherwise, any reference to 'propane', 'butane', etc. means the commercial grade of that product.

NOTES:

- 1 The characteristics of the various LP Gases, together with methods for their determination, are given in AS 4670; for the autogas sector fuel quality is regulated federally in Australia through the *Fuel Standard (Autogas) Determination 2003*.
- 2 In New Zealand, NZS 5435 sets out the composition and other requirements for the following types of LP Gas: commercial propane, commercial butane, general product and special product.

1.4.37 Mass-filling ratio

The ratio between the greatest mass of LP Gas permitted in a container and the mass of water at 15°C which would completely fill the container.

1.4.38 Maximum filling level

The highest liquid surface level permitted in a container (see Appendix B).

1.4.39 Modular direct-fired vaporizer

Vaporizer containing, within the one case/body, multiple heat exchanger modules each having separate controls (inlet valve, outlet, temperature, pilot and burner).

1.4.40 May

Indicates the existence of an option.

1.4.41 Non-return valve (check valve)

A valve which permits flow in one direction only.

1.4.42 Outdoor

An above ground, open-air situation with natural ventilation, where any leakage and products of combustion are rapidly dispersed by wind and natural convection.

1.4.43 Pigtail

A short length of small bore pipe, or a hose assembly (to allow flexibility) used for the high pressure connection between an LP Gas cylinder and the cylinder regulator or cylinder manifold.

1.4.44 POL coupling

An LP Gas union connection having a left-hand thread, as specified for Type 21 in AS 2473.2.

1.4.45 Positive shut-off

A valve in which the actuating stem is permanently connected to the flow-blocking components, and which, when actuated to the closed position, unquestionably blocks off all flow.

1.4.46 Practicable

Practicable having regard to—

- (a) the severity of the hazard or risk in question;
- (b) the state of knowledge about that hazard or risk and about ways of removing or mitigating that hazard or risk;
- (c) the availability and suitability of ways to remove or mitigate that hazard or risk; and
- (d) the cost of removing or mitigating that hazard or risk.

1.4.47 Protected place

Any of the following:

- (a) A dwelling, place of worship, public building, school or college, hospital, theatre, building or open area in which persons are accustomed to assemble in large numbers, whether within or outside the property boundary of the installation.
- (b) A factory, office, workshop, store, warehouse, shop or building where people are employed, except a building used for the storage and handling of LP Gas.
- (c) A vessel (e.g. a ship) lying at permanent berthing facilities.
- (d) Any storage facility for dangerous goods outside the property boundary of the installation, except those defined as minor storages in other Standards or regulations.

In New Zealand, the HSNO Act does not use the term 'protected place', it uses the term 'Area of High Intensity Land Use' (HILU). However, for the purposes of this Standard, the term 'protected place' and corresponding definition should be used.

1.4.48 Public place

Any place, other than private property, open to the public and including a street or road.

Parking areas for commercial and public buildings are not considered public places.

In New Zealand, the HSNO Act does not use the term 'public place' it uses the term 'Area of Low Intensity Land Use' (LILU). However, for the purposes of this Standard the term 'public place' and corresponding definition should be used.

1.4.49 Quick-closing internal valve

An internal valve designed and arranged to be closed either automatically by the operation of one or more heat-sensing devices (which may be fusible links) or manually from a remote position, or both, by the release of the means of holding the valve open.

1.4.50 Regulator

A device that automatically regulates the outlet pressure of gas passing through it to a predetermined limit.

1.4.51 Remote connection

A filling connection located remotely from the tank and which may be complemented with a vapour recovery connection for use with a tanker loading or unloading operation.

1.4.52 Safety valve

A pressure relief valve intended for the release of excessive vapour pressure, as defined in AS 1210 and AS 1271.

1.4.53 Service station

An automotive LP Gas retail outlet, where LP Gas is sold to the public as a motor fuel.

1.4.54 Shall

Indicates that a statement is mandatory.

1.4.55 Should

Indicates a recommendation.

1.4.56 Significant trafficable area

Situations/locations where there are no traffic control management procedures to minimize the risk of impact.

NOTE: Examples include service stations and commercial premises where no visual or physical barriers are installed to impede or direct traffic flow.

1.4.57 Site

The premises of an LP Gas facility, or where such a facility may be installed.

1.4.58 Site operator

A person who is responsible for the LP Gas operations.

1.4.59 Standard filling level

The highest liquid surface level permitted in a container for an LP Gas of specified density, assuming a liquid temperature of 5°C. It provides the location for the sensing point of the liquid level gauge.

NOTE: The standard filling level represents a safe but conservative filling level for any LP Gas whose density is equal to or higher than that for which the level was set, and whose temperature at the time of filling is 5°C or higher, the object being to ensure that the tank does not become liquid-full under normal operating conditions.

1.4.60 Tank

A container, other than a cylinder, that is designed for the storage or transport of LP Gas. A tank may be one of the following types:

- (a) *Static storage tank*—a tank intended to remain permanently in place once installed.
- (b) *Portable tank*—a tank intended for semi-permanent installation, but which incorporates additional features which allow it to be transported to the site, or transferred to another site, filled or partly filled with LP Gas.
- (c) *Skid tank*—a tank intended to be repeatedly transported in the full condition, and which may be moved frequently to service shifting or short-term demands. Such tanks might be used on an exchange fill basis, or might be left on-site for refilling from tankers.
- (d) *Demountable delivery tank*—a tank intended to be mounted on a vehicle to be used as an LP Gas delivery tanker, and which may be removed from the vehicle from time to time.
- (e) *Freight container tank*—a tank fitted with frames to international freight container dimensions in accordance with AS 3711.6.
- (f) *Stationary tank (New Zealand)*—a stationary container as defined in Hazardous Substances (Dangerous Goods and Scheduled Toxic Substances) Transfer Notice and amendments.

1.4.61 Tanker

A road tank vehicle that meets the requirements of AS 2809.1 and AS 2809.3, or with earlier Standards superseded by AS 2809.3, or with the New Zealand Environmental Protection Authority (EPA) *Approved Code of Practice for 'Flammable Gases and Oxygen Tank Wagons'*.

1.4.62 Thermal screen

A wall or other screen constructed and placed to protect people and the environment from the effects of short-duration fires and heat radiation. A firewall satisfies the requirements of a thermal screen.

1.4.63 Ullage

The space above the liquid level in a container.

1.4.64 Vaporizer

A device, other than a tank or cylinder, which receives LP Gas in the liquid phase and adds sufficient heat to convert the liquid to a gaseous state.

1.4.65 Vapour barrier

A wall or other barrier constructed and placed with the objective of preventing the passage of vapour from any one place to some other place.

1.4.66 Volume filling percentage

The maximum volume of liquid permitted in the container, expressed as a percentage of its capacity (see Clause 1.4.6).

NOTE: The percentage is variable, being dependent on the LP Gas density and temperature, and is the basis for determining the maximum or the standard filling level (see Appendix B).

SECTION 2 MINOR STORAGE AND USAGE

2.1 INTRODUCTION

The underlying concept of minor storage is that quantities below a certain level are generally unlikely to play a significant part in spreading a fire from place to place. The firefighting services and facilities required under normal building regulations are expected to cope adequately, provided that—

- (a) the quantity of LP Gas does not exceed that given in Table 2.1;
- (b) the storage conditions comply with Clause 2.3; and
- (c) the use of LP Gas is in accordance with AS/NZS 5601.1, AS/NZS 5601.2 as appropriate, or other relevant Standard.

NOTES:

- 1 For greater quantities of LP Gas, Section 4 applies.
- 2 Other requirements, such as workplace safety regulations and guidelines, may also apply to quantities defined as minor storage in this Standard.
- 3 Hazardous areas are not commonly defined for storages less than 25 L but may be required to be defined in some non-domestic situations, particularly where there is restricted ventilation.

2.2 APPLICATION OF SECTION

Cylinders of LP Gas in quantities not exceeding those given in Table 2.1 and complying with this Section (Section 2) may be classified as minor storage. In such cases, the requirements of this Section shall apply.

Where greater quantities of LP Gas are stored and handled, or the conditions set out in this Section cannot be met, other relevant sections of this Standard shall apply.

NOTE: Approximate volumes (water capacities) of LP Gas cylinders and the nominal masses of LP Gas contained therein are given in Appendix C.

This Section does not apply to—

- (a) cylinders used for decanting purposes; or
- (b) in situ fill cylinders.

2.3 MINOR STORAGE AND USAGE OF LP GAS

The following requirements and recommendations apply to the minor storage and usage of LP Gas:

- (a) The use and storage of cylinders of LP Gas indoors, whether full or nominally empty, should be avoided wherever practicable. Cylinders should preferably be located outdoors or in adequately ventilated areas.

NOTE: Figure 2.1 shows examples of adequately ventilated areas allowed by this Section. These examples may be different from those in AS/NZS 60079.10.1 for the purpose of minor storage covered by this Standard.

- (b) The total quantity of LP Gas kept (both in storage and connected for use) shall not exceed that stated in Table 2.1. This quantity includes LP Gas cylinders and non-refillable containers in use, whether full or nominally empty, but not those cylinders installed outdoors in accordance with Section 4 of this Standard and AS/NZS 5601.1.
- (c) Any cylinders in use shall be connected to an approved appliance and used in accordance with AS/NZS 5601.1 or other applicable Standard.
- (d) Users of LP Gas should be aware of the hazards and risks of its storage and use.

- (e) Cylinders shall be kept upright in a well-ventilated area away from any flame, heat or other ignition sources.
- (f) Cylinders shall be protected from excessive temperature rise.
- (g) Cylinders shall be protected from physical impact.
- (h) Cylinders shall be located so that they are not likely to be damaged or dislodged under normal circumstances of use. Any trolley or stand in which the cylinder is housed shall be of metal construction and of adequate stability.
- (i) Cylinders shall be kept in a location that does not hinder the escape of people, and is away from any combustible or waste materials.
- (j) Cylinders shall be kept at least 3 m from oxidizing gases, except where the cylinders of LP Gas and oxidizing gas form part of a portable oxy-fuel system used for welding, brazing, cutting and the like.

NOTE: Reference should be made to AS 4839 for portable and mobile oxy-fuel gas systems.

- (k) Cylinders shall be stored with all cylinder valves closed when not in use.
- (l) Cylinders and their fittings shall be inspected for leaks when connected and prior to the first use, e.g. by using a non-corrosive soap and water, detergent solution or leakage detection fluid to detect leaks at all joints and fittings.

CAUTION: THE AMMONIA PRESENT IN SOME SOAPS AND DETERGENTS CAN REACT WITH BRASS FITTINGS AND CAUSE SUCH FITTINGS TO CRACK AFTER A SHORT PERIOD OF TIME. CAUTION SHOULD THEREFORE BE EXERCISED WHEN USING SOAP SOLUTIONS ON BRASS FITTINGS, AND ALL CONNECTION SHOULD BE RINSED THOROUGHLY WITH FRESH WATER AS SOON AS POSSIBLE AFTER THE APPLICATION OF THE SOAP SOLUTION.

2.4 GROUPS OF TEMPORARY STRUCTURES IN WHICH LP GAS IS USED

Where temporary structures are grouped together, e.g. a group of marquees, stalls or tents at a market or event, there shall be a maximum of 10 such structures within each group if any structure is using LP Gas.

Groups of temporary structures using LP Gas shall be separated by at least 10 m. The intervening space shall be kept clear.

NOTE: For the purpose of this Clause, each structure is considered as a separate site for the application of Table 2.1 and Figure 2.1.

TABLE 2.1
MAXIMUM QUANTITIES FOR MINOR STORAGE AND USAGE

Location	Maximum quantity of LP Gas, number of cylinders and non-refillable containers
Residential properties: — A detached house or single storey attached dwelling	In Australia, indoors (including on balconies): 10 kg per dwelling In New Zealand, indoors: 20 kg per dwelling In both Australia and New Zealand, maximum total quantity (combined indoor and outdoor storage, cylinders not connected): 50 kg Maximum cylinder size: 15 kg In New Zealand, maximum cylinder size: 10 kg
— Multi-storey attached dwellings, up to and including 3 storeys	In Australia, indoors (including on balconies): 10 kg per dwelling In New Zealand, indoors: 20 kg per dwelling In both Australia and New Zealand, maximum total (combined indoor and outdoor) quantity: 20 kg per dwelling Maximum cylinder size: 10 kg
— Multi-storey attached dwellings, over 3 storeys	Maximum total (combined indoor and outdoor) quantity: 10 kg per dwelling Maximum cylinder size: 10 kg
Protected places and public places: — Outdoors and adequately ventilated areas, including open air, temporary structures (e.g. marquees, tents, booths) and under awnings. (See examples in Figure 2.1) — Outdoor and adequately ventilated areas of hotels, restaurants, cafes and take-away food shops. (See examples in Figure 2.1)	Per site: Maximum total quantity: 60 kg (see also Clause 2.4) Maximum cylinder size: 15 kg In New Zealand, maximum cylinder size: 10 kg
Protected places and public places: — Indoors, within buildings with a roof and three or more walls, e.g. hotels, bars, restaurants, cafes, take-away food shops	Per site: 10 kg per 10 m ² floor area, up to a maximum total quantity of 30 kg Maximum cylinder size: 15 kg In New Zealand, maximum cylinder size: 10 kg
Shops, offices and laboratories, indoors	Maximum total quantity: 30 kg per occupancy Maximum cylinder size: 15 kg In New Zealand, maximum cylinder size: 10 kg
Factories and warehouses, indoors	45 kg per 50 m ² floor area, up to a maximum total quantity of 180 kg per occupancy Maximum cylinder size: 45 kg

NOTES:

- 1 Outdoor areas include footpaths, enclosed gardens and public places (as defined in Clause 1.4.48).
- 2 Vehicle fuel containers on fork lift trucks or installed in motor vehicles in accordance with AS/NZS 1425, are not included in the quantities stated above.
- 3 See Appendix C for cylinder sizes and water capacities.

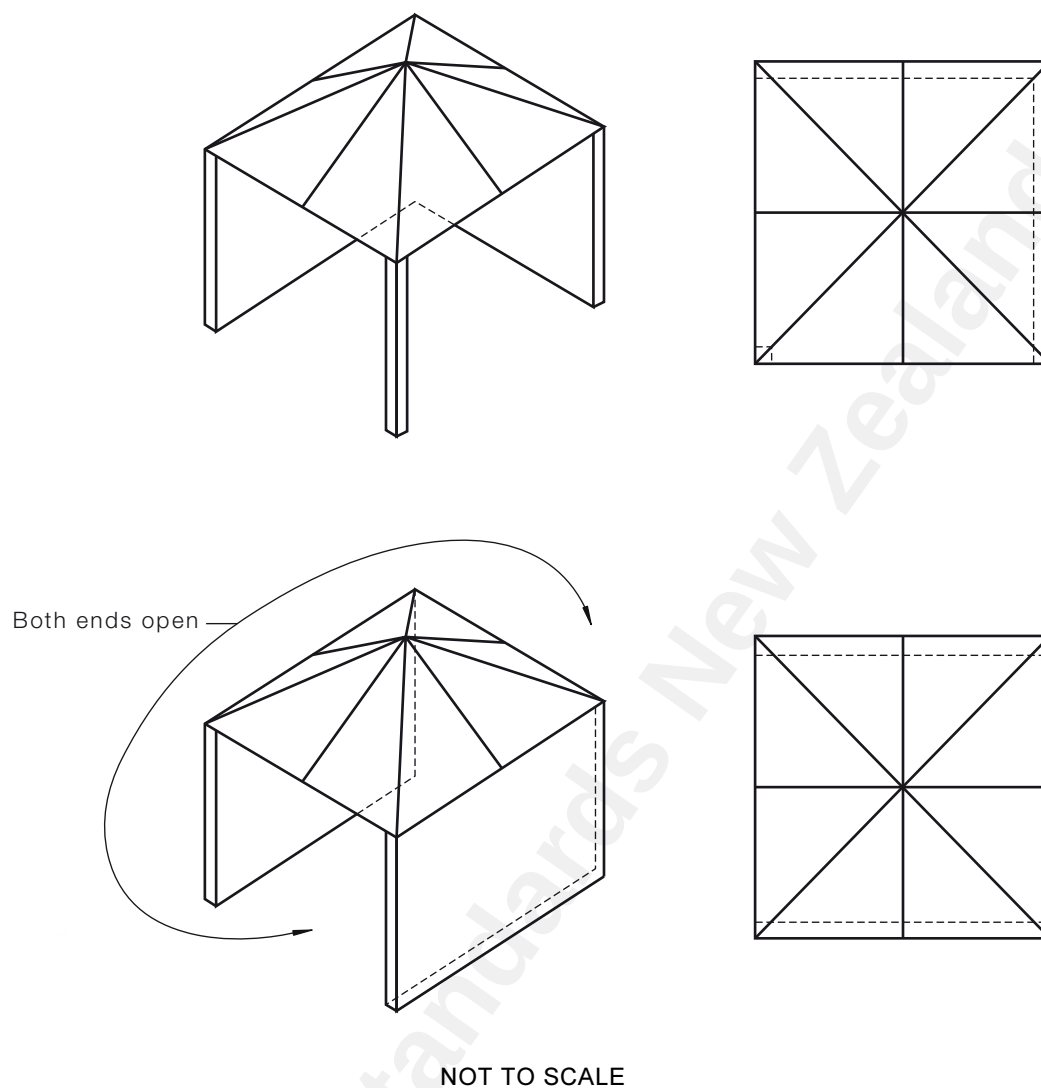
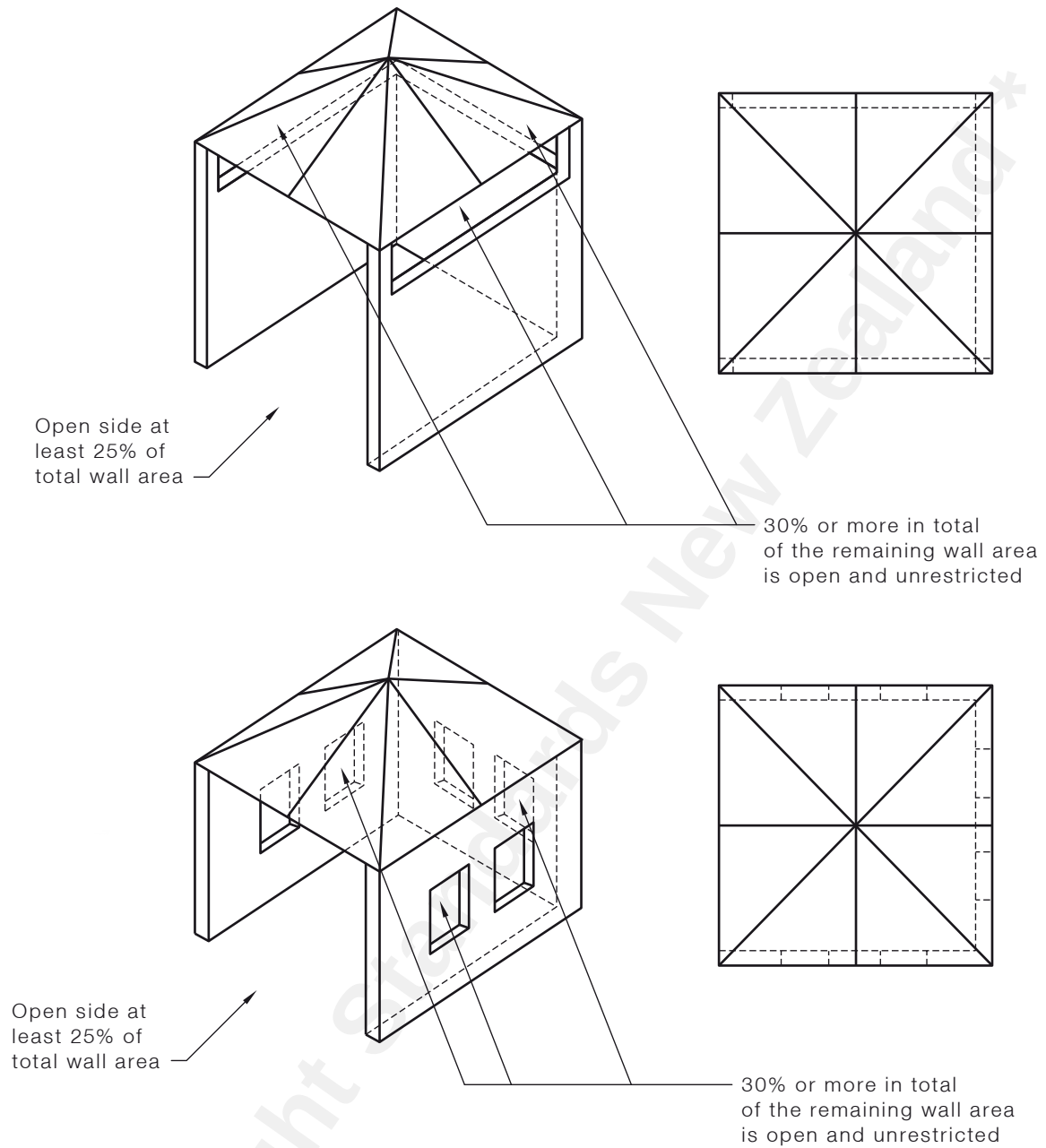


FIGURE 2.1 (in part) EXAMPLES OF ADEQUATELY VENTILATED AREAS



NOT TO SCALE

FIGURE 2.1 (in part) EXAMPLES OF ADEQUATELY VENTILATED AREAS

SECTION 3 GENERAL REQUIREMENTS

3.1 SCOPE OF SECTION

This Section sets out requirements and recommendations for the storage and handling of LP Gas that apply to both cylinders and tanks. Requirements and recommendations that are specific to certain types of storage, and to installations, piping and fittings, are given in later sections of this Standard.

3.2 RESTRICTIONS ON IGNITION SOURCES

Throughout this Standard, reference is made to hazardous areas as specified in AS/NZS 60079.10.1. Unless specifically stated otherwise, sources of ignition are not permitted within a hazardous area.

NOTE: Hazardous areas are not commonly defined for storages less than 25 L but may be required to be defined in some non-domestic situations, particularly where there is restricted ventilation.

A vehicle is not regarded as being an ignition source while it is entering or leaving a hazardous area but shall be turned off when standing or during any fuel transfers (except in the case of LP Gas road tankers).

Personal equipment, including pagers and mobile phones, shall also be considered as possible ignition sources and guidance on the requirements and use of such equipment in hazardous areas is included in AS/NZS 60079.14.

3.3 PIPES, VALVES AND FITTINGS

All pipes, valves and fittings for service with LP Gas liquid shall have a minimum design pressure rating of 2.4 MPa.

3.4 CONTROL OF ACCESS

All installations shall be appropriately secured against unauthorized access.

3.5 SEPARATION DISTANCES

3.5.1 General

Installations for LP Gas shall be separated from boundaries and protected places by the minimum distances relevant to each type of storage or use, as specified in the relevant sections of this Standard, with the additional requirements and qualifications as set out in this Clause (Clause 3.5).

3.5.2 Separation to protected places on adjoining properties

The separation distance to a protected place on another property shall be at least that given in the relevant table.

If this distance has been measured across a boundary, and alterations on the adjacent property result in a breach of the separation distance, the installation shall be modified or relocated to restore compliance, or taken out of service.

NOTES:

- 1 The use of a protected place distance measured to the boundary of the adjacent property is recommended to eliminate future problems, but such use is a commercial decision.
- 2 A formal agreement or covenant is recommended.

3.5.3 Changed site conditions

Where alterations on the installation site or an adjoining property, result in a breach of the separation distances or distances to fixed ignition sources, the affected installation (e.g. cylinders, tanks, vents, fill points, piping, etc.) shall be brought into compliance.

3.6 FIREWALLS AND VAPOUR BARRIERS

3.6.1 General

Building walls and fences shall be treated as firewalls or vapour barriers provided that they qualify as such.

For auto gas installations, see Clause 10.5.11.

Vapour barriers may be used to achieve the separation distances required to ignition sources and public places.

Separation distances shall be measured in a horizontal plane around the end of any intervening vapour barrier or firewall that complies with the following requirements as applicable:

- (a) *For ignition sources* The top of the vapour barrier or firewall shall be at least above the level of any hazardous area associated with a potential or actual LP Gas release point in normal operation. Safety valves, relief valves and drain valves are not considered release points.
- (b) *For public places* The top of the vapour barrier or firewall shall be at least 0.5 m above the level of any potential or actual LP Gas release point in normal operation. Safety valves, relief valves and drain valves are not considered release points.
- (c) The clearance between an LP Gas tank and a vapour barrier or firewall shall comply with Clause 6.3.2.

3.6.2 Vapour barriers

A vapour barrier shall comply with the following requirements:

- (a) The vapour barrier shall be impervious to LP Gas vapour over the whole of the area around which the separation distance is to be measured.
- (b) The vapour barrier shall be made of a non-combustible material. Its construction shall be sufficiently durable for the site conditions.

3.6.3 Firewalls

Where the separation distance is to a protected place, the vapour barrier shall also be a firewall. Separation distances shall be measured in a horizontal plane around the end of any intervening firewall that complies with the following requirements:

- (a) Comply with the requirements for vapour barriers; and
- (b) Have an FRL/FRR of 240/240/240 in accordance with AS 1530.4 or NZS/ISO 834.

A wall on an adjacent property shall not be used as a firewall unless by agreement between the parties concerned.

NOTES:

- 1 The use of a protected place distance measured to the boundary of the adjacent property is recommended to eliminate future problems, but such use is a commercial decision.
- 2 A formal agreement or covenant is recommended.

3.6.4 Thermal screens

Thermal screens may be provided as appropriate for the protection of facilities or personnel from heat radiation in event of a fire.

A thermal screen shall comply with the following requirements:

- (a) The thermal screen shall be constructed of non-combustible materials. It shall be firmly supported so as to be capable of withstanding short-duration LP Gas flash fires.

NOTE: Flash fires rarely exceed 20 s duration.

- (b) The thermal screen shall be of sufficient dimensions to ensure that any connections and associated pipework to the LP Gas vessel or any other operable fittings are not in line of sight from the area being protected.
- (c) For vertical tanks, the thermal screen need not be more than 0.5 m above the highest point of potential or actual release of LP Gas in normal operation. Safety valves, bleed valves and drain valves are not considered release points.
- (d) The thermal screen shall be at least 1.8 m high.
- (e) The thermal screen shall comply with the ventilation and access requirements of Clause 6.3.2.

NOTE: Blank masonry walls, or zinc-coated or pre-painted profiled steel sheeting with a minimum thickness of 0.42 mm and firmly supported on metal framework with 50 mm × 50 mm × 5.0 mm RHS uprights and rails or equivalent, are suitable.

SECTION 4 CYLINDERS AND CYLINDER SYSTEMS

4.1 SCOPE AND APPLICATION OF SECTION

This Section provides requirements and recommendations for the location, design and construction of cylinders, and for the supply of LP Gas from those cylinders, either directly from a cylinder to an individual appliance, or from single or multiple cylinders via a piping system for use away from the cylinder(s).

This Section also applies to the storage of cylinders and non-refillable containers ('cartridges') that are not connected for use.

NOTE: The design and construction of gas-burning equipment, and the installation of such equipment, is generally covered in other Standards.

4.2 CYLINDERS

4.2.1 Compliance with Standards

In Australia, cylinders shall comply with AS 2030.1.

In New Zealand, cylinders shall comply with the Hazardous Substances (Compressed Gases) Regulations.

NOTE: The New Zealand Environmental Protection Authority (EPA) Technical Guide—*Guide to Gas Cylinders* will assist in this regard.

4.2.2 Exchange-fill cylinders

A cylinder shall not be fitted with a gas release type of fixed liquid level gauge if it is—

- (a) intended to be refilled by mass (as described in Clause 8.2.1 for Australia and Clause 9.2.1 for New Zealand);
- (b) larger than 25 L capacity; or
- (c) larger than 50 L capacity automotive fuel container.

4.2.3 In situ fill cylinders

A cylinder for in situ filling applications shall be not less than 100 L capacity, and shall be provided with a filling connection that incorporates a 1¾ inch male Acme thread and one of the following:

- (a) A dual non-return valve and a fixed liquid level gauge which shall indicate a maximum liquid level equivalent to 4 L less than the standard filling level for a cylinder of that particular size.
- (b) A non-return valve and an automatic fill limiter (AFL) that stops the filling at 80% of the standard filling level. Such a device shall be appropriate for the orientation of the cylinder when being filled.

The pumping rate shall be appropriate to the cylinder size, so as to minimize the risk of overfilling.

In New Zealand, in situ fill cylinders shall conform with the New Zealand Environmental Protection Authority (EPA) Approved Code of Practice, *In-situ Filling of LPG Cylinders*, available from the Liquefied Petroleum Gas Association of New Zealand (LPGA).

4.2.4 Cylinders to be filled by decanting

Any cylinder intended to be filled by a decanting method that is monitored by means of a fixed liquid level gauge shall not exceed 25 L capacity, or 50 L capacity if used as a fuel container for a vehicle engine.

4.2.5 Liquid withdrawal cylinders

Any cylinder used for liquid withdrawal, other than an automotive fuel container, shall be marked in accordance with Appendix D.

4.2.6 Butane cylinders

Cylinders designed to contain butane exclusively shall incorporate connecting fittings that are not interchangeable with those for other LP Gas service.

4.3 REGULATORS AND PIPING FOR CYLINDER SYSTEMS

4.3.1 Regulators

Regulators shall comply with the following requirements:

- (a) Regulators for cylinder-supplied systems shall comply with Clause 5.5.8 and shall be installed in accordance with Clause 6.10.
- (b) The cylinder regulator shall be either—
 - (i) fixed rigidly to an adequate support that is independent of the cylinder and mounted with the diaphragm vertical and the vent pointing vertically downwards; or
 - (ii) for cylinders exceeding 400 L and having lockable domes, connected directly to the valve outlet, or as close as the fittings will practically allow.

Gas pressure regulators shall be located to permit drainage of any liquid back into the cylinder.

- (c) Pipework connecting the regulator to the cylinder shall comply with Clauses 4.3.3 and 5.6.
- (d) The vent terminal of the regulator shall have provision to prevent the entry of any material that could cause a blockage.

4.3.2 Location of regulator

The location of a regulator shall comply with the following:

- (a) Any first stage regulator shall be outdoors, except where the regulator is attached to a cylinder that is permitted to be used indoors (see Clause 4.4).
- (b) Second stage regulators shall be installed in accordance with AS/NZS 5601.1.
- (c) A single-stage regulator or the first stage of a multi-stage regulator shall be located so that the length of the piping that is subject to cylinder pressure is as short as practicable.

4.3.3 Piping

The following requirements apply to the piping in a cylinder installation:

- (a) The piping between a cylinder and a manifold or a first-stage regulator shall be either of the following:
 - (i) Annealed copper tube—
 - (A) complying with AS 1572;
 - (B) having an alloy designation of 122 in accordance with AS 2738; and

- (C) with a minimum wall thickness of 1.22 mm and a nominal size of 6 mm or 9 mm in industrial or commercial applications.
- (ii) A flexible pigtail—
 - (A) complying with AS/NZS 1869;
 - (B) having a maximum length of 600 mm;
 - (C) having a nominal size of 6 mm or 9 mm in industrial or commercial applications; and
 - (D) for a non-metallic lined flexible pigtail, having a maximum life of 6 years from the date of manufacture.
- (b) The piping as specified in Items (a)(i) and (a)(ii) shall be as short as practicable and not more than 1 m in length.
- (c) The cylinder and piping shall be installed such that any liquid formed in the piping will drain freely back into the cylinder. Any such liquid shall not be allowed to drain back to the regulator.
- (d) The relevant requirements of Clause 6.8 shall apply.
- (e) Manifold piping shall comply with the requirement of Clause 5.6.

4.4 CYLINDERS IN USE

4.4.1 General

This Clause (Clause 4.4) applies to the location of cylinders in use. A cylinder 'in use' is one that is connected to a system for the withdrawal of LP Gas.

4.4.2 Location of cylinders

All cylinders in use shall be located outdoors except where—

- (a) the cylinder is the fuel container for a vehicle that is used or garaged indoors; or
- (b) permitted by Section 2 or Clause 4.4.4.

4.4.3 Prohibited locations for cylinders

Any cylinder and its associated equipment shall not be installed in any of the following locations:

- (a) Within a building, except where permitted by this Standard.
- (b) Under a stairway.
- (c) In a location with restricted access, where inspection, refilling or exchange of the cylinder is restricted, obscured or hazardous to the operator.
- (d) Where nearby constructions, fences, walls or vapour barriers could prevent cross-ventilation.
- (e) Under a building, except where permitted by Clause 4.4.8.
- (f) Where the cylinder, or an incident involving the cylinder and its contents, could obstruct egress from a building.
- (g) Buried in the ground, (see Note).

NOTE: This does not limit innovative design in conjunction with appropriate authority.

4.4.4 Cylinders indoors

4.4.4.1 General

The use of LP Gas cylinders indoors is not recommended. Where an outdoor cylinder and piping system is not provided, indoor usage of cylinders shall be restricted to portable appliances and equipment as specified in Section 2 of this Standard or in Clauses 4.4.4.2 or 4.4.4.3.

The retention of reserve or exhausted cylinders indoors shall be avoided where practicable.

NOTE: In New Zealand, only the quantities set out in Table 2.1 are allowed indoors.

4.4.4.2 Volumes of LP Gas indoors—Cylinders

The following volumes and restrictions shall apply to LP Gas cylinders in use indoors in factories and warehouses:

- (a) Where the floor area is less than or equal to 200 m², Section 2 of this Standard applies.
- (b) Where the floor area is greater than 200 m², a maximum volume of 500 L (water capacity) of LP Gas per 200 m² of floor area applies.
- (c) Such groups of cylinders [as described in Item (b)] shall be separated by at least 15 m. Cylinders in such groups need not be manifolded, and may supply different consuming appliances.

4.4.4.3 Volumes of LP Gas indoors—Non-refillable containers

Where non-refillable containers of LP Gas are kept indoors, the following maximum quantities shall apply:

- (a) In buildings frequented by the public—100 L.
- (b) In warehouses—300 L in any one location. If additional storage locations are required on the same floor in the same building, they shall be separated by at least 10 m.

4.4.5 Cylinders outdoors

Cylinders that are installed outdoors shall be located in accordance with the following requirements:

- (a) Cylinders shall be separated from protected places, public places and railways lines by the distances given in Table 4.1.
- (b) Where cylinders are adjacent to a building, they may be placed next to each other in a group of up to 2500 L water capacity. Where there is more than one such group, there shall be a minimum distance of 3 m between the cylinder groups (see Figure 4.1).
- (c) Where a cylinder is adjacent to a building, and any opening into the building is below the level of the cylinder's pressure relief valve, the nearest cylinder shall be at least 1 m away, measured horizontally.

In New Zealand, if the quantity of LP Gas totals 100 kg or more, the separation distance to openings into buildings increases to 2 m.

- (d) Where a cylinder is installed beneath a window, the distance between the top of any cylinder valve and the bottom of the window's opening shall be at least that specified in Figures 4.2 and 4.3.

NOTE: Weepholes are not considered openings into the building for the purpose of this Clause.

4.4.6 Hazardous area requirements

The hazardous zone surrounding a gas cylinder, as specified in AS/NZS 60079.10.1, shall be maintained free of ignition sources.

NOTE: Appendix E illustrates hazardous areas for common cylinder situations where the aggregate exceeds 25 L. Under some circumstances, particularly in confined spaces, caution should also be taken with smaller quantities as these may also give rise to hazardous areas.

The following exceptions apply to the exclusion of ignition sources from such areas:

- (a) The presence of a door or window above a cylinder is deemed not to create a hazardous area within the building, provided that the installation complies with the requirements of Clause 4.4.5(d).
- (b) This requirement does not apply to the location of a flame or igniter of a gas appliance and its supplying cylinders where—
 - (i) the cylinder capacity does not exceed 25 L;
 - (ii) in Australia, both the cylinder and the appliance are located outdoors, either as a fixed installation or as a portable or mobile appliance unit; and
 - (iii) heat from the appliance will not cause the design temperature of the cylinder to be exceeded.

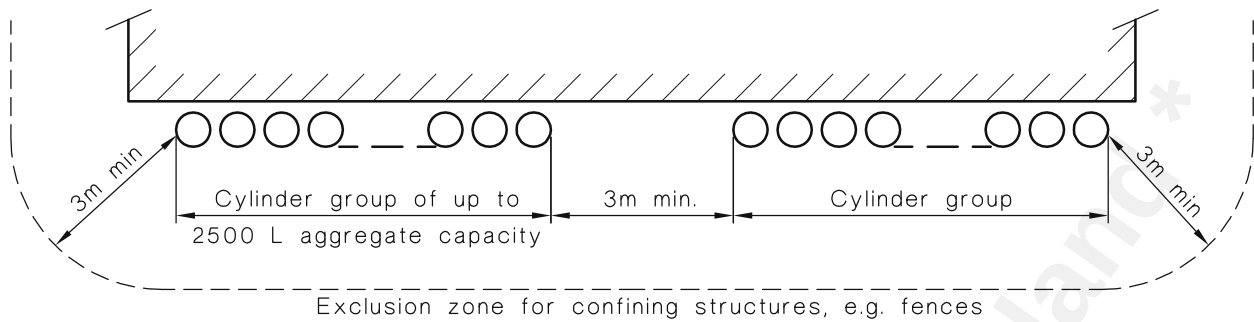
TABLE 4.1
CYLINDER LOCATIONS

1	2	3
Aggregate capacity of cylinders (water capacity)	Minimum distance from public places or railway lines	Minimum distance from a protected place
L	m	m
0 ≤1 000	0	0
>1 000 ≤2 500	1.5	3.0*
>2 500 ≤6 000	3.0	4.5
>6 000 ≤12 000	3.0	6.0
>12 000 ≤120 000	4.5	8.0
>120 000	8.0	15.0

* This distance may be reduced to zero where there are no other confining structures (other than the protected place in question), such as a solid fence or building, within 3 m.

NOTES:

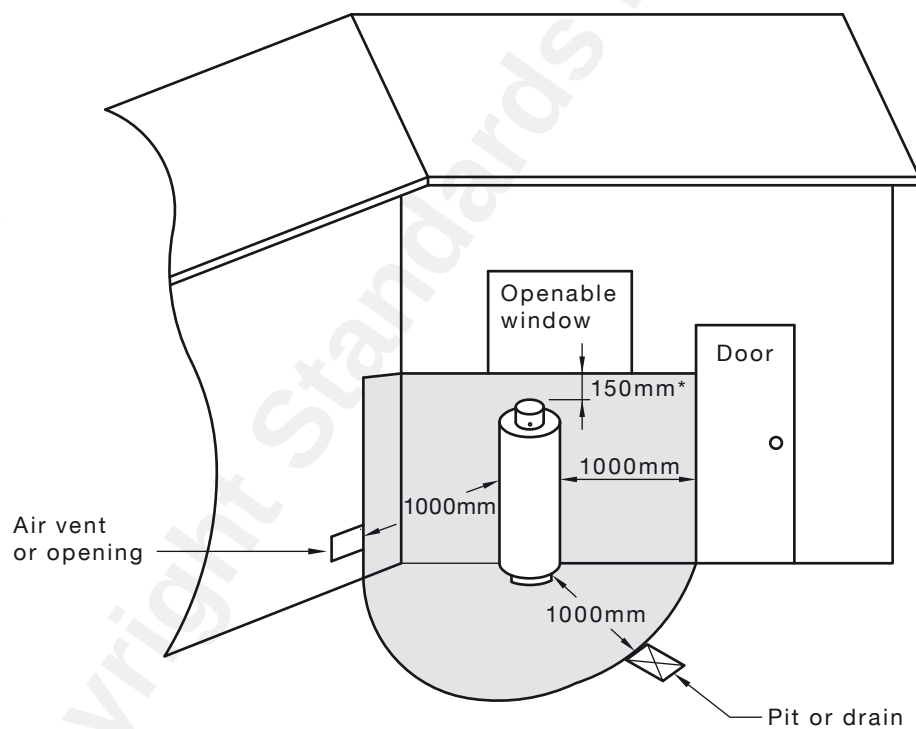
- 1 The distances are measured horizontally.
- 2 The boundary from adjacent private property is not considered to be a protected place. A separation distance may be measured across a boundary to the nearest protected place on that site, if any, but Clause 3.5.2 imposes a condition that is a potential penalty if the adjoining property is further developed.



NOTES:

- 1 Cylinders in groups need not be manifolded and may supply different installations.
- 2 Additional requirements apply if the cylinders are filled in situ. See Clause 4.4.11.
- 3 See Table 4.1 for separation distances.
- 4 Separation distances may be measured in a horizontal plane around the end of any intervening vapour barrier or firewall (see Clauses 3.5 and 3.6).

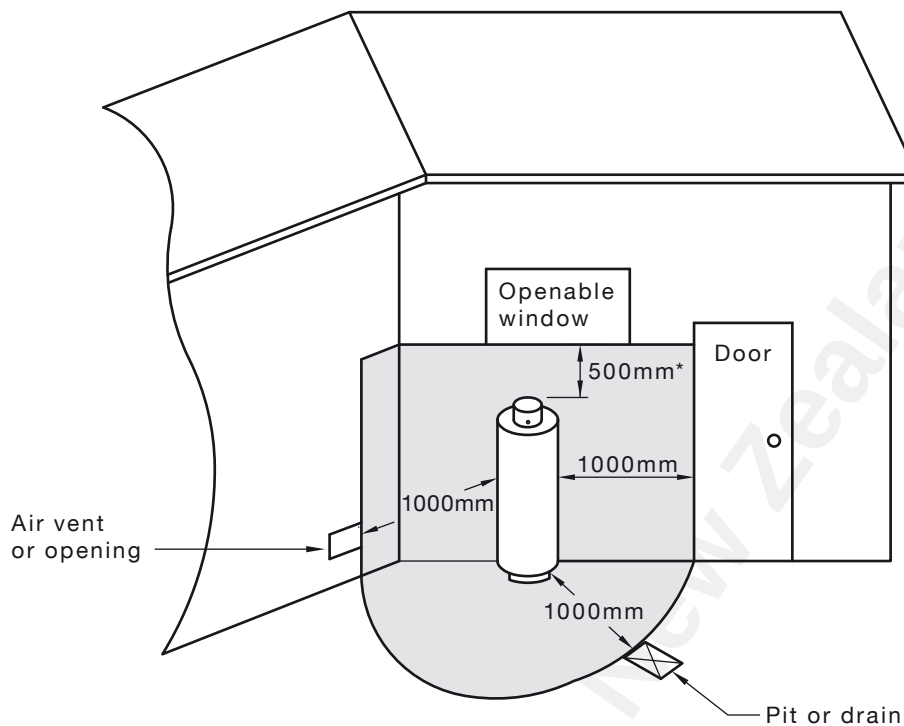
FIGURE 4.1 SEPARATION DISTANCES FOR CYLINDER GROUPS



* This distance is measured from the top of any cylinder valve

NOTE: In New Zealand, if the quantity of LP Gas totals 100 kg or more, the separation distance to openings into buildings increases to 2 m.

FIGURE 4.2 EXCHANGE CYLINDER LOCATION



* This distance is measured from the top of any cylinder valve

NOTE: In New Zealand, if the quantity of LP Gas totals 100 kg or more, the separation distance to openings into buildings increases to 2 m.

FIGURE 4.3 IN SITU FILL CYLINDER LOCATION

4.4.7 Cylinders on a verandah

Where cylinders are located on a verandah, the following requirements apply:

- (a) The location shall be nominally at ground level.
- (b) The verandah shall be open on all three sides.
- (c) The wall against which the cylinders are located shall be free of windows and doors as required by AS/NZS 5601.1.
- (d) The verandah floor shall be free of drains as required by AS/NZS 5601.1.
- (e) The hazardous area around the cylinders shall be free of fixed ignition sources, as illustrated in Appendix E.
- (f) The cylinders shall not be subject to physical damage, heat or vibration.
- (g) A maximum of 90 kg shall be kept on a verandah complying with this Clause.

The location of such cylinders is shown in Figure 4.4.

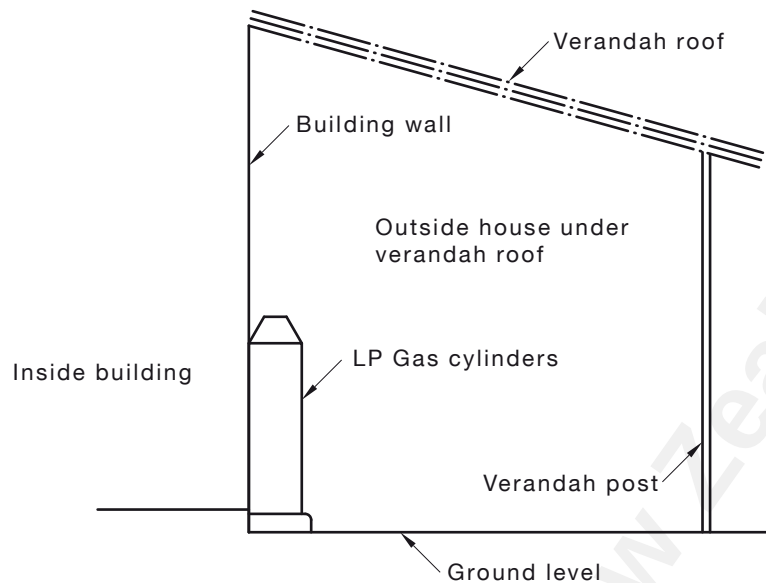


FIGURE 4.4 LOCATION OF CYLINDERS ON A VERANDAH

4.4.8 Cylinders under a building supported by piers

Where a cylinder is located under a building supported by piers, the following requirements apply:

- (a) There shall be a vertical clearance of at least 800 mm between the top of the neck ring of the cylinder and the underside of any overhanging part of the building.
- (b) No part of the cylinder shall be more than 800 mm within the perimeter of the building's walls (see Figure 4.5, side view).
- (c) The area between the piers shall be—
 - (i) open on at least three sides; or
 - (ii) enclosed by a construction through which cross-ventilation can occur (e.g. slats or battens) on at least three sides; or
 - (iii) a combination of Items (i) and (ii) above.

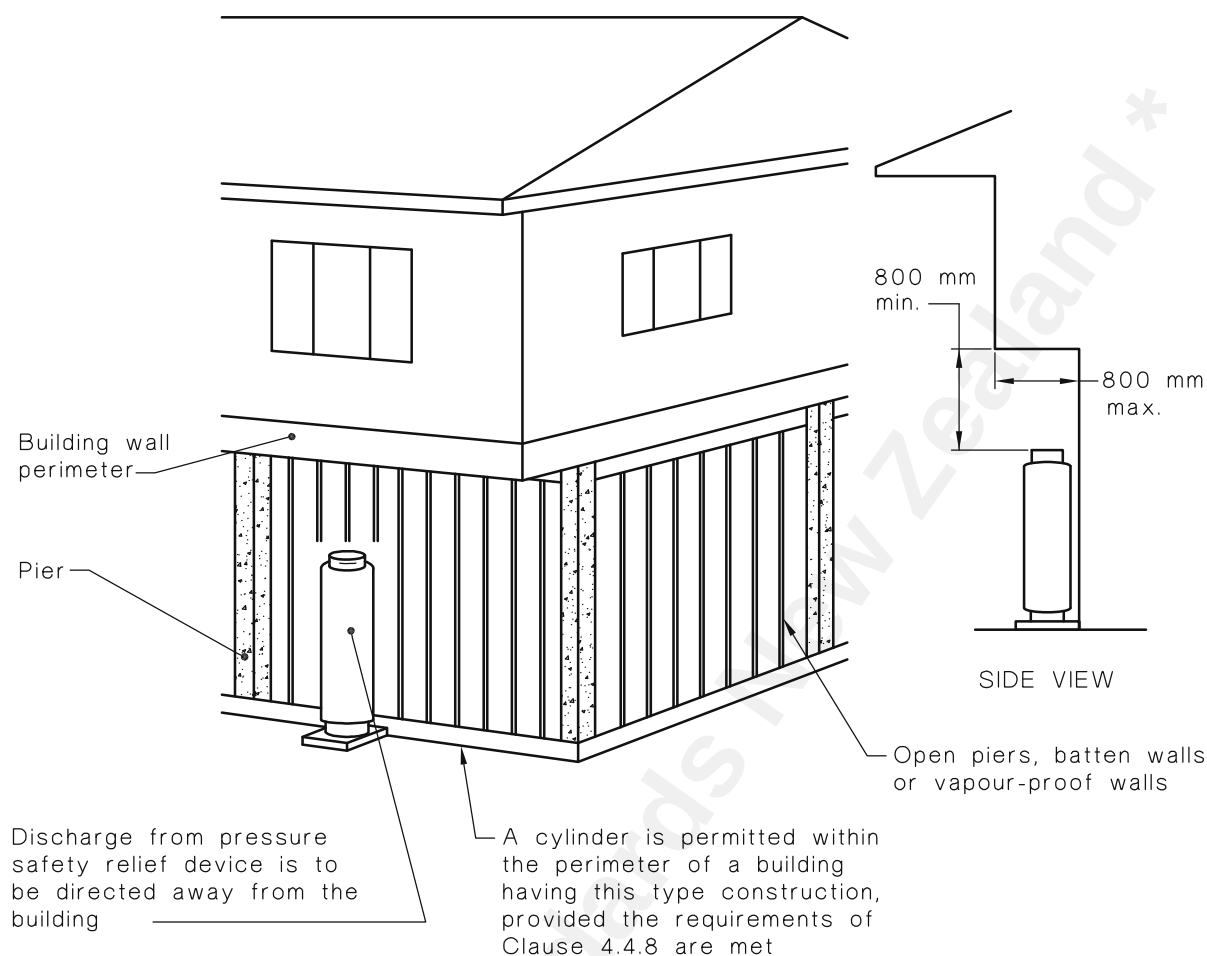


FIGURE 4.5 LOCATION OF CYLINDER UNDER A BUILDING

4.4.9 Cylinders in enclosure or recess

Where a cylinder is to be installed in an enclosure or recess, the enclosure or recess shall be designed to—

- (a) house cylinders and their associated equipment only;
- (b) allow free unimpeded discharge from each cylinder safety valve;
- (c) ensure that water will not accumulate on the base; and
- (d) ensure the cylinder and its associated equipment are clear of the surrounding soil.

Enclosures of sheet metal or similar impervious construction shall have ventilation openings at the top and bottom with each opening providing a free area of at least 0.02 m^2 for every cylinder enclosed.

4.4.10 Cylinders in public locations

Where a cylinder exceeding 12 L capacity is permanently installed for public use in a location having unrestricted public access, the following requirements shall apply:

- (a) Either—
 - (i) the cylinder and any associated regulator shall be enclosed in a lockable cage or cabinet ventilated at the top and bottom; or
 - (ii) the cylinder valves and regulator shall be guarded by a covering metal hood and the cylinder protected against accidental dislodgment.

- (b) The connecting piping shall be arranged so that it is not vulnerable to tampering or accidental impact.

Any lockable enclosure as described in Items (a)(i) and (a)(ii) shall be kept locked when not in use.

These requirements do not apply to cylinders on caravans or mobile homes, or to cylinders attached to any cabins for hire or domestic dwelling associated with the site.

4.4.11 Installing cylinders

NOTE: Authorities in most areas permit only specifically licensed persons to make, repair, alter, or open connections in any installation for LP Gas cylinders, except for their actual connection or disconnection.

Cylinders shall be installed in accordance with the following requirements.

- (a) Cylinders shall be installed on a firm, level, non-combustible base, not resting on soil. The floor or base shall be constructed so that water cannot accumulate within any enclosure or recess.
- (b) Cylinders shall not be stacked on top of each other.
- (c) Any cylinder shall be restrained to prevent from falling.
NOTE: This includes a cylinder located in an area likely to be subject to flooding or seismic activity. Cylinders greater than 200 L or less than 25 L are typically inherently stable.
- (d) In New Zealand, any cylinder larger than 25 L shall be restrained against seismic activity.
- (e) Except for domestic installations, where there are manoeuvring vehicles, cylinders shall be afforded suitable protection, e.g. by the use of bollards, or a fully contained, free-moving frame which encapsulates all cylinder components.

NOTES:

- 1 A frame may be used where the cylinder is not connected to external pipework.
 - 2 See Appendix Q for information on impact protection.
- (f) Cylinders shall not be installed below ground level unless there is sufficient ventilation to prevent the accumulation of any leaking gas.
 - (g) A cylinder shall be installed so that the pressure-relief valve is in contact with the vapour space and any discharge from this valve is directed away from any adjacent cylinders or combustible structures.
 - (h) A cylinder intended to be exchanged or removable shall be connected to a fixed piping system complying with Clause 4.3.3.
 - (i) Vapour-service cylinders shall be installed so that the point of LP Gas withdrawal is in contact with the vapour space.
 - (j) Where two or more exchange cylinders are connected for use, a means shall be provided to permit the cylinders to be isolated from the gas supply, to allow the cylinders to be disconnected. This shall be achieved by either of the following:
 - (i) A manual or automatic changeover valve shall be installed immediately upstream of the regulator serving the system.
NOTE: Such a valve may be an integral part of an automatic change-over regulator.
 - (ii) Isolating valves shall be installed on the manifold, to allow each cylinder to be individually disconnected from the manifold without depressurizing the manifold.

4.4.12 Additional requirements for cylinders for in situ filling

An installation of cylinders intended for in situ filling shall comply with the following requirements:

- (a) Cylinders other than those specifically fitted for the purpose shall not be used for in situ filling installations (see Clause 4.2.3).
- (b) In situ fill cylinders shall be located outdoors.
- (c) In situ cylinders shall only be filled by connection of the delivery hose directly onto the fill valve on the cylinder.
- (d) An opening into a building shall not lie within an area 0.5 m above and 1 m laterally from the cylinder or its valves and fittings (see Figure 4.3).
- (e) In situ filling shall be conducted in accordance with the requirements of Clause 11.12 and with regard to the hazardous areas defined in AS/NZS 60079.10.1 for in situ fill cylinder locations (see also Figure 4.3).
- (f) A cylinder having a capacity of 200 L or less shall be restrained and prevented from falling.
- (g) The cylinder shall be placed on a suitable base, raised at least 50 mm above the surrounding level, the top surface of which shall be such as to shed water from the base of the cylinder.
NOTE: Surfaces that are prone to indentation over time, e.g. bitumen (which could soften when exposed to heat), are not considered suitable because of the possibility of water accumulating in the depressions.
- (h) Where a cylinder installation is vulnerable to falling objects, the cylinder valves, regulator and piping shall be provided with protective guards. Particular attention shall be paid to the risks of impact or cylinder dislodgment due to ice and snow, where appropriate.

In New Zealand in situ fill cylinders shall conform to the New Zealand Environmental Protection Authority (EPA) Approved Code of Practice HSNOCOP 38, '*In-situ Filling of LPG Cylinders*' available from the LPGA.

4.5 CYLINDER STORAGE

4.5.1 General

This Clause (Clause 4.5) sets out requirements for LP Gas cylinders that are stored but not connected for use.

4.5.2 Site requirements

The following requirements and recommendations apply to the area in which cylinders are stored:

- (a) The location shall ensure that cylinders are not liable to physical damage, tampering, or excessive temperature rise. This requirement shall not be taken to preclude storage in an open area, exposed to the sun.
- (b) The standing area (other than a floor) shall be level, non-combustible and not prone to indentation such that water could accumulate or the cylinders could be dislodged. Any floor shall be concrete or other non-combustible material, hardwood or other suitable material, with or without suitable floor coverings.

Where there is a space between the floor and the ground, such a space shall be either completely filled with a non-combustible solid material or ventilated in accordance with Clause 8.3.3 (for Australia) and in accordance with Clause 6.2.3 (for New Zealand).

- (c) Buildings used for cylinder storage shall have non-combustible cladding for walls and roof and shall be ventilated by natural or mechanical means in accordance with the ventilation requirements of Clause 4.5.5.
- (d) Cylinders in a mixed cylinder storage area shall be segregated from any oxidizing gases by at least 3 m. This segregation distance may be measured horizontally around a vapour barrier complying with the requirements of Clause 3.6 and as illustrated in Figure 4.6.
- (e) Cylinders awaiting despatch may be kept on a filling platform in accordance with Clauses 8.3.2 and 8.3.3 (for Australia) and in accordance with Clauses 9.2.3 and 9.3.4 (for New Zealand).

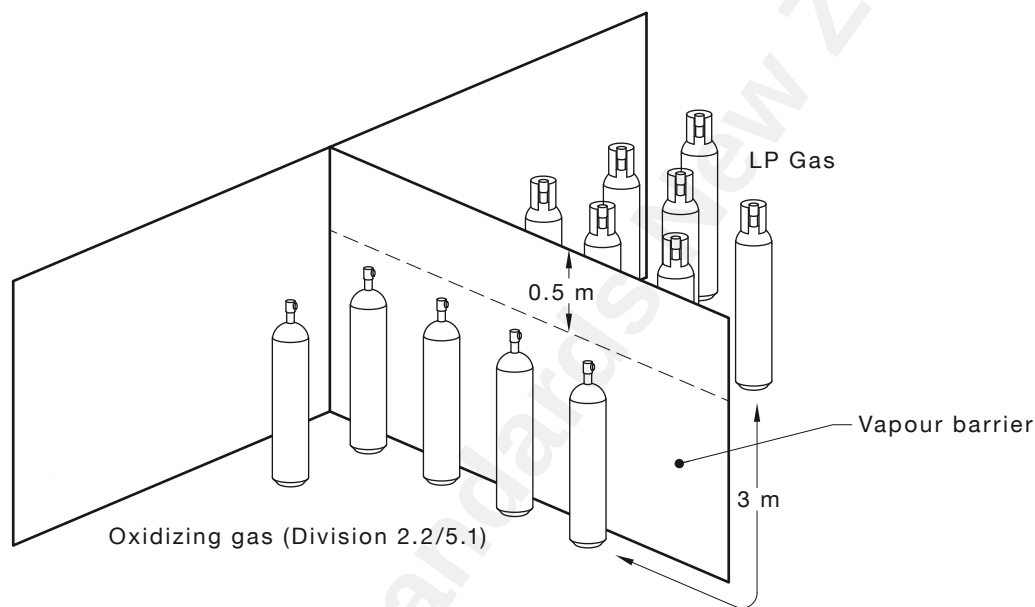


FIGURE 4.6 USE OF VAPOUR BARRIER TO ACHIEVE SEGREGATION

4.5.3 Separation distances

Cylinders stored in the open or within a building shall be located at the distances given in Table 4.1, and the following requirements shall apply:

- (a) A cylinder shall be at least 1 m horizontally away from an opening into, and shall be outside of, any building that is not used solely for storage, filling and/or handling of gas cylinders.
- (b) The distance between any cylinder and any above-ground LP Gas storage tank or flammable or combustible liquid storage that exceeds 250 L capacity shall be at least 3 m.

Cylinders shall not be stored within any compound (bunded area) for flammable liquid storages.

- (c) The distance between any two groups of cylinders shall be not less than the protected-place distance for the larger group. Otherwise, the separation distance shall be based on the aggregate volume of the two groups.
- (d) Cylinder filling locations complying with Section 8 in Australia, or Section 9 in New Zealand, shall not be treated as protected places.
- (e) Clause 3.6 shall apply.

4.5.4 Cylinder storage and handling

The following additional requirements shall apply to the storage of cylinders, including those that are empty but not gas free:

- (a) The outlet valve shall be kept closed or the cylinder opening plugged.
- (b) Where a cylinder is designed to incorporate a detachable valve cap, the cap shall be kept in place when the cylinder is not in use.
- (c) Cylinders shall be handled carefully and not allowed to fall upon one another or be otherwise subjected to undue shock.
- (d) Cylinders shall be secured to prevent movement or physical damage. Valves shall be safeguarded against physical damage in accordance with AS 2473.
- (e) Cylinders shall be placed so that the safety relief device will always be able to vent the vapour space.

4.5.5 Ventilation of cylinder stores

4.5.5.1 General requirements

Cylinder stores shall be provided with adequate ventilation complying with the following requirements:

- (a) The ventilation system shall—
 - (i) be capable of diluting and removing any vapour or gas from the store, so that such levels within the store are within workplace exposure limits and below any possible explosive levels in normal situation;
 - (ii) provide sufficient fresh air from outside, to reduce any risk of asphyxiation;
 - (iii) ventilate the atmosphere in the store, at both high and low levels relative to the floor; and
 - (iv) ventilate or mechanically exhaust directly to outside the store, to a location away from building entrances, doors or windows, airconditioning intakes, sources of ignition and areas where people are likely to congregate.
- (b) The requirements of (a) shall be satisfied by either—
 - (i) a natural ventilation system, as specified in Clause 4.5.5.2; or
 - (ii) a mechanical ventilation system, as specified in Clause 4.5.5.3.

Compliance with the above may not achieve a reduction in hazardous areas as specified in AS/NZS 60079.10.1.

4.5.5.2 Natural ventilation

A naturally ventilated store shall have one of the following:

- (a) Two opposing external sides that are open from floor to ceiling [see Figure 4.7(a)].
- (b) One external side that is open, provided that the length of the open side is at least twice the distance of that wall from the opposite side [see Figure 4.7(b)].
- (c) Vents in at least one pair of opposing external sides [see Figure 4.7(c)], provided that—
 - (i) the distance between the opposing external walls does not exceed 10 m;
 - (ii) in every two-metre length of external opposing walls, there are at least two vents, one positioned immediately above the floor and the other positioned immediately below the ceiling;

- (iii) the total area of vents per meter length of wall is at least 0.1 m^2 ; and
- (iv) vents are evenly distributed along the walls.

Care shall be taken to ensure that any nearby embankments, excavations, or retaining walls do not interfere with the operation of the natural ventilation system.

NOTE: The openings to a natural ventilation system may be covered by one of the following:

- (a) Wire mesh or grille.
- (b) Louvres.
- (c) For open, external sides only, chequered brickwork, vent bricks, slotted roller doors or equivalent.
- (d) Fire dampers, where required.

4.5.5.3 Mechanical ventilation

A mechanical ventilation system shall comprise fans, ducts, entry and exit registers and controls, and fire dampers where required. Such a system shall comply with the following:

- (a) Exhaust ventilation shall be used within the store.
- (b) The capacity of the system, in cubic metres per second for every square metre of floor space, shall be—
 - (i) for stores in which cylinders are kept but no LP Gas is decanted or filled, 0.005; or
 - (ii) for stores in which LP Gas is filled or decanted, 0.0075.
- (c) The air velocity at the register shall be at least 5 m/s.
- (d) Where localized exhaust ventilation is used, not more than 75% of its capacity shall be located at any one point. At least 25% of the capacity shall be available to ventilate the remainder of the store.
- (e) Any air intake or exhaust duct shall terminate in the open air, at distances of at least—
 - (i) 2 m from any opening to a building;
 - (ii) 4 m from the outlet of any chimney or flue; and
 - (iii) 3 m above the ground.
- (f) The ventilation system shall be designed so as to operate either—
 - (i) continuously; or
 - (ii) for sufficient time to allow two complete air changes before entry and continuously during occupation.

NOTE: Reference should be made to AS 1668.2 for requirements for mechanical ventilation systems.

4.6 EXCHANGE FACILITIES FOR PORTABLE CYLINDERS

Where portable exchange cylinders are kept in cages at service stations or other retail outlets, the requirements of Appendix F for Australia, or Appendix G, for New Zealand, respectively, shall apply.

The location of such facilities is illustrated in Figures F1 and G1 for Australia and New Zealand, respectively.

4.7 STORAGE OF LP GAS CYLINDERS ON TROLLEYS

LP Gas and oxygen cylinders may be stored together on trolleys for the purpose of LP Gas/oxygen cutting, brazing, melting, heating or similar purposes. In such situations—

- (a) the aggregate capacity of cylinders on any trolley shall not exceed 160 L; and
- (b) the aggregate capacity of LP Gas cylinders shall not exceed 110 L.

NOTE: AS 4839 should be consulted regarding the safe use of such equipment.

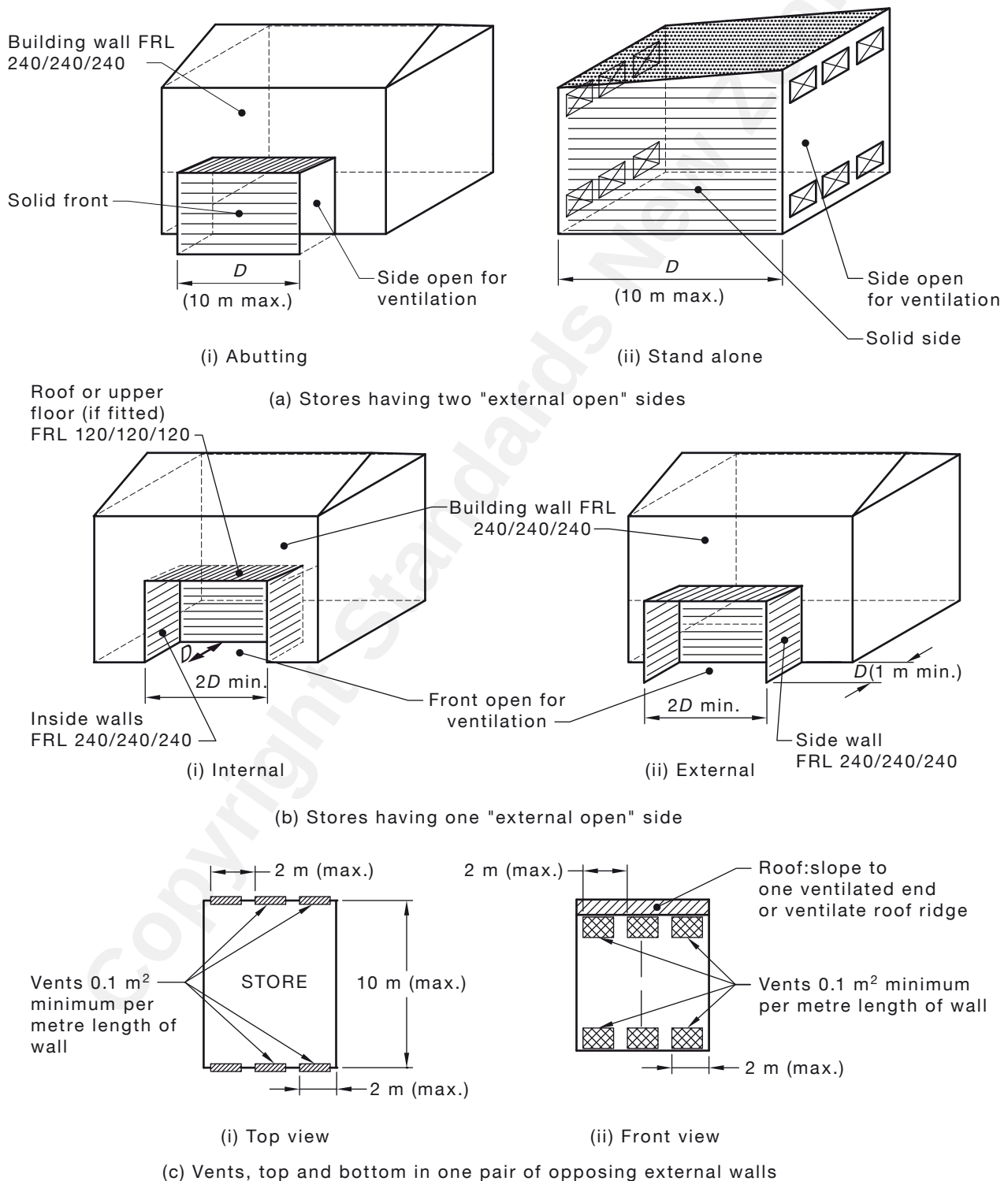


FIGURE 4.7 SCHEMATICS OF NATURALLY VENTILATED STORES

4.8 CYLINDER LOADING AND UNLOADING AREAS

4.8.1 General

An area may be established, in order to facilitate the loading and unloading of cylinders from vehicles. This area shall be outside of any hazardous area around the cylinder filling point or cylinder storage area.

Cylinders shall not remain in the loading and unloading area for longer than necessary to assemble a load for transportation or to remove them to the filling area for refilling.

The cylinder loading and unloading area shall be clearly identified by marking or signs.

4.8.2 Location of cylinders in loading and unloading areas

Cylinders shall be located in accordance with Clause 4.5.3, Items (a) and (b).

4.8.3 Ignition sources

Vehicles involved in the loading and unloading of cylinders within the area shall not be considered sources of ignition whilst engaged in loading or unloading the cylinders.

SECTION 5 COMPONENTS FOR TANK SYSTEMS

5.1 SCOPE OF SECTION

This Section specifies requirements for the design of components for LP Gas tank storage facilities.

Other Sections prescribe methods of installation for other types and locations of storages.

5.2 TANK DESIGN AND CONSTRUCTION

5.2.1 Materials of construction

An LP Gas tank shall be of steel, and shall comply with a pressure vessel design code in accordance with AS/NZS 1200.

5.2.2 New Zealand requirements

In New Zealand, if site-specific or application-specific requirements do not apply, an LP Gas vessel and its supports shall be as follows:

- (a) It shall be designed for LP Gas service with the vapour pressure being that of propane.
- (b) It shall have an above-ground service design temperature range of—
 - (i) for 1 kL to 5 kL -10°C to $+50^{\circ}\text{C}$;
 - (ii) for >5 kL to 26 kL -10°C to $+45^{\circ}\text{C}$; and
 - (iii) for above 26 kL -10°C to $+40^{\circ}\text{C}$.
- (c) It shall be designed to meet seismic loading in accordance with NZS 1170.5 (for all seismic zones).
- (d) It shall be capable of being filled with water for testing or purging purposes.

5.2.3 Australian requirements

In Australia, the following requirements apply:

- (a) Where AS 1210 is used, the design pressure shall be not less than that shown in Table 5.1, except where a design pressure for a specific condition is applied, when that condition is known. Examples are as follows:
 - (i) The service design temperature has been determined in accordance with AS 2872 and the maximum vapour pressure for the contents has been determined.
 - (ii) The tank is designed for the storage of an LP Gas for which the limits of composition and vapour pressure are known. Unless vapour pressure is calculated from the proportions of a known gas mixture, it shall be based on the use of propane.

NOTE: The more sophisticated special-purpose design techniques mentioned above are of no practical advantage for small tanks.

- (b) An underground tank shall incorporate lifting provisions, so that its protective coating is not damaged when the tank is handled.
- (c) A portable tank shall not exceed 8 kL capacity.

NOTE: Such a tank may be of Class 2A construction.

- (d) A skid tank shall not exceed 8 kL capacity. Lifting facilities shall be incorporated, and all fittings shall be internal unless a guard complying with AS 1210 is provided.
NOTE: Such a tank may be of Class 2A construction.
- (e) A demountable delivery tank shall comply with the requirements of AS 2809.3 for cargo tanks.
- (f) A freight container tank shall comply with AS 3711.6 and, if for service within Australia, the specific requirements herein for LP Gas tanks.
- (g) Any structural provisions for supporting the tank shall be designed in accordance with Clause 6.4.
- (h) Seismic loading shall be taken into account for any tank whose capacity exceeds 100 kL (see AS 1170.4). For vertical tanks, seismic and wind loading shall be taken into account for any capacity.

TABLE 5.1
DESIGN PRESSURES

Capacity-to-surface-area ratio of tank*	Design pressure, MPa	
	Propane	Butane
<0.15	1.90	0.825
≥0.15 ≤0.42	1.75	0.7
>0.42	1.55	0.525

* 'Capacity/area' as used here has the same meaning as the term 'volume/area' as used in AS 2872.

NOTE: These design pressures are valid for above-ground tanks only if the tanks are painted white. AS 1210 specifies penalties for any tank colour having a lesser heat-reflecting ability.

5.3 CONTROL OF LP GAS OUTFLOW

5.3.1 Minimization of outflow

Every opening through the tank shell shall be fitted with a means of preventing or minimizing the accidental or uncontrolled outward flow of LP Gas by the use of one of the following devices, as described in Table 5.2 and appropriate to the purpose of the opening:

- (a) A safety valve.
- (b) Primary and secondary shut-down systems of valves.
- (c) A flow-limiting orifice not greater than 1.4 mm diameter.
- (d) A plug or blank flange.
- (e) A multi-function valve (limited to openings into the vapour space of any single above-ground tank of 8 kL capacity or less). The vapour withdrawal portion shall incorporate all of the following features:
 - (i) A manual shut-off valve with attached hand-wheel or equivalent that is fitted directly to the opening, or is an integral part of or attached to a substantial fitting that is attached to the opening.
 - (ii) The opening through the multi-function valve is restricted by an orifice not exceeding 8 mm diameter, and located between the vapour space and the outlet of the shut-off valve.

- (iii) A regulator is attached directly to the multi-function valve, or is connected to it by a flexible annealed copper tube not more than 1 m long and is supported and protected on or at the tank; otherwise the vapour outlet is sealed with a metal plug.

An opening through a tank shell implies that the opening is the size of the tank nozzle as manufactured. However, reduction by means of a bush or flange is permitted, provided that no more than one such fitting is employed.

5.3.2 Safety valve

There shall be no valve or restriction of any type between a safety valve and the vapour space of the tank, except where reserve safety valves are provided under the conditions specified in Clause 5.4.3.

5.3.3 Openings for gauges

Where a gauge can be operated or serviced without opening a gas path through to the interior of the tank, the gauge may be treated as a blank flange or a plug, requiring no other protection.

Otherwise one of the following shall apply:

- (a) The opening shall be protected by—
 - (i) an orifice not greater than 1.4 mm diameter; or
 - (ii) an excess-flow valve.
- (b) The tank shall be depressurized prior to the gauge's service.

5.3.4 Shut-down system

5.3.4.1 General

Clauses 5.3.4.2 and 5.3.4.3 below should be read as specifying functions and not necessarily requiring self-contained and independent valves. The primary and the secondary functions may be incorporated in a single compound valve provided that all the functional requirements are provided.

5.3.4.2 Primary shut-down system

The primary shut-down system shall consist of a non-return valve except that, where outward flow is necessary, an excess-flow valve may be used. The design and installation of this valve shall be such that it will continue to function as a shut-down device should impact or shear-off occur to any external portion or attachment.

Where a multi-function valve complying with Clause 5.3.1(e) is used, a primary shut-down system is not required.

5.3.4.3 Secondary shut-down system

Where the primary shut-down system is a non-return valve, the secondary system shall comprise at least a manual positive shut-off valve, or, if the tank is 8 kL or less and the entry is for filling into the vapour space, another non-return valve that is gastight.

Where the primary shut-down provision is an excess-flow valve, the secondary system shall comprise—

- (a) for vapour connections, at least a manual positive shut-off valve; or
- (b) for liquid connections, an emergency shut-down system. For an opening into a tank that is not larger than 3/4 NPT and is used only for decanting into cylinders, a manual positive shut-off valve is permissible.

5.3.5 Tanks larger than 200 kL

The primary and secondary shut-down provisions may consist of a manual shut-off valve and another valve that can be shut off by remote actuation, by fire, or both. In such cases, both valves shall be located away from the tank shell and outside the horizontal outline (plan view) of the tank, provided that—

- (a) the pipe connecting to the tank is integral with the tank and designed as part of it;
- (b) the internal diameter of the attached pipe is not less than 100 mm;
- (c) the pipe is anchored and shear-protected at its outer end as for a remote connection;
- (d) the piping between the anchorage and the tank is protected from impact; and
- (e) the LP Gas installation, or the LP Gas portion of a composite installation, is continuously manned by persons able to operate the shut-off valves during that period when the tank is in service, i.e. when the valves are open.

5.4 SAFETY VALVES

5.4.1 Provision of safety valves

Safety valves shall be provided as required by the pressure vessel design code being used, with the following qualifications:

- (a) A safety valve shall communicate with the vapour space in the tank.
- (b) Safety valves shall be arranged so that the possibility of tampering is minimized. Any externally located pressure-setting adjustment shall be provided with a means of sealing or locking after setting.
- (c) Safety valves shall comply with AS 1271 or UL132 or other recognized Standard.

For Australia, the aggregate discharge capacity of the safety valve(s) shall not be less than that determined in accordance with Appendix H.

NOTES:

- 1 Care is necessary in determining the setting and capacity of safety valves for tanks with provisions for controlling the service temperature [see Clause 5.2.3(a)(i)] if the temperature of the LP Gas delivered is likely to be higher than the controlled design temperature.
- 2 Appendix H has been derived from the general formula of AS 1210, simplified to suit the particular case of LP Gas.

Where a stack pipe is fitted, a shear point shall be fitted downstream of the safety valve to ensure that impact or high wind loads cause failure at the shear point, the safety valve remains in the tank and containment is maintained.

5.4.2 Discharge provisions

Where installed, a safety valve and any discharge piping shall comply with the following requirements:

- (a) The size of any discharge piping shall be sufficient to ensure that the required discharge capacity of a safety valve is achieved. The effect of backpressure shall be taken into account in the case of changes of direction or excessive length.
- (b) The direction of discharge shall be vertically upwards.
- (c) Except for measures necessary to prevent ingress of rainwater or blockage, venting shall be direct to atmosphere.
- (d) The direction of discharge shall be such that escaping vapour will not impinge on any tank, fitting, structure or piping.

- (e) A safety valve shall not be made inoperative except under the conditions permitted in Clause 5.4.3.
- (f) For a tank of more than 8 kL capacity, the discharge shall terminate at least 2 m above the tank, except where Clause 7.5.2 applies.
- (g) Where a lockable dome or cage is provided, the arrangement shall comply with the requirements relating to free dispersal or impingement of discharge.
- (h) The discharge outlet shall be protected as far as practicable from physical damage.
- (i) A cap shall be provided to prevent the accumulation of rain, dew or foreign matter within the safety valve. Where a discharge pipe is fitted, the cap shall be fitted at the point of discharge. The cap shall not prevent the safety valve from operating effectively or prevent the safe and effective discharge of LP Gas. Where a weep hole or drain is provided in the cap, the overall design shall be such that any ignited LP Gas discharge cannot impinge on the tank, adjacent tanks, piping, or equipment.
NOTE: A deflector or a drainage discharge pipe is permissible, provided that such piping does not incorporate a valve.
- (j) The safety relief valve shall be protected from impact or high wind loading on the discharge piping that could damage the valve.

5.4.3 Reserve safety valves

Where additional safety valves are provided to permit servicing without depressurizing, the valve system used to isolate any individual safety valve shall be designed so that it is impossible to operate the system with less aggregate pressure-relief capacity than that required under Clause 5.4.1(c).

5.5 VALVES AND ACCESSORIES

5.5.1 Suitability

Any valve, fitting, or component shall be suitable for its intended purpose and service conditions.

The design pressure shall be not less than the maximum pressure of the application, e.g. tank design pressure for items subject to tank pressure, or pump outlet pressure where appropriate.

5.5.2 Metal parts

5.5.2.1 General

The melting point of pressure-containing metal parts, with the exception of the components listed in Clause 5.5.2.3 below, shall be not less than 850°C.

5.5.2.2 Additional requirements

The following additional requirements apply:

- (a) Ductile iron shall comply with ASTM A395 or equivalent.
- (b) A copper base alloy valve shall be of a material specified in AS 1271 or equivalent specification, and shall have minimum elongation of 12% in a gauge length of $5.65\sqrt{S_0}$, where S_0 = original cross-sectional area of test section.
- (c) A copper base alloy shall not be used—
 - (i) for the secondary control provision for any tank having a capacity greater than 8 kL; or
 - (ii) where the tank's capacity is 8 kL or less, and the control provision exceeds 50 mm nominal size.

- (d) Where any valve is larger than 25 mm nominal size and has no other means of being shut off, its handle or handwheel shall be a ferrous material, irrespective of the material of the valve.

5.5.2.3 Exceptions

The following are exempted from the above requirements:

- (a) Fusible elements.
- (b) Contents gauges for tanks of 8 kL or less.
- (c) Non-metallic materials used for seals, gaskets, soft seats and diaphragms.
- (d) Aluminium or zinc when used for meters or regulators.

5.5.3 Marking of valve indicators

Valve indicators shall be marked in accordance with Appendix D.

5.5.4 Filling connections on tanks

The filling connection shall be incorporated directly on the tank only under the conditions of Clause 6.6.2.

TABLE 5.2
CONTROL OF LP GAS OUTFLOW FROM A TANK

Function of opening	Features of functions required of the control provisions		Remarks
	Primary control	Secondary control	
None (opening not in use)	Plug or blank flange	None	Number of such openings should be minimized
Excess vapour pressure relief	Safety valve	None	Special conditions apply when reserve safety valves are fitted (see Clause 5.4.3)
Filling (liquid inflow only)	Non-return, internal and damage protected	Positive shut-off, but a second non-return function permitted for tanks of 8 kL or less, filled into the vapour space	See Clause 5.3.5 for alternative
Vapour withdrawal	Excess flow, internal and damage protected	Positive shut-off of any type	See Clauses 5.3.1(e) and 5.3.5 for permitted alternative
Liquid withdrawal or two-way liquid flow	Excess flow, internal and damage protected	Positive shut-off, emergency shut-down system	See Clause 5.3.5 for alternative. The use of a single connection for both filling and withdrawal is deprecated
Pump bypass (liquid return)	Non-return, or excess flow internal	Positive shut-off of any type, automatic opening for liquid return	Pump bypass should be returned to the vapour space
Gauging	Varies according to the function and construction of the gauge (see Clause 5.3.3)		—
Liquid removal, draining (infrequent use)	Excess flow, internal	Positive shut-off, metal-plugged outlet. Second positive shut-off in line if blowing down to atmosphere	The second positive shut-off valve need be attached only for the period of blowing down
Vapour return and vapour balancing lines	Non-return, internal if vapour flow is always inward, otherwise excess flow, internal	Positive shut-off	—

NOTES:

- 1 The various functions listed may be incorporated in compound fittings or valves.
- 2 'Internal' means protected against impact or shear-off as described in Clause 5.3.4.
- 3 Some valves, including some ISC valves, by their design may not be positive shut-off valves and where used additional positive protection will be required.
- 4 Specific requirements for service stations are provided in Clauses 10.5.9 and 10.5.14(b).

5.5.5 Liquid level and contents gauges

A tank shall be fitted with contents gauges that are capable of measuring the contents of the tank to allow filling to the requirements of Appendix B.

Where the standard filling level is used to determine the contents of a tank the tank shall be fitted with:

- (a) Gauge(s) that monitor the level of LP Gas in the tank and a separate gauge to indicate when the standard filling level for the LP Gas composition in the tank has been reached. This method relies on operator control of the filling process.
- (b) Electronic gauge(s) that monitor the level of LP Gas in the tank and sound an alarm at or below the standard filling level for the LP Gas in the tank to alert the operator to cease transfer. An additional separate fixed liquid level gauge that sounds an alarm and stops transfer of LP Gas into the tank at no more than 2% above the standard filling level.

Where filling by volume using temperature correction method (see Appendix B5), an electronic liquid temperature gauge shall be used in conjunction with the electronic level gauge to determine the maximum fill level. An Additional separate liquid level gauge that initiates an alarm and stops transfer of LP Gas into the tank at no more than 2% above the level determined by the volume correction method shall be fitted.

5.5.6 Pressure gauges

Each tank shall be fitted with a pressure gauge complying with AS 1349.

5.5.7 Temperature gauge

Where a temperature gauge is fitted, the sensing device shall be installed in a pocket that is located in the lower section of the tank and welded to the tank, to prevent accidental removal of the pocket when the gauge is removed.

5.5.8 Regulators

Regulators shall comply with UL 144 or other equivalent Standard (see Clause 6.10).

5.5.9 Provision for emptying tank

Every tank shall have a means for removing liquid LP Gas from the tank, should an emergency arise. Where the piping system cannot perform this function, a drain or eductor tube, fitted with an excess flow and shut-off valve, shall be provided.

The shut-off valve outlet shall be capped or plugged with a metal fitting when not in use, and Table 5.2 shall apply.

The outlet shall be directed so as to avoid impingement on the tank and fittings, supports, or other tanks.

NOTE: Bottom drain fittings on small tanks may be difficult to access and vulnerable to damage, so eductor tubes may be preferable.

5.6 PIPING

5.6.1 Design for pressure

Piping that operates at a pressure of less than 200 kPa shall comply with AS/NZS 5601.1 or AS/NZS 4645.1.

Piping for use at a pressure exceeding 200 kPa shall comply with AS 4041, or other equivalent design Standard, with the following qualifications:

- (a) The design pressure shall be not less than that of the cylinder, the tank or the pump delivery pressure, or hydrostatic relief valve pressure, whichever is the greater.

- (b) The design shall allow for expansion and contraction, both of the piping itself and of any equipment to which it is connected.
- (c) Screwed pipe shall be at least ASME B31.3 Schedule 80, except that screwed ASME B31.3 Schedule 40 pipe may be used to provide a shear-point at a remote filling connection or dispenser to comply with Clause 6.6.4.

Welded pipe shall be at least ASME B31.3 Schedule 40.

- (d) Copper pipe or tube shall not be used where the temperature is expected to exceed 120°C, or for liquid lines exceeding 10 mm in diameter or 2 m in length.

5.6.2 Joints

A joint in rigid pipework that is subject to tank pressure shall be welded, screwed taper-to-taper, flanged, or made with a ground-face union, provided that the following apply:

- (a) Where the nominal bore of the pipe exceeds 50 mm, screwed joints shall not be used unless unavoidable, e.g. for the mounting of essential auxiliary equipment having screwed connections.
- (b) Pipe jointing compounds and gaskets shall be suitable for use with LP Gas.
- (c) Spiral wound metal gaskets with a minimum temperature rating of 550°C shall be used for flanged connections on liquid phase piping.
- (d) Compressed fibrous gaskets or O-rings shall only be used on encapsulated flanges where it is impracticable to use spiral-wound gaskets.
- (e) Multiple gaskets shall not be used at any flanged joint.
- (f) Joints intended to be buried shall be welded.

NOTE: Statutory regulations in some States permit only those welders who have certification in accordance with AS 1796 to weld pressure piping.

In New Zealand, welders shall be qualified to ANSI/API 1104 or ASME Boiler and Pressure Vessel Code, Part IX.

5.6.3 Joints in copper pipe

A joint in copper piping shall be made—

- (a) with a flare compression fitting; or
- (b) with a capillary fitting using a brazing metal with a melting point not less than 540°C; or
- (c) by a spigoted joint formed from the pipe itself and brazed as above.

Flare fittings having mismatching cone angles, or olive or similar compression-sleeve type fittings, shall not be used.

5.6.4 Flexible pipe connections

A flexible pipe connection may be used where necessary to absorb vibration, or where a rigid connection is impracticable. Such a connection shall comply with the following requirements:

- (a) Individual or combined lengths shall be as short as practicable.
- (b) A flexible connection shall be of metal construction, designed for a bursting pressure not less than 12 MPa and tested to not less than 3.6 MPa.

5.6.5 Transfer hose

A transfer hose and hose assembly shall comply with—

- (a) AS/NZS 1869; or
- (b) UL 21, provided that the extractable content limits set out in AS/NZS 1869 are achieved.

The hose assembly shall be assembly-tested in accordance with AS/NZS 1869, as appropriate to the pressure to which it is subjected.

5.6.6 Transfer hose couplings

Couplings that are intended to be connected and disconnected frequently during normal operation shall be either an Acme thread form or a POL coupling. Any other hose connection shall be screwed or flanged.

NOTE: The following sizes of Acme thread are recommended:

- (a) For liquid.....3¼ or 1¾ inches.
- (b) For vapour2¼ or 1¼ inches.

5.6.7 Expansion flexibility

Any provisions for thermal expansion shall comply with AS 4041. Gland-type expansion joints shall not be used.

5.7 VAPORIZERS

5.7.1 Design and construction

Vaporizers shall comply with the following requirements:

- (a) In Australia, a vaporizer shall comply with AS/NZS 1200 or equivalent design standard if indirect-fired, or AS 1228 if direct-fired.
In New Zealand, all vaporizers should be of a type or kind approved by and registered with the New Zealand Environmental Protection Authority (EPA).
- (b) The design pressure of the vaporizer shall be not less than that of the tank to which it is connected, and shall be not less than 1.75 MPa if designed for propane, or 1.05 MPa if designed for butane.
- (c) A safety valve shall be located as close as possible to the vapour outlet, and shall have a discharge capacity calculated in accordance with the design standard.
- (d) Pressure relief shall not be achieved by means of fusible plugs.
- (e) Every connection between the tank and the vaporizer shall be provided with a manual isolating valve located as close as practicable to the vaporizer.
- (f) Tubing that carries the heating medium in an indirect-fired vaporizer shall be provided with a means for preventing LP Gas from escaping from the heat transfer unit if the tubing should fail.
- (g) It shall not be possible for LP Gas liquid phase to enter the vapour distribution system.
- (h) A fuel-fired vaporizer shall be provided with a system that shuts off both the main flame and the pilot flame in the event of a pilot flame failure.
- (i) A vaporizer shall incorporate provisions to shut off the heat source automatically when—
 - (i) the design pressure or design temperature is exceeded; or
 - (ii) the LP Gas liquid phase supply to the vaporizer fails; or

- (iii) the level of the heat transfer liquid in a calorifier-type vaporizer falls below a safe level.
- (j) A vaporizer shall be equipped with a means of draining off any heavy products that might collect in the liquid section of the system.
NOTE: Such discharge will give rise to a hazardous area and the discharge may need to be piped away to avoid proximity to an ignition source.
- (k) Heat shall not be applied to a tank in a manner that could lead to an uncontrolled release of LP Gas from the tank, or that could endanger the integrity of the tank or fittings.

5.7.2 Marking

Vaporizers shall be provided with marking in accordance with Appendix D.

SECTION 6 TANK SYSTEM INSTALLATION

6.1 SCOPE OF SECTION

This Section provides requirements for above-ground tank installations, and for underground tank installations unless varied by a specific requirement of Sections 7 or 10.

6.2 TANK SPACING AND SEPARATION DISTANCES

6.2.1 General location and spacing

Above-ground storage tanks shall be located in accordance with the distances shown in Table 6.1, with the following qualifications:

- (a) An above-ground tank shall not be installed within a building.
- (b) Where the site is used primarily for LP Gas storage and handling, or where the LP Gas facility is within a petroleum terminal, the separation distance to buildings on the same site as the LP Gas storage may be halved, provided that in no case shall a tank having a total capacity exceeding 2.5 kL be located closer than 3 m to such a building.
- (c) A tank exceeding 0.5 kL capacity shall be not less than 1 m from a boundary.

6.2.2 Tanks in groups

Tanks may be arranged in groups of up to six tanks, with each tank in the group separated in accordance with Table 6.1, Column 2. The following requirements and recommendations apply to tanks in groups:

- (a) The distance from one such group to another tank or group shall be not less than 15 m except that, where no tank in either group exceeds 2 m diameter, the distance may be reduced to 10 m.
- (b) Tanks shall not be stacked one above another.
- (c) The longitudinal axes of tanks in a group should be parallel and should be directed away from any adjacent storages of hazardous, flammable or combustible liquids or gases. Where another arrangement is unavoidable, whereby a tank could be in line with the axis of another tank, the distance between the end of any tank and the end or shell of another tank shall be not less than 3 m or twice the diameter of the larger tank, whichever is the greater.

NOTES:

- 1 For interconnected tanks, see Clause 6.8.1.
- 2 A tank that is less than 3 times the minimum separation distance from the tank (see Table 6.1) is considered 'adjacent'.

TABLE 6.1
LOCATION OF ABOVE-GROUND STORAGE TANKS

1	2	3	4
Capacity of the tank kL	Minimum distance to an adjacent LP Gas tank m	Minimum distance from the tank to a public place, or a railway line m	Minimum distance from the tank to a protected place m
≥0.5	Diameter of the largest tank	1.5	1.5
1		2	3
2		4 (3)	6 (4.5)
5		5 (3.5)	8 (5)
8		6 (4)	10 (6)
10		7	11
15		8	14
20		9	15
50		10	17
100		11	20
200		12	25
500		22	45

NOTES:

- 1 Distances for intermediate tank sizes may be interpolated.
- 2 Column 4 values above 500 kL are calculated from the formula $[100 (\log_{10} V - 2.24)]$, rounded down to the nearest 10 (V is tank capacity in kL). Column 3 values are half those of Column 4.
- 3 Distances in brackets are alternatives that are permissible only for a single tank used only for vapour withdrawal. This dispersion is NOT allowed in New Zealand. A tank that has no other tank within 8 m of it may be treated as a single tank.
- 4 If the separation distance to a protected place is measured across a boundary, see Clause 3.5.2.
- 5 The separation distance is determined individually for each tank, and is assessed on the basis of its own particular capacity, and not on any aggregation with the capacity of any accompanying tank, (see Figure 6.1).
- 6 For separation distances for underground tanks, see Table 7.1.

6.2.3 Aerial wiring systems

6.2.4 Distances to other gas storages

- (a) Cryogenic liquids in tanks or vessels: Refer to AS 1894.
- (b) Compressed gases in cylinders: Refer to AS 4332.
- (c) Gases other than LP Gas or cryogenic liquids, in tanks: 6 m separation.
- (d) Liquefied natural gas storage: Treat as flammable liquid (see Clause 6.2.5).

LP Gas tanks shall be separated from other dangerous goods and flammable or combustible materials by at least the following distance or greater if required by the relevant Standards:

- COPYRIGHT

- (e) At least 6 m to any combustible materials.

NOTE: In New Zealand, the separation between above ground tanks containing LP Gas and stationary tanks containing hazardous substances of a different class is set out in *Hazardous Substances (Dangerous Goods and Scheduled Toxic Substances) Transfer Notice*.

6.3 TANK SITE CONDITIONS

6.3.1 Ground conditions

A tank shall not be installed in or above a significant ground depression where a spill or leak could lead to the accumulation of LP Gas. Any bunding shall permit spillage to flow away from the immediate vicinity of the tank.

6.3.2 Ventilation and access

The location of the above-ground storage shall comply with the following requirements for ventilation and access:

- (a) Above-ground storage tanks shall be in the open air, outside buildings.
- (b) Nearby construction, fences, walls, vapour barriers, or the like shall permit free access around and cross-ventilation for the tank.

The clearance distances illustrated in Figure 6.2 may be used as a guide. Where the adjacent structure provides visual screening only, is open to permit air passage through not less than 30% of its area, and is completely open on at least one side except for any necessary security fencing, the minimum clearance in any direction shall be equal to the diameter of the tank.

6.4 TANK FOUNDATIONS AND SUPPORTS

6.4.1 Foundations and supports

The supporting structure and the foundations shall be designed in accordance with the vessel design Standard (such as AS 1210) and the following requirements:

- (a) The design load shall be the total mass when the tank is full of water.
- (b) Where a tank is located in an area that is likely to be subject to flooding, the tank shall be anchored to prevent its floating.
- (c) Seismic and wind loadings shall apply for foundations and supports (see Clauses 5.2.2(c) and 5.2.3).

6.4.2 Fire resistance of supports

Tank supports shall have a FRL/FRR of 120/120/120 except where—

- (a) the tank is a horizontal tank and has a capacity of 8 kL or less, and is prevented from subsiding more than 600 mm if the supports are weakened by fire; or
- (b) the tank is a horizontal tank and has a capacity of more than 8 kL, and is prevented from subsiding more than 300 mm if the supports are weakened by fire.

Piping shall be capable of withstanding any such movement without rupture.

Where the support for a vertical tank comprises a continuous steel skirt having only one opening of 450 mm or less diameter, fire protection may be applied to the outside only of the skirt.

NOTES:

- 1 Where fire-protective cladding is applied to metal supports, care must be taken to prevent the ingress of moisture between the metal and the cladding.
- 2 The ground, or a plinth or foundation of concrete, masonry, or similar fire-resistant material under a horizontal tank, would be considered as the limit to which a tank can subside.

6.4.3 Expansion provision

The supports for an above-ground tank shall allow for expansion.

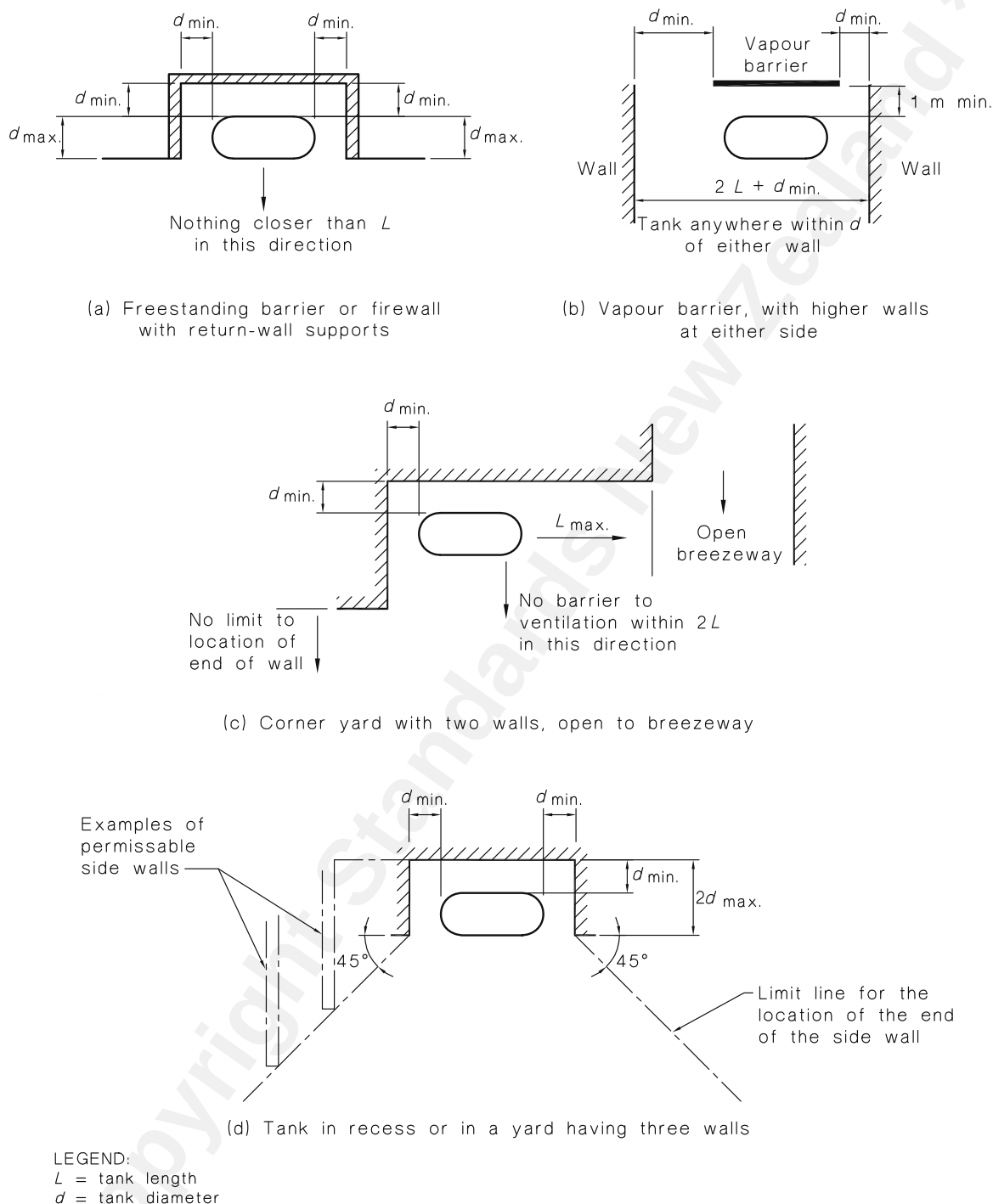


FIGURE 6.2 TYPICAL TANK VENTILATION AND ACCESS PROVISIONS

6.5 FLAME IMPINGEMENT

The designer and the installer of an LP Gas installation shall ensure that potential flame impingement on LP Gas piping, components, or tanks is minimized. Every arrangement shall be reviewed after completion of its design and before its construction to identify and remove potential impingements in a practicable manner.

Care shall be taken to avoid potential flame impingement on the tank. The following shall be considered:

- (a) The location of pumps and compressors in relation to the tank.
- (b) The orientation of piping at any shear point and filling connection so that discharge takes a safe direction.
- (c) The location of joints, fittings and flanges.

6.6 CONNECTIONS FOR TANK FILLING AND TANKER LOADING

6.6.1 Filling connection

A filling connection shall be either a direct connection on the tank or a remote connection. A connection for tanker loading shall be a remote connection.

NOTES:

- 1 Figure 6.3 explains the terminology used.
- 2 The requirements of the Clause do not apply to the emptying of tanks for maintenance purposes.
- 3 Filling into the vapour space is preferred and recommended for all tanks to reduce the possibility of LP Gas liquid escaping, and to reduce the possibility of an increase of vapour pressure during filling, which may cause the safety valve to lift.

6.6.2 Direct connection

A direct connection shall be arranged so as to fill into the vapour space of the tank.

The separation distances given in Clause 6.6.7, Items (d) and (e) shall apply.

A direct connection shall only be provided where the tank is—

- (a) an above-ground tank of 16 kL or less capacity, and the filling hose and coupling are of a type which prevents the escape of more than 0.1 L of liquid during disconnection; or
- (b) an underground tank, and the filling hose and coupling are of a type which prevents the escape of more than 0.1 L of liquid during disconnection.

Where an extension is fitted to such a direct connection for convenience in operation, the following shall apply:

- (i) All the requirements for a remote connection shall apply except Clause 6.6.7(a).
- (ii) The design and arrangement shall be such that any possibility of flame impingement on the tank in the event of a fire at the connection is minimized.
- (iii) The multi-function valve described in Clause 5.3.1(e) shall not be used for tank filling.

6.6.3 Remote connection

Remote connections with provisions for liquid, vapour, or vapour return service for either filling or withdrawal, shall comply with the following requirements:

- (a) The appropriate valve system as specified in Clause 5.3 shall be provided at the tank.
- (b) An anchorage shall be provided for the remote connection, in order to prevent its movement. Such an anchorage shall be of sufficient strength to ensure that the piping and valves between the anchorage and the tank will not be damaged before the shear point provision fails or releases.
- (c) A shear point, or the equivalent, shall be provided. It shall be designed in conjunction with the support to ensure that, if the tanker is driven away without disconnecting, a breakaway will occur at the shear point before valves or piping are damaged. The

shear point shall be located between the point of connection and the anchorage, as close as possible to the anchorage.

NOTE: As additional protection, AS 2809.3 requires a means for ensuring that the tanker cannot be moved until the hose has been disconnected.

- (d) A non-return and a manual shut-off valve shall be provided between the tank and the anchorage, and as close as possible to the anchorage. Where a non-return function is not possible because of outward flow, excess flow protection shall be used in conjunction with an emergency shut-down system.

6.6.4 Tanker loading

A remote connection shall be used for the loading of a tanker from a tank.

Where a flexible hose is used to connect the remote connection to the tanker at a point removed from the remote connection, the separation requirements for the remote connection shall also apply to the point of connection of the flexible hose and the tanker.

6.6.5 Access for contents gauging

The fixed liquid level gauge and other filling-control and contents-monitoring equipment shall be readily accessible either from ground level, or from permanently installed access ladders, walkways, or platforms complying with AS 1657 or NZS/AS 1657.

6.6.6 Hose venting

If a dry hose is used, the hose vent shall discharge via an elevated pipeaway at a position above the height of the tank and in a safe location.

6.6.7 Separation distances

A remote connection shall be located in accordance with the following separation distances:

(a) *To above-ground tanks*

A remote connection shall be not closer to the nearest above-ground tank than—

- (i) for a tank of up to and including 25 kL capacity 3 m;
- (ii) for a tank over 25 kL but not over 125 kL capacity 6 m; or
- (iii) for a tank over 125 kL capacity 9 m.

When determining the distance between a remote connection and a group of tanks (whether served by that connection or not), the distance is measured to the nearest tank and is calculated on the basis of the aggregate capacity of the group.

NOTE: For the purposes of this Clause (Clause 6.6.7), a group is considered to exist where any two tanks have less than 8 m of space between them.

(b) *To underground tanks*

Sufficient clearances for convenient access and operation.

(c) *To a public place or protected place*

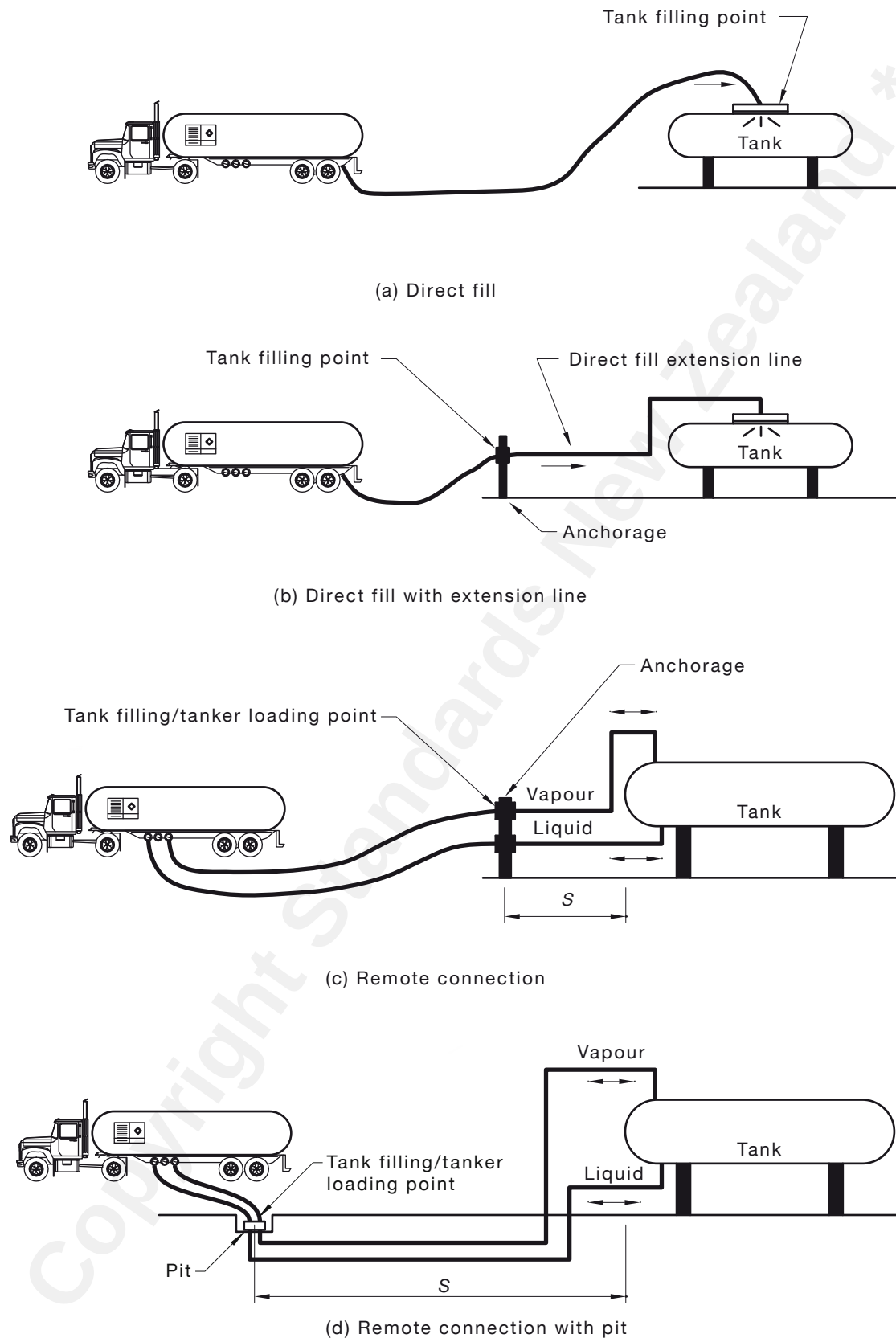
12 m, except that, where the loss of liquid on disconnecting the coupling is less than 1 L, the minimum distances shall be 3 m to a public place or 6 m to a protected place.

(d) *To an ignition source*

In accordance with AS/NZS 60079.10.1.

(e) *To an entrance to an enclosed portion of any underground drain, conduit or utilities pit*

12 m, except, where the liquid loss on disconnection is less than 1 L, the minimum distance shall be 3 m.



NOTES:

- 1 S (separation distance, in metres) is dependent on tank size.
- 2 This Figure indicates filling/loading point locations assuming tanker-transported hoses. For applications using on-site hoses, the connection would be at the tanker.
- 3 Loading arms may be used instead of hoses.

FIGURE 6.3 FILLING AND LOADING CONNECTIONS

6.7 EMERGENCY SHUT-DOWN SYSTEM

6.7.1 General design

6.7.1.1 Shut-off valve

Where an emergency shut-down system is required (see Clause 5.3 and Table 5.2), it shall incorporate a shut-off valve that—

- (a) can be manually shut off;
- (b) will shut off automatically in the event of fire; and
- (c) complies with Clauses 6.7.2 to 6.7.4 inclusive.

NOTE: Sensors may also be used to react to other fault conditions, e.g. leakage.

6.7.1.2 Use of LP Gas vapour

Where LP Gas vapour is used as a pressure source for activating the shut-off mechanism, the following requirements apply:

- (a) Actuators and pressure supply line components shall be suitable for use with LP Gas vapour.
- (b) Supply line piping shall be limited to a maximum of 9 mm outside diameter.
- (c) Supply pressure shall be controlled to prevent any condensation of the LP Gas vapour.
- (d) The maximum flow rate of the LP Gas supply to the system shall not exceed that from a 1.4 mm drill orifice.

6.7.2 Fire actuation

Fire-sensing elements of the emergency shut-down system shall be located so as to sense and respond to a fire at the filling or loading connection.

6.7.3 Manual actuation

Where the actuation point is a deadman's handle, only one actuation point is required.

Except where the actuation point is a deadman's handle, the system shall incorporate at least one provision for initiating valve closure from an actuation point that is not at the valve. A system for any tank that exceeds 6 kL capacity shall be provided with not less than two such actuation points. Actuation points shall be located as follows:

- (a) One actuation point shall be readily accessible from and within 10 m of any filling or loading connection, and any vehicle fuel dispenser. Where required, any other actuation point shall be located remote from a filling or loading connection.
- (b) The location of at least one actuation point shall be in a direction of ready egress from the installation.
- (c) Where an emergency actuation point for some other purpose exists on site, the LP Gas remote shut-down system may be operated from the same actuation point, provided that the requirements of this Clause are complied with.
- (d) The system shall require manual action to re-set the valves to resume operation.
- (e) Actuation points shall be marked in accordance with Appendix D and shall be readily accessible.

NOTE: Provision should be made to prevent condensation forming and entering emergency shut-off valve actuators.

6.7.4 Tank-to-tanker and tanker-to-tank transfer protection

Where an emergency shut-down system is required at a remote connection (see Table 5.2), the transfer system shall incorporate provisions to ensure that the flow is shut off at the tanker side and at the installation side of the transfer hose/loading arm at the same time.

Where interface provisions are required, they shall comply with the following requirements:

- (a) They shall be air-operated. Operating air pressure shall be between 500 kPa and 700 kPa.
- (b) Where a two hose air system is used, the female component of the connector shall be at the terminal end and shall be a self-sealing type 1 connector. A separate male connector that is not compatible with the terminal female connector shall be provided to receive return air from the tanker.
- (c) Where a single hose air system is used, a male connector shall be provided at the terminal end and all female connectors shall be non-self-sealing types.
- (d) The melting point of fusible links, or of air hoses that constitute fusible links, shall not exceed 180°C.

Where appropriate, the tanker actuation points may be taken into account for the purposes of complying with Clause 6.7.3.

NOTES:

- 1 Any installation that depends on terminal air and which could be unattended should be designed to minimize the possibility of unauthorized opening of the liquid outlets by a single simple action.
- 2 To optimize compatibility between systems, the same male connector should be used at the terminal end, irrespective of whether the operating air is supplied from the tanker or the terminal.

6.8 PIPING

6.8.1 Interconnected tanks

Where two or more tanks are piped so that liquid or vapour can pass from one to the other, the piping system shall be designed to ensure that no one tank can become liquid-full as the result of filling, operating, or temperature conditions. In addition, the following requirements shall apply:

- (a) The start-to-discharge pressure of all safety valves shall be the same.
- (b) The standard filling levels of all tanks shall be in the same horizontal plane unless the piping and valve system is capable of preventing uncontrolled levelling of the liquid.
- (c) Where a liquid return is provided for multiple tanks having a common withdrawal pipe, the system shall be capable of being operated so as to ensure that returning liquid does not cause overfilling of any tank.
- (d) Where tanks are connected by a common liquid line, a vapour-balancing line interconnecting the vapour space of all tanks shall be provided.
- (e) Where two or more tanks are subject to direct filling, the arrangement of pipework and auxiliary equipment shall be such that quick access and egress is not hindered.
- (f) The optional arrangement of valves described in Clause 5.3.1(e) shall not be used for any tank interconnection. Tanks shall not be manifolded by the use of multi-function valves.

6.8.2 Liquid lines between valves

Where piping can be shut off between valves when liquid full, that piping shall be provided with a hydrostatic pressure-relief valve in accordance with the following requirements:

- (a) The setting shall be not less than 2.4 MPa and be not greater than the design pressure of the piping or the lowest rated component used in the piping system.
- (b) The hydrostatic pressure-relief valve shall not be installed on the discharge side of a pump if the same or higher degree of protection could be obtained on the inlet side.
- (c) The discharge from the hydrostatic pressure-relief valve shall not terminate within a building, and shall not impinge on any tank, fitting, pipework, other equipment or on the ground, and shall be directed away from any area where persons might congregate or work.
- (d) An orifice of 3 mm maximum size shall be provided at the inlet of any hydrostatic pressure-relief valve to limit the discharge in the event of valve failure.

6.8.3 Piping system

Any protective valve external to the tank or in piping shall be installed so that any undue strain beyond the valve will not cause breakage between the valve and the tank.

6.8.4 Piping drain or bleed valves

Where a piping system is provided with a bleed valve whose only purpose is for venting a pipe during maintenance, measures to dispose of or disperse any gas released, or a flow-restricting orifice not exceeding 3 mm diameter, shall be provided.

6.8.5 Support of piping

Where piping passes from the withdrawal point of a tank to the ground or to adjacent structures, particular attention shall be paid to the support or protection of the pipework where vulnerable to impact, fire and possible use as a hand-support or step.

6.8.6 Piping within buildings

LP Gas liquid or vapour at a pressure above 140 kPa shall not be piped into, through or under a building unless unavoidable because of the type of usage, e.g. to fuel an engine, to manufacture aerosols or for similar industrial or laboratory applications. In such cases the gas or liquid shall not pass through any other room or type of occupancy.

LP Gas vapour piping in buildings shall comply with AS/NZS 5601.1 for a vapour pressure up to 200 kPa. For liquid, or higher vapour pressure, AS 4041 or ANSI/ASME B31.3 shall apply.

6.8.7 Underground piping

Underground piping shall comply with the following requirements:

- (a) The installation of underground piping and any coating or cathodic protection system shall comply with AS/NZS 4645.1, AS/NZS 4645.2, AS/NZS 4645.3, AS/NZS 2832.1, AS 2832.2, AS 2885.1, AS/NZS 5601.1, EN 13636, EN 15112 or equivalent Standard as appropriate.
- (b) The piping shall be inherently corrosion-resistant or provided with protection appropriate to the conditions.
- (c) Where cathodically protected piping is used, an insulating joint, fitting, or flange shall be installed where the piping emerges from the ground at both ends of the buried section.
- (d) Cathodic protection systems and the use of surge protection for insulating flanges shall be in accordance with the requirements of AS/NZS 60079.14.

6.8.8 Identification of pipes

Piping for LP Gas shall be identified according to its contents where both liquid and vapour phase gas are present, or different types of LP Gas are present, or where some pipes on the installation carry materials other than LP Gas.

NOTE: For New Zealand only, labelling requirements are required for installations greater than 12 kL.

The contents of each LP Gas pipe shall be identified throughout the installation by one or more of the following methods:

- (a) By printing, stencilling or labelling at critical locations, e.g. adjacent to connections, emergency shut-down system, and isolation valves.

NOTE: If additional colour coding is required, it should comply with Figure 6.4 or AS 1345 or NZS 5807.

- (b) By painting, as set out below:

(i) For liquid lines, the colour shall be Raffia No. X31.

(ii) For vapour lines, Aqua No. B25.

(iii) White along the length of the pipe, with tracer colours [as specified in Items (i) and (ii)] at critical locations.

NOTE: Colours and their code numbers are given in AS 2700.

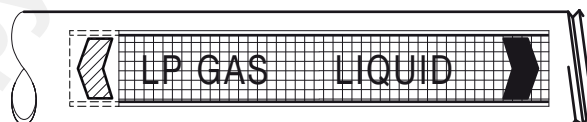
6.8.9 Pipework testing

Any piping that is within the scope of AS 4041 shall be tested in accordance with that Standard. Other piping shall be tested by the installer in accordance with Appendix I, and shall be certified by the installer to be free of leakage before being put into service.



BACKGROUND		AQUA No. B25
BORDER		RAFFIA No. X31
INNER MARGIN (where applicable)	WHITE	
LETTERING	BLACK	
ARROWS	BLACK	

NOTE: This scheme is also applicable to propane, propylene



BACKGROUND		RAFFIA No. X31
BORDER		
INNER MARGIN (where applicable)	WHITE	
LETTERING	BLACK	
ARROWS	BLACK	

NOTE: This scheme is also applicable to propane, propylene

FIGURE 6.4 IDENTIFICATION OF LIQUID AND VAPOUR LINES

6.9 VAPORIZER INSTALLATION

6.9.1 Compatibility check

Before installation is commenced, the start-to-discharge settings of the safety valves on the vaporizer and the tank shall be compared.

Safety relief valves (SRVs) for feed out (superheated) vaporizers shall be set equal to or greater than the SRVs on the supplying tank.

Where the output of a vaporizer may feed back to the tank, the SRV set point shall be not greater than the SRV set point of the tank.

6.9.2 Shelter

Any shelter structure for an undercover vaporizer shall be non-combustible, well ventilated near the floor and roof, and used exclusively for LP Gas equipment.

6.9.3 Location

The separation distance from a direct-fired vaporizer to a protected place or public place shall be not less than the following:

- (a) For each vaporizer having a vaporizing capacity of 70 L/h or less—
 - (i) In Australia—3 m.
 - (ii) In New Zealand—8 m.
- (b) For each vaporizer having a vaporizing capacity greater than 70 L/h and up to 610 L/h—8 m.
- (c) For each vaporizer having a vaporizing capacity exceeding 610 L/h—15 m.

NOTE: For a modular direct-fired vaporizer, the capacities above relate to each separate module.

A direct-fired vaporizer is an ignition source and shall not be installed within any hazardous area.

6.10 REGULATORS

6.10.1 Location

The location of a regulator shall comply with the following:

- (a) Any first stage regulator shall be outdoors, except where the regulator is attached to a cylinder that is permitted to be used indoors (see Clause 4.4).
- (b) Second stage regulators shall be installed in accordance with AS/NZS 5601.1.
- (c) A single-stage regulator, or the first stage of a multi-stage regulator, shall be located so that the length of the piping which is subject to cylinder or tank pressure is as short as practicable. For cylinder installations, see also Clause 4.4.11(h).

6.10.2 Venting

The outdoor discharge from a vent terminal, gas-pressure-relief device or terminal of a vent line shall be—

- (a) arranged to minimize the risk of vapour collecting in drains, gutters, and downpipes;
- (b) not less than 1 m in any direction from any opening into a building; and
- (c) not less than 2 m from any fixed source of ignition.

The vent terminal shall have provision to exclude rain and insects.

For design and installation purposes, vent discharge piping shall be treated as low-pressure-vapour piping. The venting requirements of AS/NZS 5601.1 shall be met.

In New Zealand, regulators used indoors and manufactured to BS 3016, EN 12864, EN 13786 or EN 13785 are deemed to comply and further compliance with this Clause (Clause 6.10.2) is not required.

6.10.3 Overpressure protection

Overpressure protection shall comply with the requirements of UL 144 (shut-off devices or type II relief valves) or AS/NZS 5601.1, or other equivalent Standard.

6.11 PUMPS AND COMPRESSORS

6.11.1 Suitability and location

A pump or compressor intended for handling LP Gas shall be designed for, and suitable for use with, LP Gas.

A pump or compressor shall be located so that its surrounding hazardous area (as defined in AS/NZS 60079.10.1) does not impinge on a protected place, public place or ignition sources.

NOTE: For service station installations, see Clause 10.5.10.

6.11.2 Pressure limitation

In addition to the specific requirements of Clauses 6.11.3 and 6.11.4, a pump or compressor installation shall incorporate an automatic system to prevent the design pressure of any tank, pipe component or the pump itself from being exceeded. If this system consists of a primary recirculation system which has in the line a manual shut-off valve, or any valve of a type that does not automatically open under pump pressure, a secondary safety recirculation system shall be provided. This secondary system shall have no means of rendering it inoperative.

NOTE: This secondary system may have a pressure setting greater than that of the primary system.

6.11.3 Pumps

A pump installation shall incorporate a provision to fit a pressure gauge on the discharge side of the pump before any external pressure relief or shut-off valve. The provision shall incorporate a flow restriction in accordance with Clause 5.3.1(c).

Where fitted, the gauge shall be liquid-damped and shall comply with AS 1349.

6.11.4 Compressors

A compressor installation shall comply with the following requirements:

- (a) A compressor shall draw from a vapour space.
NOTE: The compressor may discharge to either the liquid or the vapour space.
- (b) Means shall be provided to prevent liquid from entering the compressor suction. If such means is not integral with the compressor, a liquid trap shall be installed in the suction line as close as possible to the compressor.
- (c) A provision to fit pressure gauges complying with AS 1349 and provided with a flow restriction in accordance with Clause 5.3.1(c) shall be installed on the suction side and on the discharge side of a compressor.

NOTE: When fitted, the gauges should be liquid-damped.

6.12 SECURITY AND DAMAGE AVOIDANCE

6.12.1 Tank security

Where a tank is accessible to the public, it shall be provided with a means to prevent tampering with any tank fitting that could lead to an escape of gas, and that is able to be locked. The provision shall be one of the following:

- (a) A fully surrounding fence complying with Clause 6.12.2 or an equivalent security fence surrounding the whole of the property.
or
- (b) A fence complying with Clause 6.12.2 and surrounding only those components requiring protection.
or
- (c) A lockable enclosure complying with Clause 6.12.3, provided that the tank capacity is less than 8 kL and a tank of any size is not in a school, playground, or public place (e.g. a park or similar).

NOTE: Tampering, for the purpose of this Clause, relates to tampering without the use of tools.

6.12.2 Security fence

A security fence shall be at least a chain-wire fence of strong construction, not less than 1.8 m high and incorporating an outward opening, lockable gate not less than 1 m wide. In addition, a fully surrounding fence shall be not less than 1.5 m from the tank and shall be kept locked when not under supervision.

6.12.3 Lockable enclosure

Where a lockable enclosure is used, its design shall be such that—

- (a) the possibility of tampering with the protected fittings is prevented;
- (b) if gas discharging from a safety valve is ignited, the resulting flame will burn clear of the tank; and
- (c) moisture is not retained within it.

Any such enclosure shall be kept locked when not in use.

6.12.4 Damage avoidance

Where any tank or other part of the installation is located such that it is susceptible to impact from moving vehicles, it shall be protected from such impact.

At a service station, the requirements of Clause 10.5.15 apply for damage protection.

NOTES:

- 1 Where vehicles are involved only in low speed manoeuvring, the following would be considered to provide adequate protection:
 - (a) 75 mm steel pipes filled with and set 500 mm into concrete, spaced at a maximum of 1.3 m apart, not less than 1.5 m from the item being protected, and not less than 1.2 m high, except that if the distance from the item being protected is 4 m or more they may be 0.5 m high.
 - (b) Highway crash barriers (e.g. 'W' guardrail) not less than 700 mm high and 1.5 m from the item being protected, set into concrete or equivalent support.
 - (c) Fenced compounds with 1.8 m high chain link fencing complete with tension wires and 50 mm diameter steel posts set in concrete.
- 2 See Appendix Q for assessing the requirements, for the protection of tanks and cylinders from impact.

6.13 LIGHTNING AND BONDING PROTECTION

6.13.1 Lightning protection

An above-ground tank larger than 8 kL shall be provided with lightning protection if required by AS/NZS 1768.

6.13.2 Bonding to tanker

The filling or loading connection for a tank larger than 8 kL shall be provided with a facility for attaching the bond or static wire of a tanker in accordance with AS/NZS 1020.

6.14 LADDERS, STEPS AND PLATFORMS

Any ladder, step or platform that may be necessary to gain access to valves, fittings or gauges shall comply with AS 1657 or NZS/AS 1657.

6.15 LEAK DETECTION

Where the gas has a lower odour level than that described in Clause 11.2.2, e.g. as for aerosol manufacture, a continuously operating gas leak detection system shall be installed (see also Clause 6.18).

6.16 MARKINGS AND NOTICES

A tank or tank storage area shall be provided with signs, markings and notices as set out in Appendix D.

6.17 LIGHTING

Installations for which LP Gas transfer will occur at night shall be provided with lighting capable of providing illumination at a level of more than 50 lx at all valves, couplings, and gauges that might be involved in the transfer operation (see also the ADG Code).

6.18 UNODOURIZED GAS INSTALLATIONS

6.18.1 Emergency shut-down system

Unodourized LP Gas installations shall be equipped with an emergency shut-down system that will automatically stop the flow of gas if a gas detector senses the presence of a concentration of gas in air of more than 25% of the lower explosive limit (LEL).

6.18.2 Installation and use of gas detectors

The selection, installation and use of gas detectors shall comply with AS/NZS 60079.29.2 and AS/NZS 60079.14.

Gas detectors shall be installed—

- (a) around the installation;
- (b) at the loading point if direct fill is not being used; and
- (c) at the point of usage of the unodourized gas.

6.18.3 Function of gas detectors

Gas detectors shall—

- (a) shut off the flow of gas as required by Clause 6.18.1 above; and
- (b) initiate an audible alarm of at least 105 decibels, and a visible alarm.

6.18.4 Signage

A tank, cylinder or tanker that contains unodourized LP Gas shall bear a prominent notice to that effect.

SECTION 7 UNDERGROUND TANK INSTALLATIONS

7.1 SCOPE AND APPLICATION OF SECTION

This Section provides requirements for underground tank installations.

An underground tank installation shall comply with the requirements of Sections 5 and 6, together with the additions or qualifications given in this Section. Where any requirements of this Section differ from a similar requirement of Section 5, this Section shall take precedence.

A tank that is partly or wholly above the general ground level but is completely covered with soil, or a tank within a sand-filled chamber, shall be treated as an underground tank.

The exposed portions of a tank shell (excluding any access hole or inspection pit that is partly buried or mounded) shall be treated as an above-ground tank. Requirements for separation distances and fire protection shall be based on the capacity of the tank. Water supply for fire protection shall be based on the exposed area.

7.2 CORROSION CONTROL

7.2.1 General

Each tank shall be provided with a method of corrosion control that provides protection for the tank from the effects of the corrosive nature of the surrounding soil.

A cathodic protection system shall be designed and installed under the supervision of a specialist in cathodic protection systems.

NOTE: For guidelines on the design, installation and operation of cathodic protection systems, see AS 2832.2, EN 13636 and EN 15112.

7.2.2 Testing of coating

The electrical resistance and the continuity of the coating of the tank shall be spark-tested over the whole surface in accordance with AS 3894.1. The potential shall be suitable for the characteristics of the particular coating, as specified by the coating supplier. Any flaws detected in this test shall be repaired and retested. The coating shall be spark-tested again at the last possible stage prior to final placement and burial, and any handling damage shall be rectified and re-tested.

7.3 LOCATION AND INSTALLATION

7.3.1 Tank location

Tanks shall be located as follows:

- (a) Separation distances for underground tanks shall comply with Table 7.1.
- (b) Tanks shall not be located beneath a bunded area for flammable or combustible liquid storages.
- (c) A tank shall be not less than 3 m from an underground tank for flammable or combustible liquids, unless the two installations have compatible corrosion protection systems specifically designed to cater adequately for the area between the tanks. In such a case the separation may be not less than 1 m.

7.3.2 Tanks in groups

Tanks may be arranged in groups with no limitation on the number of tanks in a group, provided that—

- (a) tanks shall not be buried one above another; and
- (b) the standard filling level of all interconnected tanks shall be in the same horizontal plane unless the piping and valve system prevents uncontrolled levelling of the liquid.

7.3.3 Tank covering

An underground tank shall be covered as follows:

- (a) The tank shall be set on not less than 150 mm of clean inert sand and either—
 - (i) surrounded with not less than 150 mm of clean sand; or
 - (ii) supported by a concrete anchor, which shall be separated from the tank surface by a material offering adequate protection to the tank coating and preservation of cathodic protection to the total tank surface, and then covered with 150 mm of clean inert sand.

NOTE: The correct conductivity and pH levels are critical to cathodic protection.

- (b) If vehicles can pass above the tank, it shall be covered by not less than 300 mm of sand superimposed with reinforced concrete not less than 150 mm thick, but also adequate to support any likely superimposed loading, and extending not less than 300 mm in all directions beyond the outline of the tank.

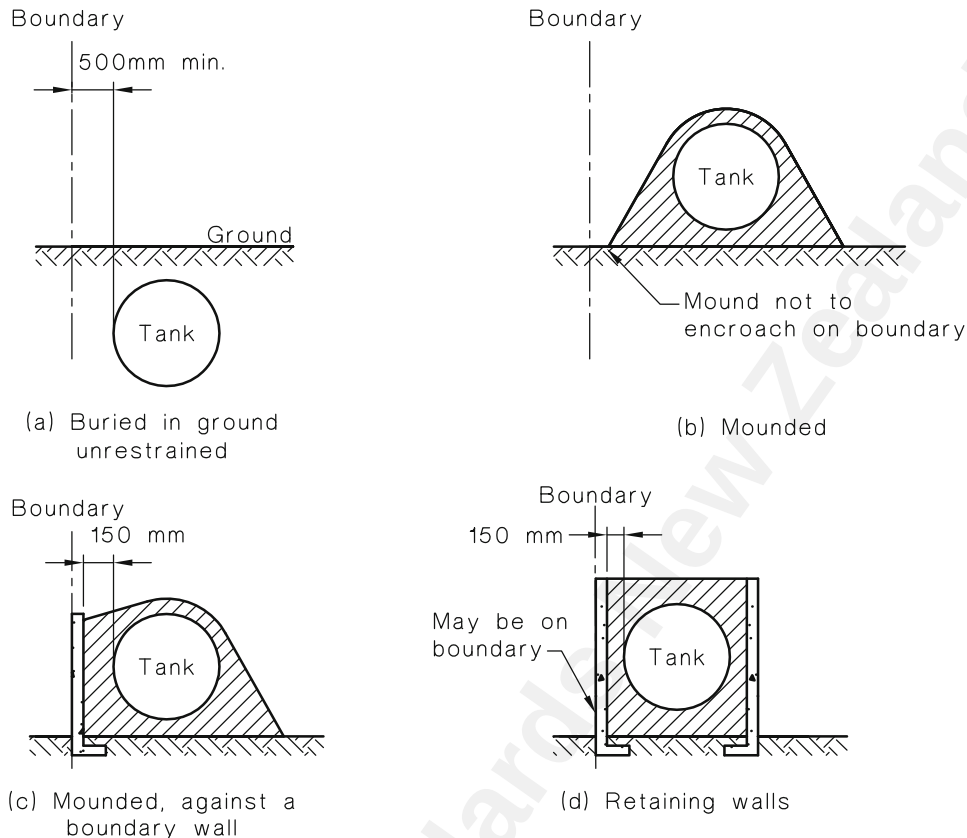
NOTE: If traffic access is prevented, a covering of not less than 600 mm of sand or earth may be substituted.

- (c) Where tanks are wholly or partly above ground level, that part above ground level shall be entirely covered to a minimum radial thickness of not less than 600 mm with either additional sand or suitable earth. For mounds without retaining walls, the slope of the cover shall be not steeper than the angle of repose of the material.
- (d) The covering for a tank shall be resistant to erosion by wind, rain, or water streams from firefighting or nearby tank cooling systems. The covering shall be maintained.

TABLE 7.1
SEPARATION REQUIREMENTS FOR UNDERGROUND
STORAGE SYSTEMS

From	To	Minimum requirement	
		Tank <8 kL	Tank ≥8 kL
Tank shell	Public place boundary Adjacent property boundary	As illustrated in Figure 7.1	
	Protected place (on or off site)	No specific requirement, limited by other requirements	
	Adjacent LP Gas tank	1 m	
Above-ground tank fitting, e.g. valve, flange, ullage gauge	Public place	1.5 m	3 m
	Adjacent property	1.5 m	6 m
	Protected place	1.5 m	6 m
Pump	See Clause 6.11.4		
Filling connection	For a wet hose delivery*, and inward flow only, distances are as for tank fittings; otherwise for remote connections as per Clause 6.6.7(b) to (e) and requirements as for Clause 6.6.3. For direct connections, Clause 6.6.2 shall apply.		
Safety valve discharge	See Clause 7.5.2		

* This term refers to a hose assembly design that limits the loss of LP Gas liquid on disconnection to 1 L or less.



NOTES:

- 1 These illustrations should not be interpreted as setting limits to the type of installation allowable. Their purpose is to indicate boundary locations only.
- 2 Despite the distances given above, the location of the fill connection (as given in Table 7.1) will dictate the location of the tank.
- 3 The dimensions are not in scale.

FIGURE 7.1 DISTANCES FROM TANK SHELL TO BOUNDARIES

7.3.4 Anchoring

The tank shall be installed in such a manner as to prevent movement in excess of the system design. Any steelwork associated with the anchoring system shall not make electrical contact with the tank or the piping system unless included in the cathodic protection system.

7.3.5 Tank branches and tappings

The provisions for tank tappings for filling, draw-off and similar use shall comply with the following requirements:

- (a) Direct filling is permitted only where liquid is discharged into the vapour space of the tank.
- (b) Any part of the manhole branch and cover that projects from the tank shell to above ground shall be heat-protected by some means having an FRL/FRR of not less than -/120/120.

- (c) Where a pipe is connected to a tank shell rather than to an access hole and cover, the protective valves shall be arranged at the first pipe joint from the tank shell, provided that—
 - (i) the joint is not more than 400 mm from the point where the pipe emerges from the ground; and
 - (ii) a manual shut-off valve, or a flanged or screwed joint, shall not be buried. (For illustrations of possible arrangements, see Figure 7.2.)
- (d) Any connection for loading a tanker shall be a remote connection.

7.3.6 Gas storages other than LP Gas

The distances specified in Clause 6.2.4 shall apply, but only from the connections to the underground LP Gas tank.

7.4 ACCESS, PROTECTION, FITTINGS AND ACCESSORIES

7.4.1 Access

Each component or fitting that may be required to be accessible, e.g. a filling connection, gauge, safety valve or access hole, shall be located within an access pit, or within a dome or housing mounted directly above such an access pit, or on or within a nozzle or branch extending from the tank to above ground level.

7.4.2 Protection for above-ground equipment

Any part of, or accessory for a buried tank which projects above ground level shall be designed, located, or protected so as to be not liable to accidental damage.

7.4.3 Access pits

Any access pit shall comply with the following requirements:

- (a) The pit opening shall be covered by an above-ground dome or housing, or by a flush cover, capable of preventing the entry of water.
- (b) Where a flush cover might be subject to traffic, it shall be of sufficient strength to withstand normal traffic loads.
- (c) Pit walls shall be of low carbon steel not less than 6 mm thick, or reinforced concrete not less than 75 mm thick, or masonry not less than 100 mm thick, or an alternative material of equivalent performance, and shall extend from the tank to the ground surface in such a manner that traffic loads are not transmitted to the tank shell by the pit wall.
- (d) A means of removing water from the pit, such as a pump, shall be provided. A pump well, if provided, shall not exceed 200 mm diameter.
- (e) A pit shall have a horizontal cross-sectional area of not more than 2 m², and shall be not more than 1 m deep to the base of the pit, excluding any pump well not greater than 100 mm deep. The top of the tank cover or lowest pipe work joint shall be not greater than 800 mm deep.
- (f) The access pit shall be provided with a means of limiting, or of detecting and dissipating, any build-up of vapour prior to access.

NOTES:

- 1 Ventilation for the access pit should discharge—
 - (a) at least 2 m above ground level;
 - (b) at least 2 m from any opening; and
 - (c) at least 1 m above any solid structure that is within 2 m of the discharge point.
- 2 Pipework for ventilation of the access pit may comprise 80 mm DWV (sewer) pipe, installed to prevent water traps, i.e. fall to the access pit or fill point.

- (g) Any fill point enclosure complying with Items (i) to (vi) below is not considered an access pit, and does not require high level (mechanical) ventilation or permanent lighting.
 - (i) The volume of the fill point enclosure shall not exceed 150 L capacity.
 - (ii) The fill point enclosure shall be of a size that prevents entry by personnel.
 - (iii) The enclosure shall be no more than 400 mm deep.
 - (iv) Any contents gauge within a remote fill enclosure shall be capable of being read at all times, without the aid of a portable light source.
 - (v) There shall be a maximum of two automatic actuated valves or rotary joints, to reduce potential leakage paths.
 - (vi) The enclosure shall not be sealed. This is to permit sufficient natural ventilation to dissipate any LP Gas vapours.

7.4.4 Contents gauging

The following requirements apply:

- (a) Where the contents gauge is obscured by the filling connection, a remote gauging system shall be used.
- (b) Any contents gauge within an access pit shall be capable of being read at all times, without the aid of a portable light source.

7.5 DISCHARGE FROM SAFETY RELIEF DEVICES

7.5.1 Discharge rate

The rate of discharge of a safety relief device on an underground tank may be reduced to not less than 30% of that required for an above-ground tank as specified in Clause 5.4.1(c).

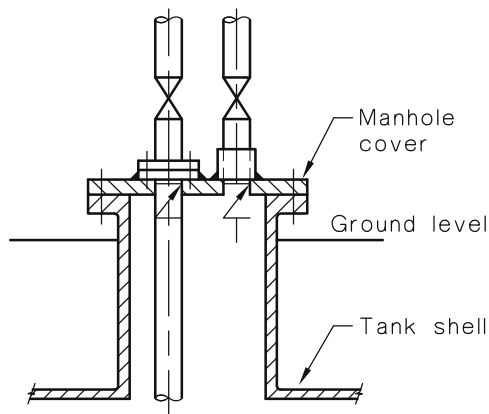
7.5.2 Point of discharge

The discharge from a safety relief device shall be directed vertically upwards. The discharge point shall—

- (a) terminate not less than 3 m above the ground, except for a stack pipe that is directly connected to the safety relief via a shear point; in such a case the discharge point shall terminate not more than 0.5 m above the ground; and
- (b) be not less than 2 m from any opening in a wall or structure which could permit vapour to enter a building, a cavity wall, a roof space, or a similar enclosure.

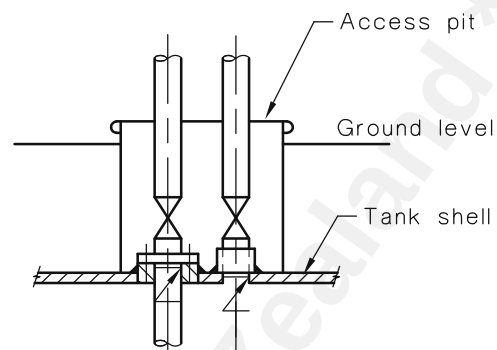
Where a vent stack that is directly connected to the safety relief via a shear point passes through a structure that provides a pivot point for the stack to act as a cantilever a second shear point shall be provided for above the structure.

NOTE: Particular care should be taken to recognize such potential entry paths as weatherboards and roofing tiles.



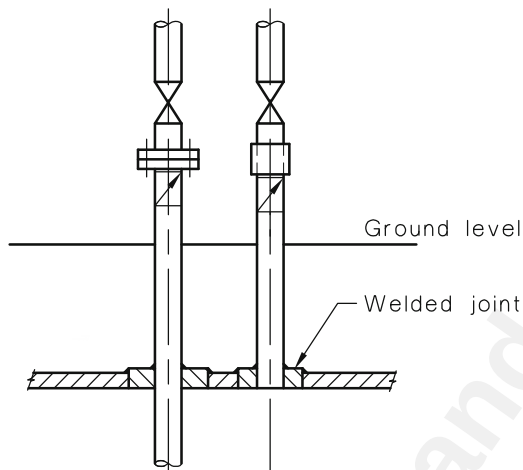
NOTE: Connections at the cover, being above ground, may be welded, flanged, or screwed. Primary and secondary protective valves are at the joint.

(a) Manhole branch, cover above ground



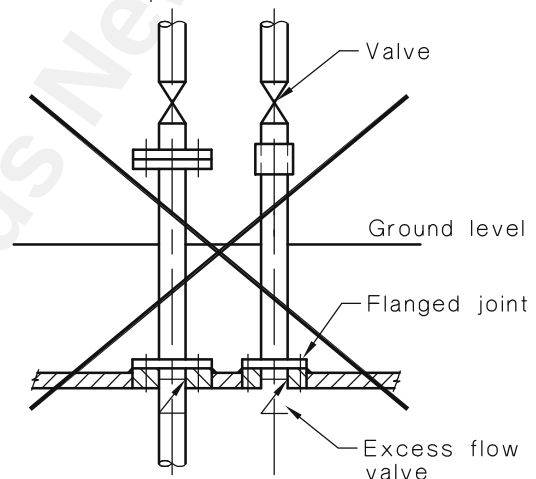
NOTE: Connections at the shell, being open to air, may be welded, flanged, or screwed. Primary and secondary protective valves are at the joint.

(b) Access pit to shell connections



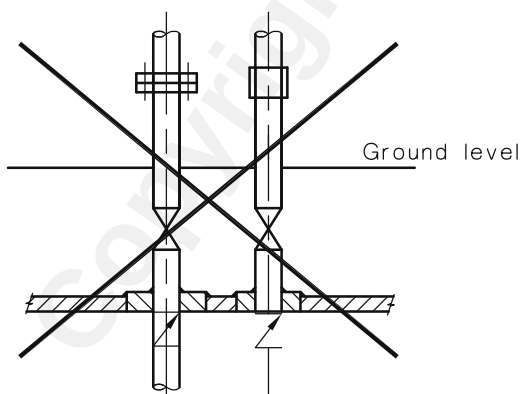
NOTE: The above-ground connections may be welded, flanged, or screwed. Primary and secondary protective valves are at the joint.

(c) Extension nozzle to above ground, welded to tank



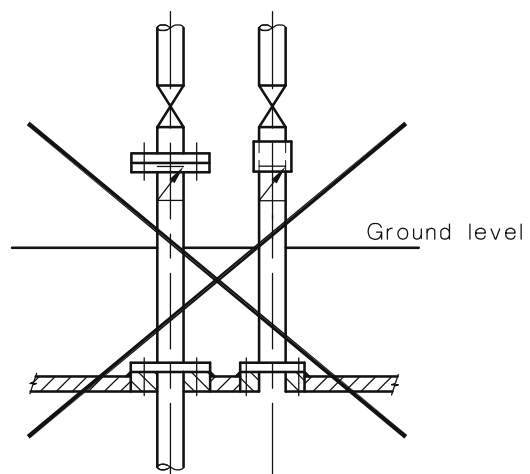
NOTE: Not acceptable because the underground flange connection is prohibited.

(d) Unacceptable arrangement



NOTE: Not acceptable because the secondary protection is buried. Arrangement (c) should be used.

(e) Unacceptable arrangement



NOTE: Not acceptable because buried mechanical connection is not protected by the primary protection valve.

(f) Unacceptable arrangement

FIGURE 7.2 PIPING CONNECTION TO BURIED TANKS

SECTION 8 CYLINDER FILLING INSTALLATIONS — AUSTRALIA

8.1 SCOPE OF SECTION

This Section provides requirements for the filling of cylinders of LP Gas within Australia.

Excluded from this Section are in situ filling installations (see Section 4) and on-vehicle automotive refuelling (see Section 10).

8.2 GENERAL REQUIREMENTS

8.2.1 Filling by mass

Where cylinders are to be filled by mass, this shall be by pumping at a cylinder filling area in accordance with Clause 8.3, except as permitted by Clauses 8.2.2 and 11.12.

When filling by pumping, the pumping rate shall be appropriate to the cylinder size, in order to minimize the risk of overfilling.

NOTE: In situ cylinders are not to be filled by mass.

8.2.2 Filling by volume

Where cylinders are to be filled by volume, the appropriate method shall be used, as set out below:

- (a) Decanting shall be limited to cylinders that are fitted with an appropriate fixed liquid level gauge, incorporating a vapour release mechanism.
- (b) Pump filling is limited to—
 - (i) cylinders that are fitted with an appropriate automatic fill limiting device; or
 - (ii) cylinders that are for automotive purposes (e.g. fork-lift trucks); or
 - (iii) cylinders that have a water capacity greater than 100 L and are fitted with a fixed liquid level gauge.
- (c) Where in situ cylinders are being initially filled prior to installation, care shall be taken to fill to less than the ullage limit.

When filling by pumping, the pumping rate shall be appropriate to the cylinder size, in order to reduce the risk of overfilling.

8.2.3 Suitability of gas supply

A tank or cylinder that supplies propane/butane blends as fuel for an engine or liquid phase burner shall not be used to fill cylinders for other applications.

8.3 CYLINDER FILLING AREA

8.3.1 Location of filling point

Where cylinders are filled by a means other than decanting, the cylinder filling point and the edge of the cylinder filling area shall be at least—

- (a) 6 m from any protected place; and
- (b) 3 m from any public place, boundary, drain, pit, basement or above-ground LP Gas storage tank.

NOTES:

- 1 Separation distances may be measured horizontally around a vapour barrier or firewall constructed in accordance with Clause 3.6.
- 2 Railing or similar, to prevent movement into the separation zone, may be taken into account in the application of this requirement.

8.3.2 Adjacent storage

Where cylinders are stored adjacent to the filling point, Clause 4.5 shall apply.

A clear means of egress, in more than one direction and at least 1 m wide, shall be maintained from the filling point clear of the area where cylinders are stored.

8.3.3 Ventilation

A cylinder filling area shall be ventilated as follows:

- (a) Where the area is covered, cross-ventilation shall be provided by openings at the top and bottom of all sides, or by leaving at least two sides open.
- (b) Where the filling area is on a platform or is elevated above ground level, the space beneath the area shall be open on at least three sides, free of combustibles and empty, or filled solid.

8.3.4 Emergency shut-down

An emergency shut-down device shall be fitted, to shut off the liquid supply to the filling point in an emergency. The actuating point for such a device shall be located so that it is convenient to the path of egress from the operating station.

Where a deadman's handle is fitted, and must be actuated in order for LP Gas to flow, this may be considered to be an emergency shut-down device.

8.4 DECANTING**8.4.1 Restrictions**

The following restrictions apply when filling cylinders by decanting:

- (a) Cylinders larger than 25 L capacity shall not be filled by decanting, except when they are cylinders used as fuel containers for vehicle engines. In such cases, this limit is increased to 50 L. Larger fuel containers for vehicle engines fitted with an automatic fill limiter (AFL) may be filled by decanting in order to permit the testing and commissioning of the installed cylinder (as required by AS/NZS 1425).
- (b) A decanting cylinder or tank shall be arranged so that only one cylinder can be filled at a time.

8.4.2 Decanting equipment

The following requirements apply to decanting equipment:

- (a) The decanting hose shall—
 - (i) comply with AS/NZS 1869;
 - (ii) be not more than 10 mm nominal bore; and
 - (iii) be not more than 1.5 m in length.
- (b) An excess flow valve shall be provided before the upstream end of the hose or rigid decanting arm.
- (c) A valve actuated by a deadman's handle shall be incorporated into the cylinder filling attachment.

8.4.3 Location of the decanting cylinder or tank

8.4.3.1 *Decanting cylinder*

The following requirements apply to the location of the decanting cylinder:

- (a) A decanting cylinder shall not be—
 - (i) stored indoors; or
 - (ii) used indoors; or
 - (iii) used in a portable application.
- (b) A decanting cylinder shall be located on the ground in a static position.
- (c) Decanting cylinders at service stations shall comply with the requirements of Clause 10.5.19.

8.4.3.2 *Decanting tank*

The following requirements apply to the location of a decanting tank:

- (a) The tank shall be installed in a fixed position in accordance with Section 6.
- (b) The flexible hose connection point to the tank shall be at least 3 m from the tank.
- (c) Provision shall be made to ensure that the hose is not used in such a manner that the decanting connection encroaches on the separation distances specified in Clause 8.4.3.3 or is closer to the tank than the hose connection point.

8.4.3.3 *Location of decanting installation*

Whether decanting from a tank or a cylinder, the design of the installation shall be such that the point of connection to the cylinder being filled cannot lie within the following separation distances:

- (a) To an opening into a building—2 m (see Note 1).
- (b) To buildings on a neighbouring property, or to any combustible materials stored above ground—5 m.
- (c) To public places—3 m.
- (d) To any above-ground tank containing dangerous goods—3 m.
- (e) To a dispenser for any type of fuel—3 m.
- (f) To the entrance to any drain, pit or basement—3 m.
- (g) To any structure limiting egress past the point of connection to the cylinder—2 m.
- (h) To the access cover of any underground storage tank—1 m.

NOTES:

- 1 Where an opening into a building is within the hazardous area and remains closed during decanting, the hazardous area is deemed not to enter the building. Where the opening does not remain closed the hazardous area extends inside the building.
- 2 The separation distances given above and the hazardous area illustrated in Figure E3 are based on a single decanting point.
- 3 The point of connection implies hose reach where a flexible decanting hose is used. Where LP Gas decanting takes place in a dedicated area clearly marked on the ground or on a fixed cylinder stand, the separation distances may be taken from the marked area or stand.
- 4 The above requirements are illustrated in Figure 8.1.
- 5 Railings or similar, to prevent movement into the separation zone, may be taken into account in the application of the above separation requirements.
- 6 Recommended decanting procedures are given in Appendix J.

8.4.4 Positioning of cylinder to be filled

The cylinder to be filled shall be positioned so that the fixed liquid level gauge and the safety valves are in communication with the vapour space.

8.4.5 Security of equipment

Where decanting equipment is on an unattended site that is accessible to the public, such equipment shall be protected from tampering by one of the following means:

- Providing a locked, ventilated enclosure for any decanting equipment and any cylinder or tank outside a building.
- Ensuring that decanting equipment is disconnected from the cylinder or tank and stored in a secure place, and providing a means to render the supply valve inoperable.
- Securing the decanting equipment and supply valve in a lockable dome or cabinet.

8.4.6 Notices

Signs and notices for cylinder filling installations shall be provided in accordance with Appendix D.

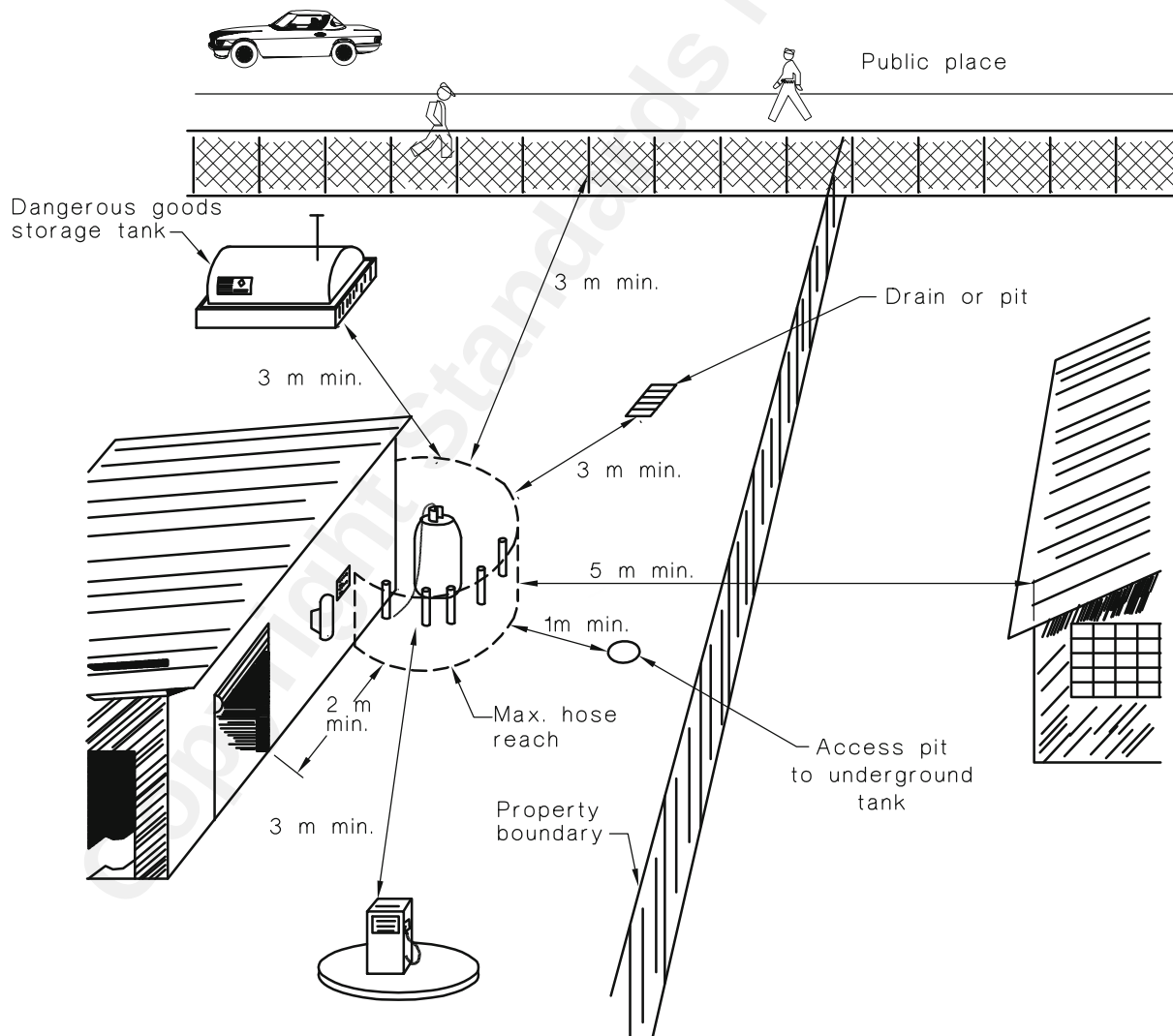


FIGURE 8.1 LOCATION OF DECANTING CYLINDER

8.5 CYLINDER TO FORKLIFT AND AUTOMOTIVE CYLINDER—FILLING BY PUMP

8.5.1 General

A typical dispensing bundle filling unit consists of one or more cylinders, nominal 500 L water capacity fixed within a framed structure. These are interconnected to a single outlet which is protected by an emergency shut-off valve and controlled by a deadman mechanism as well as a manual shut-off valve on each cylinder. A pump unit is fitted to expedite the cylinder filling process, typically the re-filling of fork truck cylinders.

8.5.2 Application

Filling shall be in accordance with Clause 8.2.2, Items (b)(i) and (b)(ii).

It does not apply to the filling of vapour service exchange cylinders or liquid withdrawal cylinders, which are dealt with elsewhere in this Standard.

8.5.3 Pump rate

The minimum pumping rate shall be no less than 5 L/min to achieve safe operation of an automatic fill limiter.

8.5.4 Frame

The frame shall retain securely all the components of the bundle and shall protect them from damage which might cause leaks. Such damage can be caused by vibration, impact loads or handling loads which can be expected in normal operation. The method of cylinder restraint shall minimize any vertical or horizontal movement or rotation of the cylinder. All cylinder displacement which would impose undue strain on the manifold shall be prevented. The total assembly shall be capable of withstanding rough handling.

The frame shall include features designed for the handling and transportation of the bundle. Bundles can be lifted typically by fork-lift, lift-jack trolley or overhead crane. Lifting eyes shall be provided on the frame, if the bundle is designed to be lifted by overhead crane.

A maximum of four (4) interconnected cylinders shall be installed in any bundle.

8.5.5 Piping

Piping shall be designed, constructed and installed so as to avoid damage due to expansion and contraction, mechanical shock and vibration. Where the manifold is made of metal, the necessary flexibility shall be achieved by the use of bends or coils. Flexible hoses shall be kept to a minimum.

Piping shall comply with the requirements of Clause 5.6.

8.5.6 System components

The design of these systems shall incorporate the following:

- (a) Liquid supply openings shall be protected with an internal excess flow valves from all cylinders.
- (b) Vapour openings greater than 8 mm shall be protected by a back check or excess flow valve valves in all supply cylinders which may be integral with the cylinder manual shut off valve.
- (c) Delivery hose shall be fitted with a breakaway coupling.
- (d) Operation shall be controlled by a deadman mechanism which must be installed in a position in which supply valves are closed other than when in operation.
- (e) A filling nozzle complying with Clause 10.3.4.
- (f) An emergency shutdown device shall be a dead man type complying with Clause 8.3.4.

- (g) The dispensing hose assembly shall comply with Clause 10.3.3.
- (h) The dispensing hose shall be protected by an excess flow valve at the end of rigid pipework and have a manual isolating valve immediately upstream of the excess flow valve.
- (i) The requirement for filler caps shall comply with Clause 10.3.5.
- (j) All piping shall be protected in accordance with Clause 6.8.2.
- (k) Pump shall comply with Clause 6.11.

8.5.7 Filling requirements

The following requirements shall apply:

- (a) The filling procedure and pre-filling inspection required in Australia by AS 2030.1.
- (b) A cylinder that does not comply with Clause 4.2.1 shall not be filled.
- (c) The filling ratio shall not be exceeded (see Appendix B).
- (d) Filling by volume shall only be carried out on a cylinder which incorporates a fixed liquid level gauge or an automatic fill limiter device.
- (e) Filling shall be stopped if any fault or malfunction causes gas to escape.
- (f) Cylinders shall only be filled by the owning organization or upon authorization of the apparent owner.

8.5.8 Hose reach zone

The location of the hose reach zone shall be compliant with Clause 10.2.2.2(a).

8.5.9 Cylinder filling area

Where cylinders are filled using this transfer method (cylinder-to-cylinder filling by pump), the filling point and the edge of the cylinder filling area shall be at least 5 m from any protected place on an adjoining property and comply with Table 4.1.

8.5.10 Fire safety

Cylinder-to-cylinder filling installations shall be provided with a minimum of a single fire extinguisher (see Clause 13.7.5).

8.5.11 Positioning of cylinder to be filled

Cylinders to be filled which are fitted with a fixed liquid level gauge shall be positioned so that this gauge and the safety valve are in communication with the vapour space.

Cylinders to be filled which are fitted with an automatic fill limiter shall be positioned in the correct orientation to avoid a malfunction with subsequent overfilling.

8.5.12 Notices

Signs and notices for cylinder filling installations shall be provided in accordance with Appendix D, Paragraph D2.3.

8.5.13 Supervision and training

A person trained and experienced in the filling operation shall be in attendance throughout the cylinder filling operation (see Section 11).

SECTION 9 CYLINDER FILLING INSTALLATIONS—NEW ZEALAND

9.1 SCOPE OF SECTION

This Section provides requirements for installations for the filling of cylinders within New Zealand. In this context a cylinder filling station may include the fill point, a scale and a specified quantity of LP Gas. It may be located in either an outdoor or an indoor location. In an indoor location, the area shall be adequately ventilated.

9.2 GENERAL REQUIREMENTS

9.2.1 Filling by weight

Cylinders for LP Gas shall be filled by weight at a cylinder filling station that complies with Clause 9.3, except as permitted in Clause 9.2.2. The weight of any charged cylinder must be checked after disconnecting the cylinder from the charging line.

9.2.2 Filling by volume

Filling by volume shall be limited to the following:

- (a) A forklift cylinder that is filled while mounted on the vehicle; or
- (b) A cylinder mounted on any other type of vehicle; or
- (c) A cylinder operating under a code of practice approved by the New Zealand Environmental Protection Authority (EPA) relating to cylinder filling.
- (d) Cylinders for in situ filling, which comply with the New Zealand Environmental Protection Authority (EPA) Approved Code of Practice—*In situ Filling of LPG Cylinders* available from the LPGA.
- (e) Cylinders of less than 4 kg (9 L) nominal capacity. When using this method, the weight of the charged cylinder must be checked after disconnecting the cylinder from the charging line.

9.2.3 Ventilation

A cylinder filling station shall be ventilated as follows:

- (a) Where the area is covered, cross-ventilation shall be provided by openings at the top and the bottom of all sides or by leaving at least two sides open.
- (b) Where filling is on a platform or elevated above ground level, the space beneath shall be either completely open on at least three sides, free of combustibles and empty or filled solid.

9.2.4 Ignition sources

An ignition source shall not fall within a hazardous area as defined in AS/NZS 60079.10.1 and illustrated in Appendix E.

9.2.5 Use of vapour barriers

The separation of ignition sources from a hazardous area may be measured around a vapour barrier.

9.2.6 Use of firewalls

Any reduction in separation distances shall require the approval of the authority.

9.2.7 Security

Where equipment in an area accessible to the public ceases to be under supervision, such as when closed for the night, it shall be protected from tampering by one of the following:

- (a) The whole of the cylinder filling equipment shall be located in a locked ventilated enclosure outside any building.
- (b) Access to cylinder filling hoses, pipes and valves shall be prevented by means of a lockable cover.

9.2.8 Signage

The requirements for signage are set out in Appendix D.

9.2.9 Emergency shut-down device

An emergency shut-down device shall be fitted, to shut off liquid supply to the filling point in an emergency. The actuation point shall be located so that it is convenient to the path of egress from the operating station.

9.3 CYLINDER FILLING AREA

9.3.1 Location of filling station with up to 100 kg of LP Gas in cylinders at the filling point

Where cylinders are filled by a means other than decanting, the cylinder filling point and the edge of the cylinder filling area shall be at least—

- (a) 6 m from an area of high intensity land use;
- (b) 3 m from an area of low intensity land use;
- (c) 3 m from an aboveground LP Gas storage tank; and
- (d) 1 m from an underground LP Gas tank turret.

All measurements shall be made from the filling point or from the cylinders stored at the filling station, whichever is the nearest to the feature described in Items (a) to (d).

NOTES:

- 1 Separation distances measured horizontally around a firewall constructed in accordance with Clause 3.6, may be taken into account (see Clause 9.2.6).
- 2 Railing or similar, to prevent movement into the separation zone, may be taken into account in the application of this requirement.

9.3.2 Location of filling station with from 100 kg and up to 1000 kg of LP Gas in cylinders at the filling point

Where cylinders are filled by a means other than decanting, the cylinder filling point and the edge of the cylinder filling area shall be at least—

- (a) 8 m from area of high intensity land use;
- (b) 8 m from area of low intensity land use;
- (c) 8 m from aboveground tank; and
- (d) 1 m from an underground LP Gas tank turret.

All measurements shall be made from the filling point or from the cylinders stored at the filling station whichever is the nearest to the feature described in Items (a) to (d).

NOTES:

- 1 Separation distances measured horizontally around a firewall constructed in accordance with Clause 3.6, may be taken into account (see Clause 9.2.6).
- 2 Railing or similar, to prevent movement into the separation zone, may be taken into account in the application of this requirement.

9.3.3 Location of filling station with 1000 kg and over of LP Gas in cylinders at the filling point

Where cylinders are filled by a means other than decanting, the cylinder filling point and the edge of the cylinder filling area shall be at least—

- (a) 15 m from area of high intensity land use;
- (b) 15 m from area of low intensity land use;
- (c) 8 m from an aboveground LP Gas storage tank; and
- (d) 1 m from an underground LP Gas tank turret.

All measurements shall be made from the filling point or from the cylinders stored at the filling station whichever is the nearest to the feature described in Items (a) to (d).

NOTES:

- 1 Separation distances measured horizontally around a firewall constructed in accordance with Clause 3.6 may be taken into account (see Clause 9.2.6).
- 2 Railing or similar, to prevent movement into the separation zone, may be taken into account in the application of this requirement.

9.3.4 Adjacent storage

Cylinders stored adjacent to the cylinder filling station shall not be closer than the HILU separation distance as set out in Table 9.3.3.

The cylinder storage area shall meet the HILU separation distance as set out in Table 9.3.3.

TABLE 9.3.3
HILU SEPARATION DISTANCE

Aggregate quantity of LP Gas in cylinders kg	Area of high intensity land use m
Up to 100	0
300	2
500	2
2000	3
5000	5
10 000	7
50 000	8
More than 50 000	15

NOTE: Where separation distances are not met, variations to the separation distances may be approved by the authority.

9.3.5 Service station forecourt filling

Where service station forecourt filling is employed, the following shall apply:

- (a) Cylinders shall not be stored or left unattended adjacent to the cylinder filling facility.
- (b) While filling, not more than 100 kg of LP Gas shall be located adjacent to the cylinder filling facility.
- (c) The separation distances as set out in Clause 9.3.1.

9.4 DECANTING

9.4.1 Capacity restrictions

All cylinders of 4 kg (9 L) nominal capacity and less shall be filled by decanting. Cylinders larger than 25 L capacity shall not be filled by decanting.

9.4.2 Decanting connection

The decanting hose shall comply with Clause 5.6.5 and shall be of not more than 10 mm nominal bore and not more than 1.5 m long. An excess-flow valve shall be provided before the upstream end of the hose, or rigid decanting arm.

9.4.3 Location of decanting area

The following separation distances from the point of connection to the cylinder being filled shall be maintained:

- (a) To an opening into a building 2 m (see Note 1).
- (b) To buildings on neighbouring property or to any combustible materials stored above ground 5 m.
- (c) To public places 3 m.
- (d) To any tank 3 m.
- (e) To the entrance of a drain, pit or basement 3 m.
- (f) To any structure limiting egress past the point of connection to the cylinder 2 m.

NOTES:

- 1 Where an opening into a building is within the hazardous area and remains closed during decanting the hazardous area is deemed not to enter the building. Where the opening does not remain closed, the hazardous area extends inside the building.
- 2 Railings, or similar, to prevent movement into the separation zone may be taken into account in the application of this requirement.
- 3 The point of connection implies hose reach if flexible decanting hoses are used.
- 4 The separation distances in this Clause (9.4.3) and ignition source exclusion zone in Appendix E are based on a single decant point.

9.4.4 Arrangement of cylinder

The cylinder being filled shall be arranged for filling so that the fixed liquid level gauge and the safety valve are exposed to the vapour space.

9.4.5 Decanting from tanks

While decanting from tanks, the decanting connection shall not be subject to pump pressure.

SECTION 10 AUTOMOTIVE FILLING INSTALLATIONS

10.1 INTRODUCTION

This Section provides additional specific requirements for a variety of types of installation for the on-vehicle refuelling of LP Gas for vehicles of any type. All the requirements of this Standard apply to such installations, except that the additions or variations below apply and take precedence in the case of conflict. Clauses 10.2 to 10.4 give basic requirements that apply to all refuelling installations, and Clauses 10.5 to 10.8 are additional requirements that apply to service stations.

If any refuelling for the general public can occur, the installation becomes a service station (as defined in Clause 1.4.53) and specific additional requirements apply. The additional requirements arise from a need to compensate for a lower level of skill and familiarity on the part of the public.

Where a service station can be used for self-service fleet refuelling during unattended periods or after hours, then the general public shall be prevented from using the self-service facility at such times.

NOTES:

- 1 The wide variety of types of LP Gas-fuelled vehicles, of drivers and owners, of supervision of refuelling, and of methods of refuelling, has given rise to a variety of types of refuelling installations and a variety of requirements for them.
- 2 Fuel containers that are removed from the vehicle for refuelling are in effect change-over cylinders, to which Sections 8 and 9 apply.
- 3 The basic installation is that which is used solely for fleet refuelling, i.e. the general public is excluded, and the requirements are graded according to the degree of supervision.
- 4 Since service stations may at times be located in commercial or residential areas having sensitive occupancies, certain planning and land use requirements apply.

10.2 COMMON REQUIREMENTS

10.2.1 Tank and pump piping

Any liquid line between the pump and the dispenser that exceeds 10 m in length shall be provided with a manual isolating valve close to the pump outlet but not before the bypass line.

NOTE: For convenience in maintenance it is recommended that a manual isolating valve be installed so that the piping between the tank and the pump can be shut off.

10.2.2 Filling area

10.2.2.1 Illumination

The area within which the filling connection can be made shall be illuminated to a level of not less than 50 lx whenever the unit is available for service.

10.2.2.2 Hose reach zone

The location of the hose reach zone on the dispenser shall permit compliance with the following requirements:

- (a) No part of the vehicle being refuelled shall be in a public place.
- (b) A public place, a protected place, an entrance to a building, a pit, or the entrance to an enclosed drain, a basement or an above-ground tank for flammable liquids, shall not be within the hose reach zone for a Style A dispenser. (See Figure 10.1 for an illustration of terms, and Figure 10.2 for the types of dispenser.)

- (c) It shall not be possible to connect a nozzle to a vehicle within 3 m of an above-ground LP Gas tank or the compound of an above-ground flammable liquids installation.
- (d) The distance between the hose reach zone and the filling connection for LP Gas delivery from a road tanker shall be not less than 6 m except where the liquid lost from a tanker hose coupling when disconnecting does not exceed 1 L, in which case the distance may be reduced to 2 m. This requirement may be waived where the site arrangement is such that the presence of either a tanker transferring or a vehicle refuelling prevents the approach of the other.

NOTE: Care should be taken to consider the interrelation of the hazardous areas in locations where LP Gas dispensers are to be installed.

10.2.2.3 Vehicle filling area

The standing area for the vehicle being filled shall not slope more than 1 in 20.

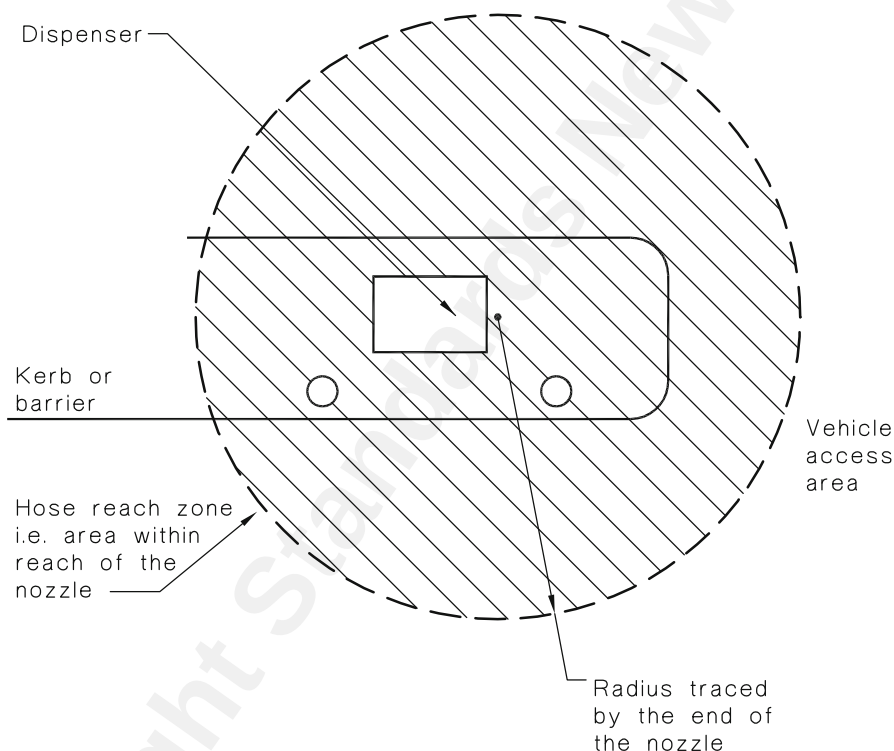


FIGURE 10.1 TYPICAL HOSE REACH ZONE

10.3 DISPENSING SYSTEMS

10.3.1 General design

The arrangement of an LP Gas dispensing system shall be one of those illustrated in Figure 10.2, as appropriate to the usage of the installation.

NOTES:

- 1 Some of these arrangements would not be permissible for vending to the public because of statutory requirements for accuracy of measurement.
- 2 Dispensers may be protected by the use of bollards or guardrails. This is considered good practice but is not mandatory.

10.3.2 Style A dispenser system

Where a Style A dispenser system is required by Figure 10.2, it shall include a dispenser that complies with AS/NZS 2229 and which is provided with an excess-flow valve, a remote-operated shut-off valve, and a pipe-shear provision, in that order, in all LP Gas pipes attached to the cabinet. The valves shall be below or as close as possible to ground level, with the shear provision above them. Where the remote-operated valve can be relied upon to shut off automatically in the event of pipe shear, the excess-flow valve may be omitted.

In addition, the system shall incorporate the following:

- (a) *Control valves* The following control valves shall be provided inside the dispenser cabinet and as close as practicable to the flexible hose between the dispenser cabinet and the breakaway coupling:
 - (i) A manually operated positive shut-off valve.
 - (ii) An excess-flow valve.
- (b) *Return line* In Australia, a return line from a dispenser shall comply with the requirements of Table 5.2 for liquid withdrawal or two-way flow.

In New Zealand, the return line shall comply with the requirements for vapour return and balancing lines.

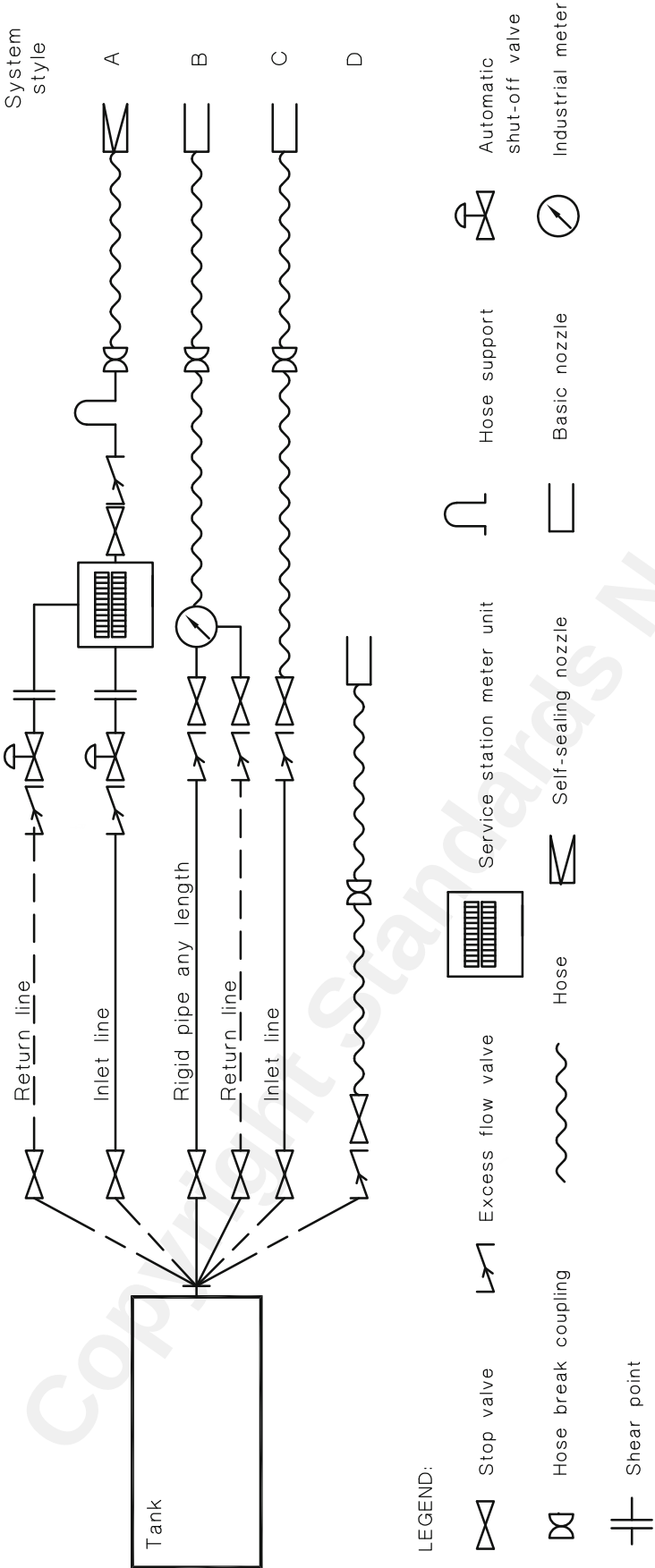
A continuous liquid bypass in the dispenser shall be limited to a flow area of less than 4 mm².

10.3.3 Dispensing hose

The dispensing hose assembly shall comply with Clause 5.6.5 and shall incorporate a self-sealing hose break coupling that complies with the following requirements:

- (a) The coupling shall disconnect in the event of undue force being applied to the hose, such as a vehicle moving off with the hose connected.
NOTE: The coupling should be capable of being re-assembled after separation, preferably without the need for draining the hose, the use of special tools, or the replacement of parts.
- (b) The force required to separate the coupling shall be less than 600 N for internal hose pressures of 700 kPa to 2500 kPa.
- (c) The coupling shall retain the liquid within each section of the hose after separation when pressures of 700 kPa to 2500 kPa are maintained in the hose. The volume of liquid lost as a result of parting shall not exceed 15 mL and the coupling shall not leak after disconnection.

NOTE: Auto gas dispensing hose is not required to be conductive.



Types of site		Options permitted
1. Service stations (Retail outlets)		Style A
Other than service stations (i.e. fleet refuelling)		} A to D but with 
2. Served by dispenser attendant		
3. Self-serve, on-site supervisor		
4. Self-serve, no on-site supervision		
NOTES :		
1. Fleet refuelling means dispensing to commercial or industrial vehicles as distinct from private vehicles in private use.		
2. Stop-valve near tank only required if pipe is longer than 10 m		

FIGURE 10.2 TYPICAL REFUELLING SYSTEMS

10.3.4 Dispensing nozzle

The dispensing nozzle shall comply with the following requirements:

- (a) The connection provision shall mate with the filler connection specified in AS/NZS 1425.
- (b) A provision to start and stop gas flow shall be incorporated and shall be quick-acting (i.e. not a screw-down valve).
- (c) Where the nozzle is self-sealing—
 - (i) it shall not be possible to discharge fuel unless the nozzle is connected and fully sealed to the vehicle's filler connection; and
 - (ii) upon disconnection, the amount of liquid released shall not exceed 15 mL.
- (d) Where a self-sealing nozzle is fitted to a Style A dispenser and is intended for public use, the nozzle lever shall be fitted with a latching device only if one of the following safety features is in place:
 - (i) The nozzle cannot be sufficiently disconnected to cause a leak while the lever is actuated.
 - (ii) The dispenser system incorporates an additional device that has to be activated by the person operating the dispenser, and that requires constant actuation by the operator to maintain the flow of fuel. When the device is released, the flow of fuel is stopped.
 - (iii) A device is provided that achieves an equivalent level of safety as Items (i) and (ii) above.

10.3.5 Filler caps

At least two spare standard metal automotive filler caps shall be kept on site for use in the event of a leak at a vehicle-filling connection.

The provision of suitable gloves with forearm protection that are impervious to LP Gas and provide thermal protection shall be provided with the filler caps.

10.4 NOTICES, SIGNS AND INSTRUCTIONS

Dispenser notices, signs and instructions shall be provided for the site in accordance with Appendix D.

10.5 SPECIFIC REQUIREMENTS FOR SERVICE STATIONS

10.5.1 Introduction

Clauses 10.5 to 10.8 provide requirements that are specific to service stations, and apply in addition to Clauses 10.2 to 10.4, Table 5.2 and other relevant requirements of this Standard.

The requirements of Clauses 10.5 to 10.8 deal with installations comprising—

- (a) a single or dual above-ground tank installation of up to 16 kL aggregate capacity; and
- (b) a single or dual underground tank installation of up to 65 kL aggregate capacity.

Storages above this size are likely to require further analysis, but all general technical principles and controls shall apply.

For other types of automotive refuelling installations, only Clauses 10.1 to 10.4 apply.

Not more than two above-ground tanks for automotive dispensing shall be located on any one site, with an aggregate capacity not exceeding 16 kL. Such dual installations are discouraged.

An additional single tank may be located on the site for uses other than automotive dispensing, subject to compliance with the provisions of the rest of this Standard and if the tank is considered a protected place.

10.5.2 Above-ground storage tanks (Australia only)

Above-ground service station tanks shall have all tank connections to pipework located at the bottom of, and at one end of, the tank (see also Clause 10.5.4).

For a dual-tank installation, a vapour balance line may be joined between the top connections of the tanks, provided that the line and isolating valves are adequately supported and protected from physical damage. The isolating valves shall be incorporated into the remote shut-down system (see also Clauses 6.7 and 10.5.14).

The requirements of this Clause (Clause 10.5.2) do not apply in New Zealand.

10.5.3 Spillage control

The ground below an above-ground tank shall be treated to prevent the accumulation of any flammable liquid or LP Gas beneath the tank. The following requirements apply to such areas:

- (a) The ground area for not less than 1.5 m beyond the tank or up to the boundary wall or barrier (if within 1.5 m) shall be paved or be resistant to saturation by flammable liquids.
- (b) It shall not be possible for any spillage elsewhere to flow under the tank. Kerbs or diversions, grading to not steeper than 1 in 40, humps, or plinths under the tank are recognized control methods (see also Clauses 6.2.5 and 6.3.1).

10.5.4 Connections for tank filling

Connections for filling points for a service station tank shall connect to the tank vapour space and incorporate a positive manual shut-off valve together with non-return valve.

10.5.5 Provisions for emptying tanks

A storage tank at a service station shall have at least two methods of removing liquid LP Gas from the tank. The dispenser piping system may serve as one method. The second method, if a drain outlet, shall be fitted with an internal excess flow protection and positive shut-off valve with the outlet plugged and the drain valve locked. An additional positive shut-off valve shall be fitted prior to use. The drain may be fitted at either the top or bottom of the tank.

In New Zealand, a fenced enclosure may be used in place of a locked drain valve.

10.5.6 Location of LP Gas dispensers

LP Gas dispensers shall not be located within 10 m of an above-ground tank unless the tank is protected from a potential fire at the dispenser by a firewall complying with Clause 3.6.

10.5.7 Dispenser service lines

The length of any dispenser service line that is above-ground shall be the minimum possible, and shall be protected against damage as far as practicable.

10.5.8 Piping for above-ground tanks

Exposed above-ground pipework interconnecting two adjacent, parallel above-ground tanks shall be located within the area encompassed by the outer perimeter of the dual tank installation.

10.5.9 Pump bypass

The pump bypass shall be a return line connected to the tank vapour space. In addition to the requirements of Table 5.2 for vapour openings, the tank outlet shall be protected with an emergency shut-off valve (see Clause 6.7).

10.5.10 Location of pumps

The location for pumps shall be as defined in Table 10.1. The centre of the pump drawing liquid from an above-ground tank shall be no nearer than 500 mm and no further than 2 m distant from the plan view of the tank.

The pump and tank supports shall be integrally constructed such that pipework interconnecting them shall only be subjected to minimal stress if either the pump or tank is subject to impact or settlement (see Figure 10.3).

10.5.11 Use of thermal screens and fire walls

Separation distances shall be measured in a horizontal plane around a thermal screen or firewall to areas classified as residential uses (other than dwellings) and commercial/recreational uses.

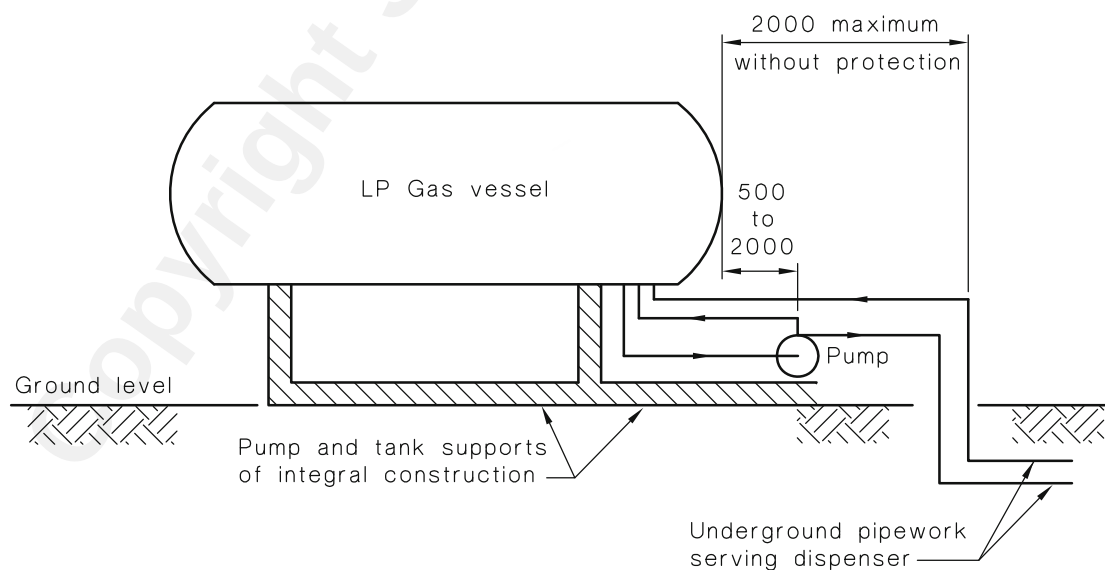
Regardless of the above, such measurement is not permitted for residential dwellings or sensitive use areas.

10.5.12 Tanker access and egress

The configuration and character of the entrance from and the exit to the roadway shall be such as to minimize the likelihood of a vehicle impacting with the tanker while it is entering or leaving the site. Features to be considered during the installation and design stage shall include road width, gradients, line of sight, traffic volumes and traffic speed.

The tanker entry and exit driveways should be located to ensure that the tanker has ready access to the site so as to reduce the time a tanker may need to be exposed to traffic while entering or leaving the site.

The tanker shall be capable of entering and leaving the site by driving in a forward direction and without reversing on a roadway.



DIMENSIONS IN MILLIMETRES

FIGURE 10.3 LOCATION OF PUMPS AND PIPEWORK

10.5.13 Tanker unloading position

The tanker unloading position shall be such that—

- (a) the tanker is positioned wholly on the site during unloading; and
- (b) the tanker standing area has the same separation requirements as the filling point (refer to Table 10.1).

Tanker unloading shall only proceed when the exit is not obstructed and the tanker is positioned so that it can leave the site without recourse to reversing.

Where the tanker stands distant from the fill point, vehicles shall be prevented from driving over the filling hose.

10.5.14 Remote shut-down system

A remote shut-down system shall be fitted and shall comply with the following requirements:

- (a) The system shall be capable of initiation at the following locations as a minimum, with at least one actuation point being accessible at all hours:
 - (i) Within 10 m from the tank filling point.
 - (ii) For self-service sites, at a point easily accessible and visible from the console.
 - (iii) For attendant-operated sites, either at the dispenser or at a point that is easily accessible to the attendant.
 - (iv) At a remote location external to any building and distant from the tank filling point and console, in a direction to allow ready egress from the site.
- (b) It shall be of a fail-safe design that will close all liquid supply, pump bypass and vapour return tank connections and shut down the pump systems as quickly as possible with a response time of no more than 12 s when operated from any point.
- (c) It shall be checked in accordance with the requirements of the site safety management system (see Clause 10.8), with test results recorded and the records retained on site. If any defect compromising safety is identified, the tank shall be manually isolated until such time as the system is returned to full function.
- (d) The system shall be designed and installed so that—
 - (i) manual action is required to re-open the valves to resume operation; and
 - (ii) it achieves a high degree of reliability.
- (e) Instructions on how to actuate the system shall be provided. These shall be as illustrated in Appendix D.
- (f) Clear access shall be available to the actuation points and such access shall be maintained.

NOTE: The remote shut-down system and the emergency shut-down system (see Clause 6.7) may use common components, provided the function of each system is satisfied.

10.5.15 Damage avoidance

Where impact protection is required, it shall be at least the equivalent of the W guardrail-type highway crash barrier, installed in accordance with the supplier's instructions.

NOTE: An example of such an installation would be a guardrail manufactured from grade 250 steel of at least 300 mm finished width and mounted with the rail top approximately 700 mm above ground and a minimum of 1.5 m from the item being protected. The following construction may also be used:

- (a) The guardrail may be secured with M16 bolts without washers to C-section posts constructed from at least 6 mm thick steel. Standard terminal sections should be used. The posts should be spaced at no more than 2 m centres in augured holes of at least 900 mm depth with at least 150 mm of concrete between the post and surrounding soil (see Figure 10.4).
- (b) The holes may be backfilled with concrete having a minimum compressive strength of 20 MPa. No formwork should be used.
- (c) Where the soil into which the posts are placed is not natural, undisturbed soil, the soil may be compacted throughout the full depth of the posts and for a radius of at least 2 m surrounding each post. The compaction should achieve at least 95% of the maximum dry density obtained by the standard compaction test of AS 1289.5.1.1 or NZS 4402.4.1.1.

10.5.16 Dispenser system

A dispenser system shall be a Style A dispenser system complying with Clause 10.3.2. Adaptors shall not be used in conjunction with dispenser nozzles.

10.5.17 Locking provisions

A dispenser that is open to the public shall have a provision for locking the nozzle to the dispenser when the unit is not available for operation.

The dispenser shall be locked when it is not intended to be available for operation.

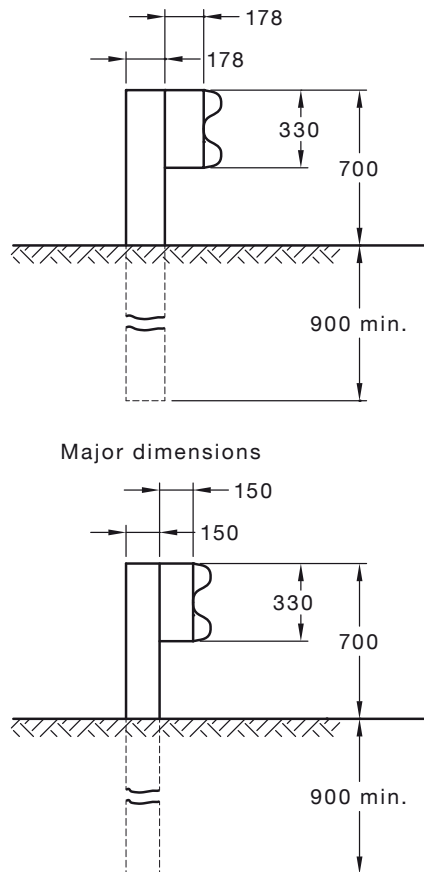
10.5.18 Self-service

10.5.18.1 General design

Any service station at which a dispenser is operated by a customer shall be designed to be supervised from a control station (console) by an attendant who is specifically in charge of dispensing operations.

10.5.18.2 Limitations

It shall not be possible for general-public customers to self-serve under conditions permitted only for fleet refuelling.



SYSTEM NAME: Old style W-beam
GENERAL DETAILS

'C' posts of 6 mm plate at 2000 centres.
Block out of same section.

Rails are nominal 2.7 mm thick, made from
grade 250 steel.

Block outs are the same section as
the posts and are optional.

SYSTEM NAME: AASHTO G4 strong post,
blocked out steel W-beam

GENERAL DETAILS

Rails are grade 350 steel, manufactured from
2.7 mm plate and then galvanized.

Posts are 150 x 110 'Charlie' posts rolled from
4.5 mm plate and then galvanized.
Posts are at 2000 centres.

Block outs are the same section as the posts
and are optional.

DIMENSIONS IN MILLIMETRES

FIGURE 10.4 IMPACT PROTECTION FOR SERVICE STATIONS

10.5.18.3 Control and supervision facilities

A self-serve service station shall incorporate the following features:

- A provision to prevent the start of each dispensing operation until switched on from the console.
- A provision to stop any dispenser from the console.
- A clear view of each dispenser from the console, either directly or by other means.
- A communication system such that the console operator can speak to each user.

10.5.18.4 Additional requirements for card-operated dispensers

Any dispensing system designed to be operated without site supervision via the use of a personal key or card shall comply with the requirements of this Section (Section 10) and the following additional requirements:

- Each user of the system shall be adequately trained in the proper use and safe handling of LP Gas and in the various safety systems incorporated in the installations they wish to access.
- When the site is unattended, all valves shall be in the off position and designed so that a member of the public cannot use the dispensing system without authorization.

NOTE: This authorization may be in the form of a current key card issued to appropriately trained users.

- (c) The installation shall include an emergency device having the following functions or features:
- (i) When activated, the device shall shut off the dispenser pump and transmit an alarm signal to some person or organization capable of responding.
NOTE: It may also be used to release access to the fire extinguishers.
 - (ii) The device shall be readily accessible and integral with or adjacent to the currency or card receptor.
 - (iii) The device shall be protected from vandalism or unwarranted operation by a break-glass screen or equivalent.
 - (iv) The device shall be provided with a notice giving instructions on how to operate the device in the event of a major spill or fire.

10.5.19 Decanting cylinders

Any LP Gas cylinder provided for the filling of other LP Gas cylinders by decanting shall be located not less than 5 m from any above-ground LP Gas storage tank, unless the tank is protected from a fire at the decanting cylinder. A firewall constructed in accordance with Clause 3.6 is deemed to provide suitable protection. The outlet and safety relief valves of the decanting cylinder shall be directed away from the storage tank.

Apart from the provisions of this Clause (Clause 10.5.19), all provisions of the relevant Sections of this Standard shall apply to the use of decanting cylinders.

10.6 SITE SUITABILITY FOR SERVICE STATIONS

The suitability of a site for development as a service station shall be assessed with regard to all criteria necessary to minimize the loss of containment due to vehicle impact on the tank, pump or road tank vehicle and to comply with the requirements of Clauses 10.5, 10.7 and 10.8.

10.7 LAND USE CONTROLS FOR SERVICE STATIONS

10.7.1 General

The installation and operation of a service station shall be subject to two types of land use controls, as follows:

- (a) Separation distances.
- (b) Population limit areas.

These controls, when used in conjunction with each other and the separation distances of Table 6.1, define where LP Gas equipment can be located, and safe separation distances on a site as defined in this Section.

10.7.2 Separation distances

10.7.2.1 General

Specific items of LP Gas equipment shall be located so as to achieve the minimum separation distances from defined uses as shown in Table 10.1. The three types of defined usage are sensitive uses, residential uses and recreational/commercial uses.

NOTE: In defining these areas of usage the major factor to be taken into account is the vulnerability of the persons using the area and their ability to readily evacuate the area.

10.7.2.2 *Sensitive uses*

Sensitive uses include establishments and other uses where vulnerable people are concentrated such as schools, hospitals, aged persons' accommodation and child-care facilities (including crèches).

NOTE: When considering sensitive uses, the possibility of easy evacuation by the people using the area should be considered. For example, doctors', dentists' or other medical surgeries, where no general anaesthesia occurs, could be considered as commercial properties provided they are not providing specialist care to vulnerable people.

When considering large establishments, the actual use of the buildings and areas within the defined radius should be evaluated. For example, in a large school or hospital surrounded by a garden, playing field or car park, the required distance should be maintained from the school or hospital buildings and places of regular occupancy as a sensitive use. However, the garden, playing field or car park should be considered as a recreational/commercial use, provided vulnerable people are not normally present in these areas. Similarly, service buildings associated with the establishment, such as a boiler house, maintenance workshop or mechanical plant storage garage, should be considered as recreational/commercial uses.

10.7.2.3 *Residential uses*

Residential uses include dwellings and places of regular occupancy on residential properties.

10.7.2.4 *Recreational/commercial uses*

Recreational/commercial uses include commercial uses, recreational uses and places of regular occupancy on commercial properties.

10.7.2.5 *Place of regular occupancy*

A place of regular occupancy is an area (not necessarily a building) where people may be present on a regular basis.

Places of regular occupancy are areas which are likely to be accessed. For example, areas such as accessways, gardens, lawns, work sheds, swimming pools and barbecue areas would be classified as places of regular occupancy.

Areas subject to less frequent access, such as private ornamental rockeries, storage sheds and boat or wood storage areas, would generally not be classed as places of regular occupancy.

Car parks for commercial or industrial properties are not considered to be places of regular occupancy, given the normal hours of operation and limited vehicular use of these areas. However, car parks associated with areas of consumer activity such as supermarkets, drive-in food outlets and the like where people are present a significant proportion of the time would normally be classified as places of regular occupancy.

10.7.2.6 *Ancillary services*

Ancillary services are services or activities that are subsidiary to fuel retailing at the site and may involve servicing of vehicles and retailing of spare parts and foodstuffs. These subsidiary activities are exempt from area requirements although an operation drawing a greater number of people to the site for primary purposes other than the purchase of fuel, servicing of vehicles or purchase of spare parts and foodstuffs is not exempt from area requirements and would be assessed as though it were not part of the site operation.

10.7.3 **Population limit areas**

10.7.3.1 *General*

Population limit areas are circular areas around the service station, centred as described in Table 10.1, within which an equivalent population level should not be exceeded.

10.7.3.2 *Equivalent population factor*

Application of an equivalent population factor within the population limit area makes allowance for permanent and transient occupancy. Multiplication factors applicable to various land use categories are shown in Table 10.2. The occupancy rate, as calculated in Clause 10.7.3.3, multiplied by the factors given in Table 10.2 shall not exceed the population limit given in Table 10.1.

NOTE: If an applicant or approval authority believes that the techniques used to calculate the equivalent population within the population limit area are not applicable to a particular case and may significantly understate or overstate the population, they may conduct their own survey to obtain actual data.

10.7.3.3 *Occupancy rates*

Occupancy rates are used to determine the expected population in each area. The calculation should include the population for residential or commercial activities carried out on the site which are not directly associated with the sale of fuel or ancillary services described in Clause 10.7.2.6.

The following methods may be used to determine the number of residents or occupants within the population limit area:

- (a) For dwellings, the local government area (LGA) average residential occupancy rate for dwellings should be used. Alternatively, documented census or population survey results may be used.

Where the population limit area partly encroaches on a dwelling, the population attributed to the dwelling should be proportionally reduced unless—

- (i) the only entry to or egress from the dwelling falls within that area; or
 - (ii) there is no means of egress from the dwelling which would allow a person to move to a safe distance on or off the site.
- (b) For shops, offices, factories and other places of work, the LGA average commercial population density or specific available information on the workforce should be used.
Where there is partial population limit area encroachment, the proportioning and access principles of Item (a) should apply.
- (c) For restaurants, bars, sports fields, swimming pools and other facilities with varying occupancy, the likely maximum capacity or peak occupancy should be established. The principles of proportioning in Item (a) should apply where there is partial population limit area encroachment.
- (d) Where customers purchase fuel or use ancillary services, and in circumstances where people are present on roadways and footpaths, these individuals should be excluded from the population count as the value allows for their presence. The population count should consider and include people congregating for a specific reason, e.g. at a bus stop.

10.7.3.4 *Separation distances and equivalent population values*

The separation distances and equivalent population limit values for an installation are summarized in Table 10.1.

Application of these values is dependent on initial and continued compliance with all the requirements of Clause 10.5.

10.8 SAFETY MANAGEMENT SYSTEM FOR SERVICE STATIONS

A safety management system shall be prepared and implemented.

TABLE 10.1
SEPARATION DISTANCES AND EQUIVALENT POPULATION LIMIT AREAS

Capacity kL	Zone	Sensitive m	Residential m	Commercial recreational m	Equivalent population limits
≤8 kL Single/dual above-ground tank	Dispenser	15	15	15	90 persons within 55 m radius of the tank*
	Pump	55	15	10	
	Tank	55	15	10	
	Remote fill point	55	15	10	
	Road tank vehicle standing area†	55	15	10	
8 kL to 16 kL Single/dual above-ground tank	Dispenser	15	15	15	90 persons within 85 m radius of the tank*
	Pump	55	15	10	
	Tank	85	15	10	
	Remote fill point	55	15	10	
	Road tank vehicle standing area†	55	15	10	
≤65 kL Underground tank	Dispenser	15	15	15	110 persons within 55 m radius of the pump casing and fill point
	Pump (above-ground)	55	15	10	
	Pump (in tank)‡	55	15	10	
	Fill point	55	15	10	
	Road tank vehicle standing area†	55	15	10	

* Population limit radii and separation distances are positioned on each tank centre for both single and dual tank installation (see Clause 10.5.1).

† Separation distances are positioned on the road tank vehicle tank centre for road tank vehicle standing areas.

‡ Where an in-tank pump with no exposed pipework is involved, separation distances do not apply. Where pipework is exposed above ground, the separation distance is measured from the exposed pipework.

TABLE 10.2
EQUIVALENT POPULATION FACTORS

Establishment category	Occupancy	Factor
Dwellings	Residents	1.0
Shops, offices, factories operating normal hours	Average number of occupants	0.3
Shops and factories operating 24 hours	Average number of occupants	1.0
Restaurants, taverns, bars	Peak usage occupants*	0.3
Active open spaces, sport/ entertainment centres	Peak usage occupants	0.2

* Refer to Clause 10.7.3.2.

SECTION 11 OPERATIONS

11.1 SCOPE OF SECTION

This Section sets out procedures for the operation and upkeep of an LP Gas installation as distinct from its construction.

Procedures are dependent on the complexity of the site: what is necessary and desirable at a major installation may be excessive at the simpler type of installation, and is not warranted or intended. Clauses such as Clause 11.3 are therefore checklists of what might need to be considered and are not necessarily universal requirements.

11.2 GENERAL PRECAUTIONS

11.2.1 Requirements

Where LP Gas is to be transported, handled, or stored, the requirements of the ADG Code/The New Zealand Land Transport Rule Dangerous Goods and this Section shall be satisfied.

11.2.2 Odour

Except if stored in a facility complying with Clause 6.18, LP Gas shall have an odour which is distinctive and unpleasant with an intensity which indicates the presence of gas down to 20% of the lower flammability limit.

NOTES:

- 1 Because of the possibility of olfactory fatigue, over-odourization is undesirable.
- 2 In Australia, the addition of 25 g/t of ethyl mercaptan odourant in LP Gas has been found to meet this requirement.
- 3 In New Zealand, refer to—
 - (a) NZS 5263:2003 (Gas detection and odourization); or
 - (b) odourized in accordance with a code of practice approved by the authority.
- 4 The possibility of odour fade should be considered particularly when equipment is entering or re-entering service.

11.2.3 Initial pre-delivery check

Before the initial delivery of LP Gas to a new installation, the relevant statutory approvals and certificates and the installer's test certificates shall be obtained, and the installation shall be visually examined to ensure that, as far as is reasonably practicable, it complies with the requirements of this Standard.

11.2.4 Subsequent pre-delivery checks

Prior to delivery, all components forming part of the delivery system shall be checked to ensure that they are in serviceable condition. If they are not, delivery should be refused.

11.2.5 Post-delivery checks

The storage system shall be visually checked after filling is completed, to ensure the facility is left in a safe condition.

11.2.6 Tanks out of service

Where a tank is being decommissioned permanently, the authority shall be notified and the tank shall be gas-freed.

The fill opening shall be plugged and a notice painted on the tank indicating its condition.

NOTE: Particular care should be taken when LP Gas vapour is being removed to ensure that the withdrawal rate does not exceed the vaporization rate. If vapour is vented too rapidly, the refrigeration effect will cause a pressure drop which could be taken erroneously as an indication that no liquid remains in the tank.

11.2.7 Changed usage of tanks

Where the conditions of tank usage are altered, e.g. different types of gas, or altered draw-off system, the appropriate alterations shall be made to procedures, instructions, notices and markings, vapour/liquid colour codes and marking.

11.2.8 Post-installation site work

No work shall be carried out in the vicinity of an LP Gas installation that would present a hazard to the installation, or would result in the contravention of any requirement of this Standard.

11.3 ESTABLISHMENT OF PROCEDURES

11.3.1 Procedures required

Procedures shall be established appropriate to the scale and nature of the installation, as follows:

- (a) Operating procedures, sufficient to ensure safety and reliability in the day-to-day operation of the facility.
- (b) Maintenance procedures, covering testing, inspection, monitoring and documenting of the equipment, its repair and general upkeep.
- (c) Emergency procedures, covering action to be taken in the event of fire, spillage, accident, equipment failure or other abnormalities or emergencies.
- (d) Training procedures, covering operating, maintenance, and emergency functions.

NOTE: For procedures specific to service stations, see Clause 10.5.

11.3.2 Documentation of procedures

The procedures shall be documented in a form appropriate to the particular installation in notices, manuals, or other recorded instructions on view or readily available on site.

11.3.3 Review and upkeep of procedures

The procedures shall be modified when necessary because of equipment or organizational changes, and reviewed as required.

11.3.4 Operating procedures

The operating procedures shall be appropriate to the particular installation and shall take into account, amongst other things, the following:

- (a) Identification of safe areas.
- (b) Initial commissioning procedures.
- (c) Normal handling procedures.
- (d) Monitoring of essential functions and equipment.
- (e) Recognition of fault conditions.
- (f) Product transfer.
- (g) Housekeeping and site upkeep.
- (h) Any manufacturer's operating instructions for equipment.

- (i) Equipment not in use (i.e. isolation, deactivation, identification).
- (j) Maintaining clear spaces for access.
- (k) Personnel safety.
- (l) Personal protective equipment.
- (m) Control of ignition sources.
- (n) Earthing and bonding.
- (o) Control of access, security and lock-up.
- (p) Vehicle movement.

11.3.5 Construction and maintenance procedures

The construction and maintenance procedures shall be appropriate to the particular installation and shall take into account, amongst other things, the following:

- (a) Periodic testing of protective devices, alarms, and personal protective equipment.
- (b) Regular testing of hoses.
- (c) Establishment of safe areas.
- (d) LP Gas purging.
- (e) Isolation and tagging.
- (f) Control of contractors.
- (g) Work authorization.
- (h) Confined spaces.
- (i) Hot work permits
- (j) Firefighting equipment.
- (k) Pipelines and equipment.
- (l) Storage tanks.
- (m) Electrical equipment.
- (n) Fences and security measures, signs, notices.
- (o) Lighting.
- (p) The needs of any other individual item or plant components, e.g. pumps, equipment.

11.3.6 Emergency procedures

The emergency procedures shall be appropriate to the particular installation and shall take into account, amongst other things, the following:

- (a) List of contacts, emergency services, phone numbers.
- (b) Chain of command.
- (c) List of duties.
- (d) Any mutual aid (inter-company) arrangements.
- (e) Cooperation with relevant authorities, and compatibility with equipment.
- (f) Fire teams (initial response).
- (g) Fire drills.
- (h) Evacuation provisions.

- (i) Assembly points.
- (j) First aid.
- (k) Debriefing and reporting.
- (l) Vehicles on site.
- (m) Contractors.
- (n) Access control.
- (o) Site plan.

For large sites storing dangerous goods in addition to LP Gas, the total site shall be considered when preparing on-site and off-site emergency plans.

NOTE: Appendix K provides information on emergency plans.

11.3.7 Training

11.3.7.1 Employees

Each employee shall be trained in such duties as have been individually assigned to that person in connection with LP Gas. In addition, the training shall include the following, as appropriate:

- (a) Procedures to be followed in the event of an incident (e.g. accident, fire).
- (b) Layout of the installation.
- (c) Location of firefighting equipment.
- (d) Basic principles of first aid firefighting, and the use of fire hose reels and fire-extinguishers.
- (e) Regulations relevant to the employee's tasks.
- (f) Properties, characteristics and hazards of any materials kept or handled.
- (g) Correct use of any personal protective equipment provided.
- (h) Area housekeeping.

NOTE: Procedures and precautions for motor vehicle refuelling are provided in Appendix L. Notices illustrating such procedures are shown in Appendix D.

11.3.7.2 Contractors

Contractors and their staff shall be trained in the following as appropriate to the site:

- (a) Properties, characteristics and hazards of any materials likely to be encountered and appropriate for the site.
- (b) Correct use of any necessary personal protective equipment provided.
- (c) Hazards likely to be encountered.
- (d) Procedures to be followed in the event of an incident (spillage, accident, fire).
- (e) Layout of the installation.
- (f) Location of firefighting equipment.
- (g) Basic principles of first aid firefighting, and the use of fire hose reels and fire extinguishers.
- (h) Any regulations relevant to their tasks.
- (i) Area housekeeping.
- (j) Safety rules of the installation.
- (k) Conditions and obligations associated with work permits and confined-space entry permits.

11.3.7.3 *Visitors*

Procedures shall be established for the control of entry to a restricted area and the prevention of unauthorized access to the facilities and equipment, and the supervision of such visitors as are admitted. A restricted area shall be locked whenever it is unattended.

11.3.7.4 *Specialists*

Emergency response personnel having assigned duties shall be given specific training in those duties. Such special duties may include the following:

- (a) Firefighting.
- (b) Evacuation control.
- (c) First aid.

Training drills shall be conducted at regular intervals to ensure that persons have adequate opportunity to become familiar with their tasks. Local emergency authorities should be involved in joint exercises wherever possible.

11.3.7.5 *Refresher training*

Any training procedures shall be repeated at sufficient intervals to ensure that capability is maintained.

11.4 COMPLIANCE, SUPERVISION AND RECORDS

11.4.1 *Compliance with procedures*

The procedures established for the particular installation shall be complied with.

NOTE: In most occupational safety legislation there is also a general responsibility placed on employers, employees and visitors to be familiar with and to observe established procedures.

11.4.2 *Supervision*

All operational and maintenance work shall be planned and supervised. All necessary precautions shall be observed.

11.4.3 *Records*

Records of the following shall be kept. (Suggested periods for retention of records are indicated in parentheses.)

- (a) Training and retraining. (Permanent record.)
- (b) Routine testing of safety equipment and fire protection equipment. (As required by this Standard and other referenced Standards.)
- (c) In-service inspections. (Life of installation.)
- (d) Training drills. (Six years.)
- (e) Maintenance and repairs, and equipment deficiencies. (Life of equipment.)
- (f) Abnormal incidents, including 'near-miss' occurrences. (Life of installation.)

11.5 CONSTRUCTION, INSPECTION, TESTING AND MAINTENANCE

11.5.1 *Installation certification*

Before an installation is put into service, it shall be certified in writing that the complete installation was installed, tested and proved to be leak-free and in accordance with the installation specifications and the provisions of this Standard.

11.5.2 Scheduled checks

A routine check procedure shall be prepared for each installation appropriate to that installation. Such a procedure shall cover all operating valves and systems, instruments, emergency shut-down systems, safety valve protective caps and discharge provisions, and fire protection equipment. Tank-emptying provisions are excluded.

In particular—

- (a) emergency shut-off valves shall be operated at intervals not exceeding 1 month, from each available operation point;
- (b) pumping and supplementary fire protection systems shall be tested in accordance with AS 1851; and
- (c) records of tests and inspections shall be kept.

NOTE: Where an emergency system installation can be actuated from any of a number of points, it is not necessary to test all such points on the same day, i.e. operation may be tested from the various points in rotation, provided that each has been checked within the stated time.

11.5.3 Tanks

Tanks shall be inspected according to regulatory requirements. The following additional requirements apply to underground tanks:

- (a) Where tests prove the effectiveness of the corrosion-resistance system, periodic external inspection of a tank need not be made. Inspection of test plates or probes buried close to the tank and connected to it may be used as evidence of satisfactory cathodic protection.
- (b) Tanks shall be inspected at the end of the effective life of sacrificial anodes, or when tests indicate the permanent loss of effectiveness of the cathodic protection system or coating.
- (c) When any fault is found in the protective coating, it shall be stripped in the immediate vicinity of the fault and the underlying metal surface examined. Such defective coating shall be repaired and tested to ensure compliance with Clause 7.2.
- (d) Cathodic protection systems shall be tested at intervals not exceeding 6 months and record of these tests shall be kept and shall be available for 10 years.

NOTE: AS 2832.2 provides guidelines for operation and maintenance of cathodic protection systems.

11.5.4 Piping

Piping for LP Gas, whether it is new, modified, or added piping, shall be tested in accordance with the relevant procedures detailed in Appendix J.

11.5.5 Hoses

Transfer hoses (other than decanting hoses) shall be periodically inspected and tested as follows:

- (a) Visually inspected for damage over the whole length in use, at intervals not exceeding 1 month.
- (b) Hoses and couplings that are assembled and tested in accordance with the manufacturers' guidelines, and in compliance with AS/NZS 1869, shall, at the end of five years' service, be either—
 - (i) replaced; or
 - (ii) tested at a pressure of 2.4 MPa at intervals not exceeding one year thereafter.

Delivery hose for in situ filling shall be inspected visually over its entire length weekly.

Hoses that fail inspection or testing shall be discarded or repaired.

11.5.6 Flexible connections

Metal flexible connections shall be inspected visually at intervals not exceeding one year and shall be hydrostatically tested to 3.6 MPa at intervals not exceeding 5 years. Elastomeric flexible connections shall be inspected and tested as for transfer hose.

Metal flexible connections that fail inspection or test shall be replaced.

11.5.7 Protective flow-control valves

Excess-flow valves and non-return valves shall be checked immediately after installation to ensure correct functioning. Hydrostatic relief valves shall be checked in accordance with AS/NZS 3788.

NOTE: AS/NZS 3788 requires that pressure relief valves for LP Gas be tested at intervals not exceeding the internal inspection interval for the vessel, provided that weatherproof external protection prevents dust, rain or other contamination from affecting the valve's performance.

11.5.8 Authorization of work

No work shall be carried out on any LP Gas installation without authorization in one of the following forms:

- (a) A verbal instruction, which shall be limited to non-hazardous actions or to work for which the person has been trained.
- (b) Procedures in accordance with Clause 11.5.10 for which the person has been trained.

11.5.9 Safe work procedures

Work on or near any components containing LP Gas shall be undertaken only after necessary steps have been taken to ensure safety. Safe work procedures shall be followed, which may be in the form of compliance with the following:

- (a) AS/NZS 2865.
- (b) A recognized safe work system.
- (c) A recognized work permit system, including a hot work permit in accordance with AS 1674.1.

11.5.10 Work preparation

Appropriate procedures shall be adopted and responsibilities assigned to ensure that the relevant site area and plant are properly prepared to prevent injury to people, damage, fire or explosion.

Particular attention shall be paid to the following, as appropriate:

- (a) Observance of any written safety procedures.
- (b) Liaison with site operator.
- (c) Testing for flammable gas and oxygen levels, particularly for any confined spaces, pits, trenches and other potentially hazardous areas.
- (d) Removal of protective devices unless substitute protection has been provided.
- (e) Prevention of inadvertent operation during maintenance.
- (f) Prevention of contamination of equipment by water, dirt, foreign matter, or other materials.
- (g) Identification of both the equipment to be worked on and all other equipment affected, including underground piping or equipment.
- (h) Depressurization and disconnection.
- (i) Isolation from other equipment.

- (j) Blanking or capping opened connections.
- (k) Purging of the equipment.
- (l) Removal of combustible materials.
- (m) Sealing off of sewers, drains and gutters.
- (n) Provision of additional protection, fireguards, and firefighting equipment.
- (o) Tagging where required.

NOTE: Pits for underground LP Gas tanks and pipework should be treated with heightened caution. The probability of flammable gas mixtures and depleted oxygen levels is high. Suitable procedures should be implemented to ensure a safe atmosphere prior to work in pits including the positioning of the head in or close to a pit opening. The use of a safety man observing all work on pits should be part of such procedures. The potential release of gas and volume of such gas releases from the tank or pipework should be considered and vented to a safe location or minimized. Forced ventilation should be implemented to achieve and maintain the safe atmosphere during any operations that may result in gas release.

11.5.11 Completion of work

Upon completion of work, a check shall be made to ensure that—

- (a) any written safety procedures have been followed;
- (b) the work has, in fact, been completed;
- (c) any temporary arrangements have been discontinued, e.g. bypass lines or blank flanges have been removed or fuses replaced;
- (d) all personnel and equipment are accounted for;
- (e) relevant work permits or other documentation have been signed off; and
- (f) the site operator is notified of status.

11.5.12 Site upkeep

The area around any storage and handling facility and the installation itself shall be maintained in a safe condition. The following requirements apply in particular:

- (a) Any accumulation of flammable or combustible materials, of a type and quantity sufficient to constitute a significant heat radiation hazard to the installation in the event of fire in those materials, shall be removed.
- (b) Specified clearance distances shall be maintained.
- (c) Vegetation that could become a fire hazard shall be kept short.
- (d) The area shall be kept clear of all extraneous material.
- (e) Lighting shall be kept operational and effective.
- (f) The surface coating of any tanks shall be maintained in good condition.
- (g) Signs and notices shall be legible.
- (h) Water levels in drainage system vapour traps shall be maintained at the correct level.

11.6 SPECIFIC OPERATING PROCEDURES

11.6.1 Commissioning of new or modified equipment

No installation or part thereof shall be brought into service until operating and emergency procedures and associated equipment are available, appropriate to the facility being brought into service, and adequate personnel are available to cope with any likely fault condition.

11.6.2 Outlets not in use

Any outlet that is not in use shall be capped or otherwise blanked off to prevent inadvertent release of LP Gas, irrespective of the fitting of a valve.

11.6.3 Personnel safety

Protective clothing and equipment as is necessary for normal operations and foreseeable emergency situations shall be provided at each facility.

11.6.4 Security measures

Security measures appropriate to the installation shall be in writing, and shall be available on site. Such measures shall include the following:

- (a) Where limitations to access are necessary, a means of identifying persons authorized to enter.
- (b) A list of persons and their responsibilities and duties.
- (c) A list and description of any routine inspections or patrols.
- (d) Instructions on action to be taken over any attempted or actual breach of security.

11.7 PREPARATIONS FOR THE TRANSPORT OR REMOVAL OF STORAGE TANKS

Storage tanks shall be prepared for transport to or from their installation sites under the following conditions:

- (a) A tank of 8 kL capacity or less may contain LP Gas but shall be nominally liquid-free.
- (b) A tank over 8 kL and up to 125 kL capacity may contain LP Gas, provided that the pressure does not exceed 200 kPa.
- (c) A tank exceeding 125 kL capacity shall be purged and LP Gas-free.

The requirements for permanent decommissioning of tanks are given in Clause 11.2.7.

Portable tanks and skid tanks may be transported in accordance with the ADG Code or NZS 5433, and are not subject to the requirements of this Clause (Clause 11.7).

11.8 TANKER OPERATIONS

11.8.1 Delivery vehicle

Deliveries of bulk LP Gas to on-site containers shall be carried out by means of a tanker complying with AS 2809.3 or NZ *Hazardous Substances (Tank Wagons and Transportable Containers) Regulations* or an ISO container that complies with the IMDG Code.

NOTE: A skid tank or a wheeled tank trailer not exceeding 5 kL capacity may be used at an agricultural, mining, or construction site to refill on-site LP Gas containers or engine fuel tanks without dismounting or uncoupling.

11.8.2 Tanker loading procedures

The requirements of Clauses 11.8, 11.9, and 11.10 shall apply to the filling of the cargo space of a tanker as appropriate.

11.8.3 Check of accessories

Before moving the vehicle on any delivery run it shall be ascertained that the accessories required are being carried, e.g. extinguishers, emergency procedure guides, warning signs and gloves. Refer to the ADG Code or NZS 5433.

11.8.4 Refuelling

When LP Gas is being transferred from the cargo tank into the engine fuel tank, the tanker shall be stopped and the precautions set out in Clause 11.12 shall be taken.

11.8.5 Parking and garaging

Vehicles that carry or contain LP Gas, other than in the vehicle's engine fuel tank, shall not be parked within any building other than a shelter which has at least three sides open, or is otherwise designed so as to permit complete dispersal of any LP Gas that may escape.

11.8.6 Tanker servicing and repair

Tanker servicing and repair that can be conducted on a scheduled basis at pre-arranged times shall be carried out in workshops in which the staff have had prior training in the precautions required for LP Gas tankers. Where, because of breakdown at a remote location, a workshop without such trained staff must be used, the owner's representative shall instruct the workshop staff of the nature of the cargo and precautions to be taken and shall remain in attendance while the work is being done.

NOTE: This instruction would normally be provided by the tanker driver.

Any such work shall be conducted under the following conditions:

- (a) No work of any nature shall be conducted indoors on the tank or any gas-containing components unless the component is gas-free.
- (b) No hot work shall be conducted either indoors or outdoors on the tank or any gas-containing component unless the tank and the component are gas-free [see Item (h)].
NOTE: Where hot work is involved, particular care is necessary to ensure that the gas-free condition is maintained, as vapours can be released by the heating of deposits or sediments, or from gas pockets in the tank or pipework.
- (c) Work on the vehicle itself, which does not involve any gas-containing components, shall be done indoors or outdoors, provided that the precautions in Items (d) to (h) are observed.
- (d) The vehicle shall not be parked near a source of heat of sufficient intensity to risk causing a discharge from the safety valve due to heating of the cargo.
- (e) All secondary shut-off valves shall be closed before the vehicle is moved into the building. All hose connections shall be capped.
- (f) The system shall be checked for leaks and any found shall be repaired before the vehicle is moved into the building.
- (g) The tank shall be checked to determine that it is not filled beyond the maximum filling level. Any overfilled condition shall be corrected before the vehicle is moved into the building.
- (h) If hot work on the vehicle is unavoidable—
 - (i) all gas outlets shall be capped gastight;
 - (ii) the surroundings shall be checked and shown to be gas-free;
 - (iii) fire extinguishers shall be deployed in the area; and
 - (iv) a water hose of sufficient capacity to quench sparks or minor fires that might result from the hot work shall be within reach.

11.9 GENERAL PRECAUTIONS FOR TANKER TRANSFER

11.9.1 Suitability of gas

Before transfer operations commence, it shall be ascertained that only those gases for which the system is designed and approved are delivered into a tank.

11.9.2 Condition of installation

No tank, road tanker, rail tanker, or cylinder shall be filled with LP Gas unless it is in a serviceable condition.

11.9.3 Advice of delivery

Where a site is not staffed at the time of delivery, a check shall be undertaken to establish whether there is still gas in the tank or cylinder.

If the tank or cylinder is nominally empty, the tank or cylinder service valve shall be closed before filling is commenced. Advice to this effect shall be left at the site. This requirement does not apply to automotive outlets.

If the tank or cylinder is part filled, filling may proceed.

11.9.4 Safety of hose

Transfer hose shall not be run through any building or across any carriageway for transfer operations.

NOTE: This does not apply to a carport or verandah.

11.9.5 Use of gloves

Protective gloves shall be worn when handling equipment that could involve contact with LP Gas.

11.9.6 Tanker location

For delivery into storage tanks exceeding 8 kL aggregate capacity at any one location, a tanker shall be located off the public street and as far as practicable from open doorways, and shall not obstruct entrances to buildings or fire escapes. In no case shall a tanker be located closer than 3 m to a storage tank nor left unattended if delivery is being carried out from a public street, except in accordance with the requirements of Clause 11.10.

The tanker shall be positioned so that in an emergency it can be driven or towed out without recourse to reversing.

11.9.7 Multiple tanker transfer operations

Product shall not be transferred into or out of a tanker at an automotive filling installation within 15 m of—

- (a) another tanker transferring or containing a flammable or combustible product; or
- (b) a filling point at which another tanker is transferring a flammable or combustible product.

11.10 SUPERVISION OF DELIVERY

At least one person shall be in attendance during delivery in circumstances where all the following requirements are satisfied:

- (a) There is a line of sight between the filling point and the tanker. The filling point can be up to a maximum deviation of 1.5 m from the direct line of sight.
- (b) There is direct unobstructed access between the filling point and the tanker.
- (c) There is no more than 50 m of hose length between the tanker and the receiving vessel.

During this period the person shall maintain surveillance of the in situ fill cylinder or tank and the tanker.

Where these conditions are not satisfied, two persons shall be in attendance, one at the tanker and one at the filling point.

11.11 PROCEDURES FOR FILLING AND EMPTYING TANKS

11.11.1 Valve handles

Where any valve is open to transfer liquid, any operating handle or wheel shall not be removed.

11.11.2 Venting

Gas or liquid shall not be vented to the atmosphere to assist in transferring contents of one tank to another; however, the blowing-down of hoses after conclusion of transfer operations is permitted.

11.11.3 Procedures prior to transfer

The following procedures and precautions shall be effected prior to the commencement of LP Gas transfer:

- (a) Where two or more tanks are filled from a common filling point, steps shall be taken to ensure delivery personnel become familiar with the valve system and the correct filling procedure for the installation before commencing delivery.
- (b) The tank data plate shall be checked to ensure that the tank is suitable for the type of LP Gas being delivered and that the tank owner's authorization is present.
- (c) The tank contents shall be checked to ensure that the tank has sufficient ullage to receive the intended delivery.
- (d) The area shall be checked to ensure that ignition sources do not exist within the transfer area.
- (e) The antistatic (bonding) lead shall be connected when the tank being filled has a capacity greater than 8 kL.
- (f) Connections shall be made gastight.
- (g) In Australia, where the filling provision for the tank does not incorporate a manual shut-off valve, a removable adaptor shall be attached to the filling connection. The adaptor shall incorporate a non-return valve arranged to prevent outward flow from the tank when the filling connection is removed. The adaptor shall not be removed unless it has been ascertained that the non-return valve of the tank has closed correctly after filling. If not, the adaptor shall be left in place until the fault is rectified.

In New Zealand, appropriate valves/adaptors shall be carried, when filling, to enable a leaking fill valve to be safely capped off.

11.11.4 Procedures during transfer

The following procedures and precautions shall be effected during gas transfer:

- (a) All necessary valves shall be opened, and the pump started.
- (b) Filling shall be stopped if any ignition source becomes active within the transfer area or if any fault or malfunction causes gas to escape.
- (c) The person conducting the transfer shall remain in the immediate vicinity of the transfer operation, and shall ensure that the receiving tank is not overfilled.
- (d) The filling ratio for the tank shall not be exceeded. The appropriate procedure described in Appendix B shall be followed.

- (e) Where filling without vapour return or transferring with vapour compression is being undertaken, care should be taken to ensure that the safety valve does not discharge.

11.11.5 Procedures after transfer

The following procedures and precautions shall be followed when gas transfer is complete:

- (a) The pump or compressor shall be stopped.
- (b) All necessary valves shall be closed.
- (c) If dry hoses are used, pumps and engines shall be stopped prior to the venting of the hose. Venting shall be completed prior to uncoupling the hose.
NOTE: Where a compressor is available, the liquid in the liquid line should preferably be blown into the receiving tank(s) before venting takes place.
- (d) Hose couplings shall be disconnected and the terminal caps of pipework and hose shall be replaced immediately.
- (e) Hoses shall be stowed away. Long hoses which are to be wound up under power shall first be cleared of all obstructions that could impede the movement of the hose or cause damage.
- (f) Electrical connections and antistatic leads shall be disconnected and stowed away.
- (g) The operator shall inspect the area by walking around the vehicle. If the engine has been stopped, it shall not be started until all vapour has been safely dispersed.

11.12 CYLINDER FILLING

11.12.1 General requirements

The filling of all cylinders shall be carried out in accordance with the following requirements:

- (a) The filling procedures and pre-filling inspection shall be in accordance with—
 - (i) Australia, AS 2030.1; and
 - (ii) in New Zealand, New Zealand Environmental Protection Authority (EPA) Technical Guide—*Guide to Gas Cylinders*.
NOTE: In Australia, NOHSC:1010 nominates employer responsibilities for the inspection and maintenance of cylinders, the operation of gas cylinder test stations and gas cylinder filling stations.
- (b) A cylinder that does not comply with Clause 4.2.1 shall not be filled.
- (c) The filling ratio for the cylinder shall not be exceeded (see Appendix B).
- (d) Filling by volume shall not be carried out on any cylinder that does not incorporate a fixed liquid level gauge.
- (e) Filling shall be stopped if any fault or malfunction causes gas to escape.
- (f) Cylinders shall only be filled by the owning organization where the owner's name is clearly stamped on the cylinder or upon the authorization of the apparent owner.
- (g) In Australia, where the filling provision for the in situ fill cylinder does not incorporate a manual shut-off valve, a removable adaptor shall be attached to the filling connection. The adaptor shall incorporate a non-return valve arranged to prevent outward flow from the cylinder when the filling connection is removed. The adaptor shall not be removed unless it has been ascertained that the non-return valve of the cylinder has closed correctly after filling. If not, the adaptor shall be left in place until the fault is rectified.

In New Zealand, appropriate valves/adaptors shall be carried, when filling, to enable a leaking fill valve to be safely capped off.

11.12.2 Supervision

A person trained and experienced in the filling operation shall be in attendance throughout the cylinder filling operation.

11.12.3 Leakage check

Every cylinder shall be checked for leakage after filling.

11.12.4 In situ filling

In situ filling shall be conducted in accordance with the requirements of Clauses 11.9 and 11.10 and with regard to the hazardous areas defined in AS/NZS 60079.10.1 for in situ fill cylinder locations (see Appendix E).

11.12.5 Filling at automotive fuel dispensers (Australia only)

Only cylinders that are permanently mounted on a vehicle to fuel an internal combustion engine or liquid phase burner shall be filled from an automotive fuel dispenser at a service station. Gas shall not be vented to atmosphere during filling.

Samples of LP Gas may be drawn from an automotive fuel dispenser into a cylinder or other container for calibration, quality or regulatory purposes.

Any vehicle that has a burner operating (being an ignition source) shall not be allowed to enter the dispenser area.

NOTES:

- 1 Automotive LP Gas may comprise blends for which appliances have not been tested or approved.
- 2 Since October 1990, AS/NZS 1425 requires a cylinder installed for in situ filling to be provided with an automatic fill limiter.

11.12.6 Decanting

The filling of cylinders by decanting shall be carried out in accordance with Clauses 11.12.1, 11.12.2, 11.12.3 and the following requirements:

- (a) Protective gloves shall be worn.
- (b) The lever of the cylinder filling nozzle shall not be held open by any means other than by hand.
- (c) Only one cylinder shall be filled at any one time on any decant installation.
- (d) The cylinder being filled shall rest on the ground or on an electrically earthed surface. Plastic crates are not to be used as a stand, due to their non-conductive nature.

NOTE: The recommended decanting procedure is set out in Appendix J.

11.13 FILLING FROM TANKERS TO CYLINDERS OTHER THAN IN SITU

11.13.1 Application

Clauses 11.13.2 to 11.13.7 specify procedures for the filling of cylinders of not less than 100 L capacity directly from a tanker, by arranging a temporary filling installation adjacent to the tanker. It does not apply to in situ filling, nor to the filling of permanently mounted automotive fuel containers on vehicles, which are dealt with elsewhere in this Standard.

11.13.2 General requirements

The requirements of Clause 11.12 shall apply unless varied herein, except that the procedure shall be limited to filling by mass.

11.13.3 Preliminary safety check

The transfer area shall be checked for potential ignition sources and any such hazards shall be removed before filling is started.

11.13.4 Notices

A notice shall be displayed near the cylinder filling point indicating 'NO SMOKING—NO FLAME' in letters not less than 50 mm high.

11.13.5 Tanker egress

The tanker shall be positioned so that in an emergency it can be driven or towed straight out without recourse to reversing.

11.13.6 Cylinder handling

Cylinders shall be placed so that access to the tanker cabin and pump controls and the egress of personnel or the tanker are not impeded.

In Australia, the requirements of Clauses 8.3.2 and 8.3.3 shall be observed.

In New Zealand, the requirements of Clauses 9.3.2 and 9.3.4 shall apply.

11.13.7 Separation distances

The distances specified in Clause 8.3.1 for Australia, or Clause 9.3.1 for New Zealand, shall apply to both the tanker and the cylinder filling point.

SECTION 12 EMERGENCY MANAGEMENT

12.1 SCOPE OF SECTION

This Section sets out requirements and recommendations on planning for emergencies.

NOTE: The fire authority and other emergency service agencies should also be consulted.

12.2 PLANNING FOR EMERGENCIES

12.2.1 Consideration in designing premises

The likelihood of an incident occurring in an area used for storage and handling of dangerous goods can be minimized by good design and layout, sound engineering, good operating practices, and proper instruction and training of personnel in the performance of their duties.

The design and layout of the facility should consider the following provisions:

- (a) Sufficient space between bund walls, storage areas and other structures as will allow access for maintenance and during emergencies.
- (b) A means for reducing emission of vapours to the outside atmosphere.
- (c) Alarms connected directly to the fire brigade or 24 h monitoring services.
- (d) Water supplies.
- (e) Fire protection equipment.
- (f) A means of evacuation and assembly points.
- (g) Protection of personnel responding to an emergency.
- (h) Access routes for fire brigade appliances.
- (i) Containment of leaks, spills and run-off of firefighting water.
- (j) The location of the emergency plan and safety information.

NOTE: Appendix K provides guidance on the preparation of the emergency plan.

Where so required by the relevant regulatory authority, the fire authority and other emergency service agencies should be consulted with respect to the above matters.

12.2.2 Emergency plan

12.2.2.1 General

A detailed plan for combating emergencies that could occur on-site shall be prepared. Such a plan should be developed in consultation with the emergency services and relevant regulatory authorities.

The emergency plan shall—

- (a) take into account any potential for the occurrence of fire, explosion, natural disaster, reaction or release of dangerous goods;
- (b) be appropriate to the size and complexity of the installation, its resources and personnel; and
- (c) be regularly reviewed and updated as necessary.

Plant personnel shall be familiar with the contents of the emergency plan.

NOTES:

- 1 Guidelines for emergency plans are given in Appendix K.
- 2 The emergency services may require a copy of the plan to be kept in a secure location at the site entrance.

12.2.3 Review of emergency plan

The emergency plan shall be kept up-to-date and reviewed and revised where necessary. For example, a review should be undertaken whenever—

- (a) new substances are to be kept on the premises, which could impact upon the LP Gas;
- (b) the quantity or composition of LP Gas or any other hazardous substance being kept is changed significantly;
- (c) a change is made in the way the LP Gas or any other hazardous substance is stored or handled;
- (d) a change is made in a process or procedure, which might result in a change of risk;
- (e) new information regarding the hazardous properties of a substance or system is established;
- (f) deficiencies are identified through testing of the plan, through training, after an incident, or any other means; or
- (g) change of occupancy or operator.

12.3 MANIFEST

A manifest shall be provided and maintained in accordance with regulatory requirements.

12.4 PLACARDING

Stores shall be placarded in accordance with regulatory requirements.

SECTION 13 FIRE SAFETY

13.1 INTRODUCTION

This Section sets out requirements for the protection of LP Gas installations from fire, and for the fighting of fire in such installations, based on the fundamental concepts described in Appendix N. The type and nature of the fire protection provided for the installation depend upon the nature of the surroundings and any associated structures, hazards and activities, that may threaten the LP Gas facility, rather than solely on the quantity of LP Gas being stored. Similarly, consideration of protection of LP Gas installations from fire should have regard to potential threats to nearby populations and structures.

Fire protection provisions are intended to fulfil two distinct functions. The first is *firefighting*, to control and extinguish any fire that might occur. The other is *heat protection*, to protect tanks and auxiliary fittings from overheating from a nearby fire. The same fire protection equipment is used for both functions and a particular item of equipment might be used in several situations, e.g. a stream of water from a hose may be used to fight a fire, to cool a threatened tank, or to dissipate escaping gas.

Firefighting The requirements for firefighting are based on the surroundings and less on the need of the LP Gas installation, as a gas fire is most often terminated by stopping the gas flow, and almost never by extinguishing the fire. So the extent of the firefighting equipment depends on the needs of the whole of the site and not just that of the LP Gas storage. The actual LP Gas installation may not require a great deal of firefighting equipment if the engineering fire safety requirements of this Standard are in place. Any associated buildings and the like will need to have firefighting equipment to comply with building regulations and this should be counted as an important part of the overall protection of the site, including the LP Gas installation.

Heat protection Heat protection is essential when there is a fire risk that could present a significant threat of heat radiation to the tank. The need for heat protection also depends on the surrounding structures, hazards and activities rather than the quantity of LP Gas or the size of the containers.

13.2 APPLICATION OF SECTION

For low capacity storages, no specific fire protection requirements are given, based on the assumption that there is almost always a building or activity for which there will be at least a basic tap-water supply capable of connecting a garden hose. If there is no associated structure or activity nearby, then there may be insufficient risk to the LP Gas storage to justify the installation of otherwise-mandatory fire protection.

Some installations are at such a distance from people or buildings that neither can be considered a threat to the other, e.g. some Class A sites. In such cases there are no mandatory requirements and the operator may choose whether or not to provide some level of protection. Figure 13.1 shows the dividing line, based on thermal radiation.

This Section may be used as a guide to determining fire protection measures for tank installations in Class B situations, as derived from Figure 13.1. In some situations the fire protection measures given in this Section may not be appropriate, due to the complexity or topography of the installation, the associated structures, hazards or types of activities. In such situations the fire authority may need to be consulted on the type of fire protection equipment to be provided.

13.3 GENERAL REQUIREMENTS FOR FIRE PROTECTION EQUIPMENT

13.3.1 Integration with existing facilities

Where there is fire protection equipment already on site (e.g. as required by the BCA/NZBC), such equipment shall be taken into account when determining the fire protection necessary for the LP Gas installation.

13.3.2 Variations to this Section

Where the requirements of the Section need to be varied because of a reduced fire risk, a fire safety analysis shall be prepared.

13.3.3 Equipment compatibility

All fire protection equipment, including hoses, connectors and booster connections, shall be compatible with that of the local fire authority.

13.3.4 Location of firefighting equipment

Firefighting equipment shall be located so as to be reasonably adjacent to the facility or equipment to be protected and shall be accessible without undue danger in an emergency.

13.3.5 Variations to requirements

Where it is desired to propose variations from this Standard for a particular installation because of reduced risk, a specific analysis of the safety of the site shall be prepared to the satisfaction of the authority.

13.4 CYLINDERS

13.4.1 Cylinder storage and use

Where cylinders are stored (but are not gas-free) or are in use, the fire protection facilities set out in Table 13.1 shall apply.

TABLE 13.1
FIRE PROTECTION FACILITIES FOR CYLINDER STORAGE

Aggregate capacity L	Requirements
≤1 000	—
Greater than minor storage, outdoors	No specific requirements
Greater than minor storage, indoors	At least one hose reel or one extinguisher*
>1 000 ≤12 000	At least one hose reel or one extinguisher
>12 000 ≤60 000	At least one hose reel and one extinguisher, or two hose reels
>60 000	At least two extinguishers and two hose reels with one on-site hydrant system, or monitors, or a sprinkler system

* This requirement does not apply to domestic portable cylinders used in a residential situation.

NOTE: In New Zealand, one fire extinguisher is required where there is more than 50 kg of LP Gas stored or used and two fire extinguishers if the quantity exceeds 200 kg. The fire extinguishers may be substituted by a hydrant system incorporating a 20 mm diameter hose fitted with a spray nozzle.

13.4.2 Cylinder filling areas

An extinguisher shall be provided adjacent to the cylinder filling area where cylinders are filled by mass or decanting.

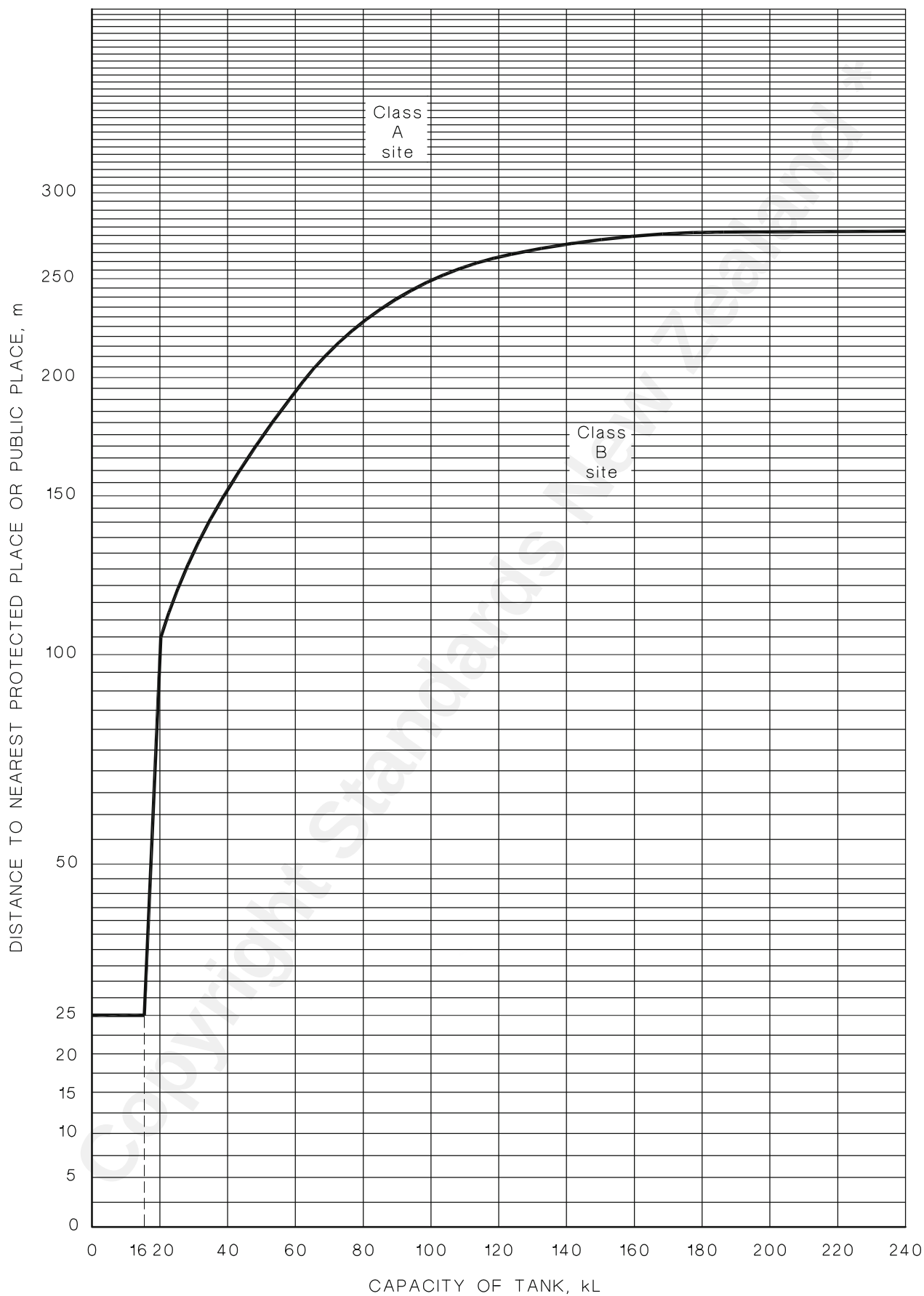


FIGURE 13.1 SELECTION CHART

13.5 TANK INSTALLATIONS

13.5.1 Site evaluation

Where an above-ground storage tank is located in a Class B site in relation to a protected place or public place (see Figure 13.1), the firefighting requirements for the whole of the site shall be determined from an evaluation of the needs and the available facilities of the particular site, conducted on the basis of the following principles:

- (a) The provisions of Clause 13.2 shall apply to any installation.
- (b) For a single tank of 8 kL or less, where there is a basic tap water supply for any adjacent structure that is capable of operating a garden hose, such a water facility supplied with a suitable hose will be taken to provide adequately for minor fire risks in the vicinity of the tank. Where there is no such structure, the installation will be deemed to be not effectively a fire risk.

NOTE: A building that is less than 3 times the minimum separation distance from the tank (see Table 6.1) is considered 'adjacent'.

- (c) For all other tank installations, at least a hose reel installation in accordance with Clause 13.7.2 shall be available for the tank. However, for a single tank of 16 kL or less, a fire extinguisher may be substituted where water availability requirements cannot readily be met.
- (d) Where the capacity of an individual tank or a group of tanks exceeds 50 kL, the installation shall be assessed for heat protection in accordance with Appendix M and treated in accordance with Clause 13.5.2.

In New Zealand, a tank or a group of tanks with a capacity of more than 12 000 L shall require firefighting facilities unless the requirement is waived by the authority.

- (e) Any above-ground tank exceeding 50 kL capacity that has been assessed as not requiring heat protection as in Item (d), shall be provided with a hydrant system complying with Clause 13.7.1 as supplementary protection.
- (f) Where the aggregate capacity of a tank installation exceeds 300 kL, the firefighting facilities for the installation shall be determined from a specific site assessment.

13.5.2 Heat protection

Where an installation has been assessed as requiring heat protection in accordance with Appendix M, due to insufficient separation distance, one of the following shall be used:

- (a) Burying or mounding in accordance with Section 7.
- (b) Insulation applied to the tank or component.

NOTE: Insulation can sometimes cause corrosion, or conceal it. Therefore it should only be used after careful consideration of its effect on steel, and in conjunction with appropriate inspection procedures.

- (c) Use of heat radiation barriers.
- (d) Use of cooling water by means of—
 - (i) fixed or traversing monitors; or
 - (ii) a fixed water spray system.

NOTE: Of the various options, permanent physical protection is preferable, as cooling water systems are vulnerable to interruption and require procedures or automatic devices for their operation. Also cooling water will not be effective if there is jet fire impingement onto the tank.

Where a tank has been assessed and found to require heat protection, hand-held hoses from a hydrant system shall not be used as an alternative form of basic heat protection, but can perform a useful supplementary function.

13.6 AUTOMOTIVE FILLING INSTALLATIONS

All automotive filling installations shall be provided with fire extinguishers as follows:

- (a) Service stations—two extinguishers.
- (b) Sites other than service stations—one extinguisher.

Fire extinguishers that are provided for other reasons may be counted for the purposes of this requirement.

13.7 DETAILED REQUIREMENTS

13.7.1 Hydrant system

Where an on-site hydrant system is specified, the following requirements shall apply:

- (a) The system, its equipment, and its supporting water supply, shall comply with AS 2419.1 or NZS/BS 750 as appropriate, and as qualified in this Section.
- (b) Hydrants shall be located so that each area requiring protection is within 30 m of but not less than 10 m from a hydrant under any conditions of fire and wind, and the cooling effect is optimized.

NOTE: Best results are achieved from hydrants at diagonally opposite locations relative to the tank.

- (c) Where an on-site fire team is provided, for each hydrant, at least one hose and hose fitting, as well as one adjustable spray-type hose nozzle, shall be provided.
- (d) Hoses and fire hydrants shall be maintained in accordance with AS 1851 or NZS 4503.

13.7.2 Hose reels

Any hose reel installed shall comply with the following requirements:

- (a) Hose reels shall comply with AS/NZS 1221 and shall be installed in accordance with AS 2441 or NZS 4503.
- (b) The water supply to a hose reel may be provided by any available on-site reticulated water supply system, or from any form of storage system, provided that the hose reel is able to deliver at least 0.33 L/s. Where the supply is from a storage system the duration shall be at least 15 min.
- (c) The number and location of hose reels shall be such as to ensure that a hose nozzle will reach every point in an area bounded by a line around, and 5 m distant from, any tank and tanker standing area.
- (d) Maintenance shall be in accordance with AS 1851 or NZS 4503.

13.7.3 Water spray system

Any water spray system shall be a fixed system complying with NFPA 15 that is automatic and is able to apply cooling water to the tanks at a rate not less than 10 L/min per square metre of total surface area of the tank and tank support to be cooled.

The system shall be adequate to supply the three largest tanks of a multiple tank installation. In the event of a tank being affected by heat impingement, then that tank and the two adjacent tanks shall be simultaneously protected by the system.

In addition, the supplementary protection equipment shall comply with the following:

- (a) The supports of a water spray system shall be effectively cooled by themselves being the water carriers, or by being covered by water spray cooling, or by being effectively insulated.
- (b) An automatically operated spray system shall include provision for manual initiation and shut-off, and the means of manual initiation shall be accessible under all fire conditions.
- (c) Automatic systems shall be designed to turn on in response to a 'fire detected' signal but shall not be operated by the vapour pressure of the LP Gas tank.
- (d) An automatic system shall be designed to fail-safe with the water supply on.
- (e) The installation shall be such that run-off water from a cooling provision does not endanger foundations by scouring.

13.7.4 Monitors

Monitors shall comply with the following requirements:

- (a) Where monitors provide the only means of cooling, they shall comply with Clause 13.7.3, where relevant, and shall be numbered and located such that all surface areas of the tanks and tank supports are able to be protected.
- (b) Cooling monitors shall be capable of applying the required water quantity at the target area under adverse wind conditions.
- (c) The means for manually starting and stopping flow, and for controlling the direction if the monitor is adjustable, shall be from a location determined safe and following an assessment of possible fire scenarios.

NOTES:

- 1 For design purposes, 'adverse wind conditions' will be taken as requiring at least 50% more water than the quantity calculated as being required at the tank shell.
- 2 It is considered that, to provide significant cooling effect in adverse wind conditions, monitors would need to be so close to the tank that any design that relies on manual adjustment of direction or spray pattern could be inoperable because of proximity to the heat source.

13.7.5 Fire extinguishers

Where the term 'extinguisher' is used without any other qualification it shall mean a portable powder type fire extinguisher having a rating of at least 2A 60B(E).

Fire extinguishers shall comply with AS/NZS 1841.1, AS/NZS 1841.5 or AS/NZS 1841.6, and with AS/NZS 1850, as appropriate.

Maintenance shall be in accordance with AS 1851 or NZS 4503.

Any extinguisher required by this Standard shall be located in a prominent and readily accessible position when the equipment or facility is in use.

NOTES:

- 1 These Standards are limited to portable extinguishers, and therefore do not deal with extinguishers of the wheeled type which are addressed by AS 4265. This omission should not be interpreted as precluding the use of such larger units.
- 2 Extinguishers should for preference be located in accordance with AS 2444 or NZS 4503, as the needs of the site permit.

13.7.6 Water supply

Where water is required, the water supply shall comply with the following requirements:

- (a) The water pressure, flow rate and the water reserves shall be adequate for the needs of the installation and for any possible simultaneous needs of nearby buildings or facilities for firefighting water.

NOTE: Where an assessment of the site indicates that operation of the whole of the water system at the one time is unlikely, an appropriate diversity factor may be applied.

- (b) The required water shall be provided by town mains wherever practicable. Where the mains cannot provide the available pressure and flow rate, provision shall be made for boosting. Where the supply conditions are inadequate for boosting, a static water supply system shall be provided.

- (c) A static water supply and pump shall be capable of providing not less than 2 h of running time for the whole system as determined under Item (a), or 15 min if supplying only a hose reel system.

NOTE: Return water, or additional make-up water, may be taken into account in the calculation of reserves and a bore, dam, river, lake or the sea may be utilized if sufficiently reliable in terms of seasonal or tidal fluctuations.

APPENDIX A

LIST OF REFERENCED DOCUMENTS

(Normative)

AS	
1170	Structural design actions
1170.4	Part 4: Earthquake actions in Australia
1210	Pressure vessels
1228	Pressure equipment—Boilers
1271	Safety valves, other valves, liquid level gauges, and other fittings for boilers and unfired pressure vessels
1289	Methods for testing soils for engineering purposes
1289.5.1.1	Method 5.1.1: Soil compaction and density tests—Determination of the dry density/moisture content relation of a soil using standard compactive effort
1319	Safety signs for the occupational environment
1345	Identification of the contents of pipes, conduits and ducts
1349	Bourdon tube pressure and vacuum gauges
1530	Methods for fire tests on building materials, components and structures
1530.1	Part 1: Combustibility test for materials
1530.4	Part 4: Fire-resistance test of elements of construction
1572	Copper and copper alloys—Seamless tubes for engineering purposes
1657	Fixed platforms, walkways, stairways, and ladders—Design, construction and installation
1668	The use of ventilation and airconditioning in buildings
1668.2	Part 2: Ventilation design for indoor air contaminant control
1674	Safety in welding and allied processes
1674.1	Part 1: Fire precautions
1796	Certification of welders and welding supervisors
1851	Maintenance of fire protection systems and equipment
1894	The storage and handling of non-flammable cryogenic and refrigerated liquids
1940	The storage and handling of flammable and combustible liquids
2030	The verification, filling, inspection, testing and maintenance of cylinders for the storage and transport of compressed gases
2030.1	Part 1: Cylinders for compressed gases other than acetylene
2419	Fire hydrant installations
2419.1	Part 1: System design, installation and commissioning
2441	Installation of fire hose reels
2444	Portable fire extinguishers and fire blankets—Selection and location
2473	Valves for compressed gas cylinders (series)
2700	Colour Standards for general purposes

AS	
2738	Copper and copper alloys—Compositions and designations of refinery products, wrought products, ingots and castings
2809	Road tank vehicles for dangerous goods
2809.1	Part 1: General requirements for all road tank vehicles
2809.3	Part 3: Road tank vehicles for compressed liquefiable gases
2832	Cathodic protection of metals
2832.2	Part 2: Compact buried structures
2872	Atmospheric heating of vessels containing fluids—Estimation of maximum temperature
2885	Pipelines—Gas and liquid petroleum (series)
3711	Freight containers
3711.6	Part 6: Tank containers
3814	Industrial and commercial gas-fired appliances
3894	Site testing of protective coatings
3894.1	Part 1: Non-conductive coatings—Continuity testing—High voltage ('brush') method
4041	Pressure piping
4265	Wheeled fire extinguishers
4332	The storage and handling of gases in cylinders
4670	Commercial propane and commercial butane for heating purposes
4839	The safe use of portable and mobile oxy-fuel gas systems for welding, cutting heating and allied processes
4983	Gas fuel systems for forklifts and industrial engines
AS/NZS	
1020	The control of undesirable static electricity
1200	Pressure equipment
1221	Fire hose reels
1425	LP Gas fuel systems for vehicle engines
1677	Refrigerating systems (series)
1768	Lightning protection
1841	Portable fire extinguishers
1841.1	Part 1: General requirements
1841.5	Part 5: Specific requirements for powder type extinguishers
1841.6	Part 6: Specific requirements for carbon dioxide type extinguishers
1850	Portable fire extinguishers—Classification, rating and performance testing
1869	Hose and hose assemblies for liquefied petroleum gases (LP Gas), natural gas and town gas
2229	Fuel dispensing equipment for explosive atmospheres
2832	Cathodic protection of metals
2832.1	Part 1: Pipes and cables
2865	Safe working in a confined space

AS/NZS	
3000	Electrical installations (known as the Australian/New Zealand Wiring Rules)
3788	Pressure equipment—In-service inspection
4645	Gas distribution
4645.1	Part 1: Network management
4645.2	Part 2: Steel pipe systems
4645.3	Part 3: Plastics pipe systems
5601	Gas installations
5601.1	Part 1: General installations
5601.2	Part 2: LP Gas installations in caravans and boats for non-propulsive purposes
60079	Explosive atmospheres
60079.10.1	Part 10.1: Classification of areas—Explosive gas atmospheres
60079.14	Part 14: Electrical installations design, selection and erection (IEC 60079-14, Ed 4.0 (2007) MOD)
60079.29.2	Part 29.2: Gas detectors—Selection, installation, use and maintenance of detectors for flammable gases and oxygen
NZS	
1170	Structural design actions
1170.5	Part 5: Earthquake actions—New Zealand
4402	Soil compaction tests
4402.4.1.1	Part 4.1.1: Determination of the dry density/water content relationship— Test 4.1.1: New Zealand standard compaction test
4503	Hand operated fire-fighting equipment
5258	Gas distribution networks
5263	Gas detection and odorization
5433	Transport of dangerous goods on land
5435	Specification for liquefied petroleum gas (LPG)
5807	Code of practice for industrial identification by colour, wording or other coding
NZS/AS	
1657	Fixed platforms, walkways, stairways, and ladders—Design, construction and installation
NZS/BS	
750	Specification for underground fire hydrants and surface box frames and covers
NZS/ISO	
834	Fire-resistance tests—Elements of building construction
BS	
3016	Specification for pressure regulators for liquefied petroleum gases
EN	
12864	Low-pressure, non-adjustable regulators having a maximum outlet pressure of less than or equal to 200 mbar, with a capacity of less than or equal to 4 kg/h, and their associated safety devices for butane, propane or their mixtures
13636	Cathodic protection of buried metallic tanks and related piping

- EN
13785 Regulators with a capacity of up to and including 100 kg/h, having a maximum nominal outlet pressure of up to and including 4 bar, other than those covered by EN 12864 and their associated safety devices for butane, propane or their mixtures
- 13786 Automatic change-over valves having a maximum outlet pressure of up to and including 4 bar with a capacity of up to and including 100 kg/h, and their associated safety devices for butane, propane or their mixtures
- 15112 External cathodic protection of well casing
- AUSTRALIAN BUILDING CODES BOARD
BCA Building Code of Australia
- ANSI/API
1104 Welding pipelines and related facilities
- ASME
Boiler and Pressure Vessel Code, Section IX: Welding and Brazing Qualifications
- B31.3 Process piping
- ASTM
A395 Standard specification for ferritic ductile iron pressure-retaining castings for use at elevated temperatures
- NATIONAL BUILDING TECHNOLOGY CENTRE
Technical Record 517—Building Separation and Fire Source Features
- NATIONAL TRANSPORT COMMISSION, AUSTRALIA
ADG Code Australian Code for the Transport of Dangerous Goods by Road and Rail
- OFFICE OF THE AUSTRALIAN SAFETY AND COMPENSATION COMMISSION
NOHSC
1010 National Standard for Plant
Hazardous Substances (Tank Wagons and Transportable containers) Regulations
Fuel Standard (Autogas) Determination 2003
- UN Manual of Tests and Criteria and the Recommendation on the Transport of Dangerous Goods—Model Regulations
- NFPA 15 Standards for water spray fixed systems for fire protection
- UN Manual of Tests and Criteria and the Recommendations of the Transport of Dangerous Goods—Model Regulations
- UL UNDERWRITERS LABORATORIES
21 Standard for LP Gas Hose
- 132 Standard for Safety Relief Valves for Anhydrous Ammonia and LP Gas
- 144 Standard for LP Gas Regulators
- IMO INTERNATIONAL MARITIME ORGANIZATION
IMDG Code International Maritime Dangerous Goods Code
- US DEPARTMENT OF TRANSPORTATION
Hazardous Materials Regulations

NEW ZEALAND**Statutes**

Hazardous Substances and New Organisms Act 1996 (HSNO Act)

Regulations

Hazardous Substances (Classification) Regulations

Hazardous Substances (Compressed Gases) Regulations 2004

Hazardous Substances (Tank Wagons and Transportable Containers) Regulations 2004

Land Transport Rule: Dangerous Goods 2005

Approved Codes of Practice

HSNOCOP 29—Flammable Gases and Oxygen Tank Wagons

LPG Compliance 100 kg to 300 kg

HSNOCOP 38—In-situ filling of LPG Cylinders

Technical guide

New Zealand Environmental Protection Authority (EPA) Technical Guide—Guide to Gas Cylinders

Transfer notices

Hazardous Substances (Dangerous Goods and Scheduled Toxic Substances) Transfer Notice

APPENDIX B

TANK OR CYLINDER FILLING LEVEL

(Normative)

B1 INTRODUCTION

An LP Gas container shall never be allowed to become liquid-full, because any subsequent thermal expansion will cause liquid to discharge from the safety valve and could subject the container to hydrostatic pressure. It is necessary to stop filling before the tank space is completely filled, and the resulting ullage space must be sufficient to allow for the expected liquid temperature fluctuations and the relatively high coefficient of expansion of LP Gas liquid. This Appendix describes alternative methods of calculating filling levels that will ensure that normal variations are catered for, so that a vapour space will remain above the liquid surface in all but the most abnormal circumstances.

Basic to the determination of the filling level is an assumption of the temperature likely to be reached in the liquid when a container stands in the sun. Heating is caused by both ambient air temperature and solar radiation effects. Experiments in various locations have established that the temperature will stabilize at a value which will vary according to two features of the tank itself, firstly the heat sink effect of its size and secondly, but to a lesser extent, its aspect ratio. Both these features are a function of the ratio between the volume and the surface area, and this V/A ratio has become the basis for determining the maximum likely temperature of the tank contents.

Theoretically, since every tank will have an individual V/A ratio, each will reach an individual temperature, therefore each tank should have an individual filling ratio. However, it is considered that this degree of complexity is not justified, so a simple convention has been adopted to the effect that tanks up to 5 kL capacity may reach 55°C, but tanks over 5 kL are not likely to exceed 46°C. Filling ratios have been calculated to leave a small ullage of 1% to 2% when the liquid is at these temperatures.

A tank may be filled by mass, using a mass-filling ratio method, or by volume, using a volume-filling percentage method. The former is generally considered to be the more exact method, because the only variable is the composition of the particular LP Gas being loaded and the temperature is not a factor, so that temperature observations and corrections are not necessary. However, weighing is rarely practicable for stationary tanks, so filling is normally controlled by observation of the level of the liquid surface, i.e. the filling level.

Two methods of filling level control are available. The simpler is to fill to the standard filling levels as indicated by the fixed liquid level gauge. This level is chosen by assuming a specific LP Gas composition whose temperature at the time of filling is 5°C.

The standard filling level is fairly conservative to allow for variability, so a more complex procedure is available, which permits filling to the maximum filling level. This technique requires knowledge of the exact liquid temperature and the exact LP Gas composition.

The procedures described in this Appendix apply to uninsulated and unshaded tanks or cylinders exposed to normal weather cycles. It is recognized that in a specific installation some or all of these conditions may not apply, but any variation to filling ratios should be negotiated individually with the authority. The Hazardous Materials Regulations of the Department of Transportation (USA) make provisions for special cases and may be used as a guide.

B2 DETERMINATION OF STANDARD FILLING LEVEL

B2.1 Application

The standard filling level gives the location point for the fixed liquid level gauge, necessary for the filling procedures outlined in Paragraphs B4 and B5.

B2.2 Principle

The standard liquid level is calculated on the basis that a liquid at 5°C is loaded into a tank and then allowed to warm to either 46°C or 55°C (according to size), at which temperature the tank is one to two percent less than completely filled. It is considered that temperatures outside this range are unlikely to occur in practice.

B2.3 Procedure

The procedure shall be as follows:

- (a) Determine the following:
 - (i) Whether the tank is to be above-ground or underground.
 - (ii) The capacity of the tank.
 - (iii) The type of LP Gas which the tank will contain, i.e. commercial propane, commercial butane, or a specified LP Gas, or possibly a variety of LP Gases, in which case, assume propane.
- (b) Read the volumetric filling percentage (V) from Table B6, or calculate it from the following equation:

$$V = \frac{R}{\rho F} \quad \dots B1$$

where

V = volumetric-filling percentage

R = mass-filling ratio for the nominated LP Gas (see Table B1)

ρ = liquid density at 15°C of the nominated LP Gas, in kilograms per litre

F = correction factor for 5°C (see Table B5)

- (c) Calculate the location of the liquid surface from (V) and the tank diameter by geometric methods.

NOTE: The Tables are heavily rounded, so a higher degree of accuracy is gained by using the calculation.

B3 PROCEDURE—FILLING BY MASS

B3.1 Application

Since it is not normally practicable to provide weighing facilities for a fixed stationary tank, filling by mass is generally usable only for cylinders, or for tankers if they can be filled on a weighbridge, or cross-checked on a weighbridge after filling by volumetric monitoring.

B3.2 Filling procedure

The mass-filling ratio is read directly from Table B1 according to the characteristics of the particular LP Gas and the size of the tank. The ratio is multiplied by the tank capacity to give the maximum permissible cargo loading, in kilograms or tonnes.

B4 PROCEDURE—FILLING BY VOLUME—STANDARD LIQUID LEVEL METHOD

B4.1 Application

The standard liquid level method may be used for any cylinder, tank or tanker that is not to be filled by mass, provided that the liquid being transferred is neither—

- (a) unusually cold, i.e. below 5°C; nor
- (b) less dense than the LP Gas for which the gauge is calibrated.

If the liquid density is substantially different from that for which the gauge is calibrated, the methods set out in Paragraph B5 shall be used if the gauge cannot be recalibrated for the particular LP Gas.

B4.2 Filling procedure

The volume-filling percentage represented by the fixed liquid level gauge is first read from the tank marking, and checked that it is appropriate for the liquid to be filled. The tank is filled to the standard filling level indicated by the fixed liquid level gauge. No further checking is necessary. Where a variable liquid level gauge is used to monitor the rising liquid level during filling, it shall not be used as the final arbiter of when to stop filling.

B5 PROCEDURE—FILLING BY VOLUME—TEMPERATURE CORRECTION METHOD

B5.1 Application

The temperature correction method may be used to fill above the standard liquid level to the maximum filling level for the particular liquid being transferred. A variable liquid level gauge and a liquid-temperature gauge are necessary.

B5.2 Determination of maximum permitted level

The following information is needed for determination of the maximum permitted level:

- (a) The density at 15°C of the particular LP Gas being delivered. If not available, the lowest likely value is assumed.
- (b) The temperature of the liquid as it is filled into the tank.
- (c) The size of the tank, i.e. whether it is above or below 5 kL capacity.
- (d) Whether the tank is above-ground or underground.

The volume-filling percentage may be read from Table B2, Table B3, or Table B4, as appropriate. Interpolations are permitted when using the relevant table. As an alternative to reading values from the table, the volume-filling percentage may be calculated from the following equation:

$$V = \frac{R}{\rho F} \quad \dots B2$$

where

V = volumetric-filling percentage

R = mass-filling ratio (see Table B1)

ρ = density at 15°C of the LP Gas being loaded, in kilograms per litre

F = liquid volume correction factor from Table B5, in which the observed temperature is the actual temperature of the LP Gas liquid in the tank

NOTE: The Tables are heavily rounded, so a higher degree of accuracy is gained by using the calculation method.

B5.3 Filling procedure

The method for controlling filling shall be as follows:

- Fill to the standard filling level as in Paragraph B4. As filling proceeds, monitor the temperature, and determine the value of V (see Paragraph B5.2).
NOTE: Standard filling levels are given in Table B6.
- As the liquid level approaches the fixed liquid level gauge point, prepare to start using the variable liquid level gauge.
- As the liquid level reaches the standard filling level as indicated by the fixed liquid level gauge, check the calibration of the variable liquid level gauge.
- Continue filling until the calculated maximum filling level, corrected for any gauge error, has been reached.

TABLE B1
MASS-FILLING RATIO

Density at 15°C kg/L	Maximum mass-filling ratio, percent		
	5 kL total capacity or less	Over 5 kL capacity	Underground, all capacities
0.473 to 0.480	38	41	42
0.481 to 0.488	39	42	43
0.489 to 0.495	40	43	44
0.496 to 0.503	41	44	45
0.504 to 0.510	42	45	46
0.511 to 0.519	43	46	47
0.520 to 0.527	44	47	48
0.528 to 0.536	45	48	49
0.537 to 0.544	46	49	50
0.545 to 0.552	47	50	51
0.553 to 0.560	48	51	52
0.561 to 0.568	49	52	53
0.569 to 0.576	50	53	54
0.577 to 0.584	51	54	55
0.585 to 0.592	52	55	56
0.593 to 0.600	53	56	57
0.601 to 0.608	54	57	58
0.609 to 0.617	55	58	59
0.618 to 0.626	56	59	60
0.627 to 0.634	57	60	61

TABLE B2
VOLUME-FILLING RATIO (CAPACITY 5 kL OR LESS)

Liquid temperature °C	Maximum volume-filling ratio, percent						
	Density at 15°C, kg/L						
	0.500	0.510 Commercial propane	0.520	0.540	0.560	0.580 Commercial butane	0.590
-20	74	75	77	78	79	82	82
-10	76	77	79	80	81	83	84
0	78	79	81	82	83	85	85
5	79	80	82	83	83	86	86
10	80	81	83	84	84	87	87
15	82	82	84	85	85	87	88
20	83	84	85	86	86	88	89
25	84	85	87	87	87	89	89
30	86	86	88	88	88	90	90
35	87	88	89	89	89	91	91
40	89	89	91	91	91	92	92
45	91	91	92	92	92	93	93
50	93	93	93	93	93	94	94
55	95	95	95	95	95	96	96

TABLE B3
VOLUME-FILLING RATIO (CAPACITY GREATER THAN 5 kL)

Liquid temperature °C	Maximum volume-filling ratio, percent						
	Density at 15°C, kg/L						
	0.500	0.510 Commercial propane	0.520	0.540	0.560	0.580 Commercial butane	0.590
-20	80	80	83	84	84	87	87
-10	82	82	84	85	86	88	89
0	84	84	86	87	88	90	90
5	85	85	88	88	89	91	91
10	86	87	89	89	91	92	92
15	88	88	90	90	91	93	93
20	89	89	91	91	91	94	94
25	91	91	92	93	93	95	95
30	92	92	94	94	94	95	95
35	94	94	95	95	95	96	96
40	95	95	97	97	97	98	98
45	97	97	99	98	98	99	99
50	99	99	99	99	99	—	—

TABLE B4
VOLUME-FILLING RATIO—UNDERGROUND TANKS (ALL SIZES)

Liquid temperature °C	Maximum volume-filling ratio, percent						
	Density at 15°C, kg/L						
	0.500	0.510 Commercial propane	0.520	0.540	0.560	0.580 Commercial butane	0.590
–20	82	82	84	85	86	88	89
–10	84	84	86	87	88	90	90
0	86	86	88	89	89	92	92
5	87	87	90	90	90	93	93
10	88	88	91	91	91	93	94
15	90	90	92	92	92	94	94
20	91	91	93	93	93	95	95
25	92	92	94	94	94	96	96
30	94	94	96	96	96	97	97
35	96	96	97	97	97	98	98
40	98	98	99	99	99	99	99
45	99	99	—	—	—	—	—

TABLE B5
LIQUID VOLUME CORRECTION FACTORS

Liquid temperature °C	Volume correction factor						
	Density at 15°C, kg/L						
	0.500	0.510 Commercial propane	0.520	0.540	0.560	0.580 Commercial butane	0.590
–40	1.145	1.139	1.133	1.121	1.111	1.101	1.096
–30	1.120	1.116	1.110	1.101	1.091	1.082	1.080
–20	1.095	1.091	1.087	1.080	1.072	1.066	1.063
–10	1.070	1.067	1.064	1.058	1.053	1.048	1.045
0	1.044	1.042	1.039	1.035	1.032	1.029	1.027
5	1.030	1.028	1.025	1.024	1.022	1.019	1.018
10	1.015	1.014	1.013	1.012	1.011	1.010	1.009
15	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	0.985	0.985	0.986	0.988	0.990	0.990	0.990
25	0.967	0.970	0.972	0.975	0.978	0.980	0.981
30	0.951	0.954	0.957	0.962	0.966	0.970	0.972
35	0.935	0.938	0.942	0.949	0.955	0.960	0.962
40	0.917	0.922	0.927	0.935	0.943	0.949	0.951
45	0.899	0.905	0.910	0.921	0.933	0.938	0.941
50	0.880	0.887	0.894	0.907	0.919	0.927	0.930
55	0.861	0.868	0.876	0.891	0.906	0.916	0.920
60	0.840	0.850	0.860	0.880	0.889	0.904	0.908

TABLE B6
STANDARD FILLING LEVEL

Type of LP Gas	Maximum volume-filling ratio, percent			
	Above-ground tank		Underground tank	Cylinder
	5 kL or less	Over 5 kL		
Commercial propane	80	85	88	80 (Australia only)
Mixture (0.55 kg/L)	83	87	91	
Commercial butane	87	89	93	

APPENDIX C

APPROXIMATE CAPACITIES OF CYLINDERS FOR LP GAS

(Informative)

The approximate masses and volumes (water capacities) of LP Gas cylinders are given in Table C1.

TABLE C1
APPROXIMATE CAPACITIES OF CYLINDERS
FOR LP GAS

Nominal mass of LP Gas in cylinder kg	Approximate volume (water capacity) of cylinder L
3	7
4.5	11
5	13
9	22
10	26
13.5	32
15	36
18	44
45	108
90	200
190	454
210	499

APPENDIX D MARKINGS, SIGNS AND NOTICES

(Normative)

D1 TANK SYSTEMS

D1.1 Valve indicators and marking

A quarter-turn valve shall be provided with indicators and marking in accordance with the following requirements:

- (a) A means for indicating clearly the open and closed position shall be provided. Where a lever valve is fitted with a fixed handle, the valve shall be open when the handle is in the direction of the pipe. Where a valve handle is removable, a separate metal plate shall be provided, and this plate shall be clearly and indelibly marked to indicate the open and closed positions of the valve.
- (b) Globe and angle valves shall have handwheels permanently marked with the direction to open.
- (c) Any valve other than for a safety valve, pressure gauge or fixed liquid level gauge shall be marked to indicate whether it connects to liquid or vapour space.

D1.2 Vaporizer marking

A vaporizer shall be marked with the following information:

- (a) Manufacturer's name or identification mark.
- (b) The number of the Standard to which the vaporizer is constructed.
- (c) Design pressure, in kilopascals.
- (d) Design temperature, in degrees Celsius.
- (e) The maximum vaporizing capacity in litres per hour.
- (f) The surface area of the heat exchanger in square metres, as detailed in Paragraph H2 of Appendix H.
- (g) If direct-fired—
 - (i) the rated input of the heat source; and
 - (ii) the outside surface area of the vaporizer in square metres, as detailed in Paragraph H2 of Appendix H.

D1.3 Tank data plate

In addition to the markings required by AS 1210, the tank data plate shall include the following:

- (a) Capacity in cubic metres.
- (b) Any limitation on permissible contents [see Clause 5.2.3(a)].
- (c) The design colour (for an above-ground tank only).
- (d) The last date of internal tank inspection and the last date of inspection of the safety valves.

D1.4 Liquid and vapour lines

Liquid and vapour lines shall be identified as illustrated in Figure 6.4.

D2 TANK INSTALLATIONS

D2.1 Emergency shut-down system marking

Every actuation point for the emergency shut-down system except the provision on the valve itself shall be provided with a sign—

- (a) marked 'LP GAS EMERGENCY STOP' in red letters not less than 40 mm high on a white background; and
- (b) giving instructions on how to actuate the system, in red or dark lettering not less than 20 mm high on a white or light contrasting background.

D2.2 Tank or tank storage area placarding

An above-ground tank or tank storage area shall be provided with a placard in accordance with Paragraph D2.3. Where the tank and pump are underground, the placard shall be in accordance with Figure D1.

D2.3 Placard

The placard shall display the following information:

- (a) Phone number of the organization having ownership of the equipment. The nominated phone number shall provide contact with 24-hour specialist advice via the supply company or a contractor.
- (b) Standard symbols of at least 100 mm diameter, or words, for—
 - (i) 'flammable gas' and 'no smoking'; or
 - (ii) a warning notice in letters not less than 50 mm high prohibiting smoking and ignition sources.

Where the area is isolated by a fence, the sign shall be visible from outside the fence and also from points of access.

NOTE: Suitable wording for such signs would be—

- (a) NO SMOKING, NO IGNITION SOURCES.
- (b) NO SMOKING, KEEP FIRE AWAY.
- (c) FLAMMABLE GAS, NO SMOKING.
- (c) Emergency information panel complying with, in Australia, ADG Code (see example in Figure D1), or in New Zealand, NZS 5433.

D2.4 Construction of signs

Signs shall be constructed in accordance with AS 1319. Class labels shall be a minimum of 250 mm square unless permitted to be reduced by the ADG Code or NZS 5433.

D3 CYLINDER INSTALLATIONS

D3.1 Liquid withdrawal cylinders

The following markings shall be used:

- (a) If the cylinder is for liquid withdrawal only, the cylinder shall be marked with the words 'LIQUID WITHDRAWAL ONLY' in letters not less than 30 mm high. In addition, the valve protection collar if permanently attached, or an area not less than 150 mm high around the circumference of the cylinder immediately below the valve, shall be painted blue.
- (b) If the cylinder is equipped for both liquid and vapour withdrawal, each valve opening shall be marked 'VAPOUR' or 'LIQUID' as appropriate, in letters not less than 10 mm high.

D4 CYLINDER FILLING

D4.1 Notices for cylinder filling

The following notices shall be prominently displayed as closely as possible to the decanting point:

- Standard symbols of at least 100 mm diameter for 'flammable gas' and 'no smoking' or a warning notice in letters not less than 50 mm high, reading—
FLAMMABLE GAS, NO SMOKING
- Instructions for decanting or filling, including a warning to stop filling as soon as the fixed liquid level gauge indicates that the maximum permitted liquid level has been reached.

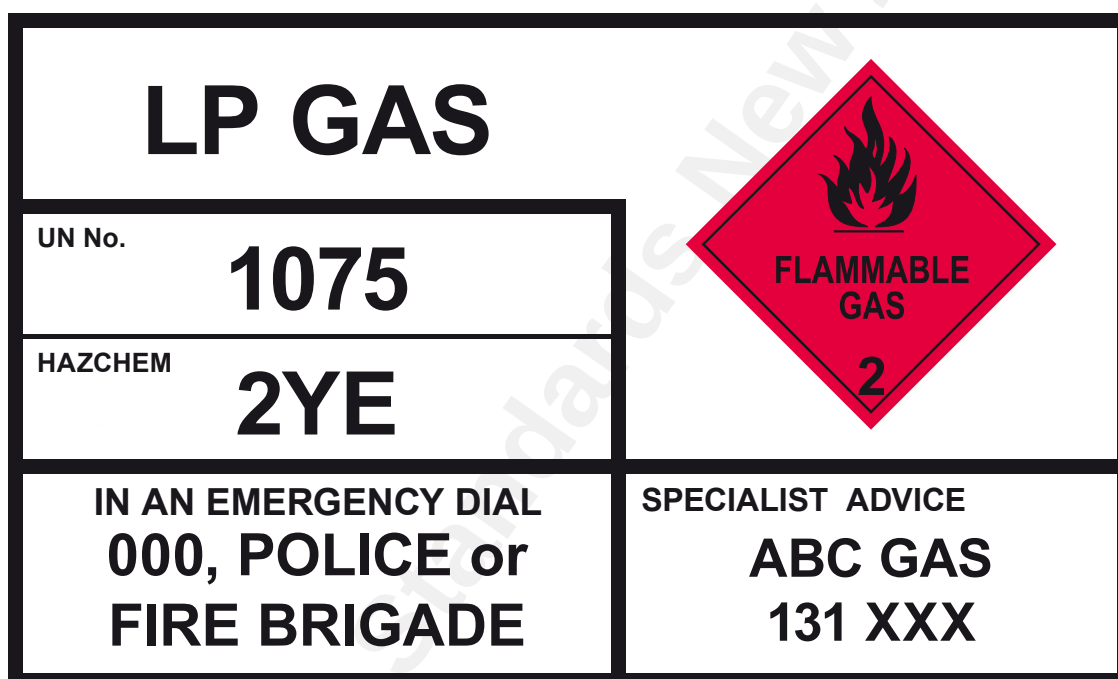


FIGURE D1 TYPICAL EXAMPLE OF PLACARD

D5 AUTOMOTIVE FILLING STATIONS—VEHICLE FILLING AREAS

D5.1 Precautionary sign

Standard symbols of at least 100 mm diameter for 'stop engine—no smoking' shall be displayed at the filling area or adjacent to it and readable from the filling area. Alternatively a prominent sign reading 'STOP ENGINE—NO SMOKING' in red or dark lettering not less than 50 mm high on a white background may be substituted.

D5.2 Driver instructions

D5.2.1 General

In Australia, the signs specified in Paragraphs D5.2.2, D5.2.3 and D5.2.4 shall be displayed on, or adjacent to and readable from, each dispenser.

D5.2.2 Driver instruction signs

Driver instruction signs shall have a minimum size of 125 mm wide × 270 mm high, and shall be printed in the colours indicated in Figures D2 and D4 on a white background. Such signs shall be displayed in a readily visible location on LP Gas dispensers.

D5.2.3 *Warning sign*

The warning sign shall be substantially of the form illustrated in Figure D2.

D5.2.4 *Filling instructions*

The sign displaying filling instructions shall be substantially as illustrated in Figure D4.

NOTE: Liquefied petroleum gas may be abbreviated to 'LP Gas' or 'LPG'.

D5.3 **Emergency procedures**

A notice reading as shown in Figure D3 shall be prominently displayed outside the principal building or office and another such notice shall be displayed inside and at the focal point of the building or office.

The notice shall have a white background; the heading 'LP GAS EMERGENCY PROCEDURE' shall be in red letters not less than 40 mm high; subheadings 'Gas Leak' and 'Fire' shall be in red or dark lettering not less than 10 mm high; and the text shall be in red or dark lettering not less than 7 mm high.

Where the tank or dispenser is not associated with any specific office or building, e.g. in a depot or an industrial location, the notice should be displayed in a location appropriate for the particular installation.

LP GAS WARNING
HIGHLY FLAMMABLE
LIQUID UNDER PRESSURE

NOTES:
WARNING — RED
(Signal Red R13)
TEXT — BLACK



Stop engine

CIRCLE & SLASH — RED
SYMBOL — BLACK
INSTRUCTION — BLACK



No mobile phones

CIRCLE & SLASH — RED
SYMBOL — BLACK
INSTRUCTION — BLACK



No flames or pilot lights

CIRCLE & SLASH — RED
SYMBOL — BLACK
INSTRUCTION — BLACK



No smoking

CIRCLE & SLASH — RED
SYMBOL — BLACK
INSTRUCTION — BLACK



Avoid contact-LP Gas
may cause cold burns

CIRCLE & SLASH — RED
SYMBOL — BLACK
INSTRUCTION — BLACK



Do not fill cylinders.
For vehicles only

CIRCLE & SLASH — RED
SYMBOL — BLACK
INSTRUCTION — BLACK

FIGURE D2 WARNING SIGN

LP GAS EMERGENCY PROCEDURE <i>(Provide the address of this site on or adjacent to the notice.)</i>
GAS LEAK 1 Shut emergency stop. 2 Close all valves of tank. 3 Keep bystanders away. 4 No smoking. No naked lights. 5 No engine to be started. 6 Phone fire brigade <i>(insert telephone number of local station, or general emergency number)</i>. 7 Phone <i>(insert name and all-hours number for LP Gas distributor or site 24-hour emergency response contact)</i>. 8 Isolate electricity <i>(by supply authority if main switchboard unsafe)</i>.
FIRE 1 Shut emergency stop. 2 Phone fire brigade <i>(insert telephone number of local station)</i>. 3 Phone <i>(insert LP Gas distributor's name and all-hours number)</i>. 4 Close all valves of tank. 5 Keep bystanders away. 6 Isolate electricity <i>(by supply authority if main switchboard unsafe)</i>. 7 Follow firefighting instructions.

NOTE: Text in italics needs to be completed by the site operator.

FIGURE D3 LP GAS EMERGENCY PROCEDURE NOTICE

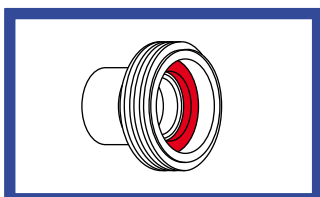
LP GAS FILLING

INSTRUCTIONS

**STAY AT VEHICLE FILL POINT DURING FILLING.
DON'T OVERLY STRAIN THE HOSE.**

NOTES:

INSTRUCTIONS — BLUE
(Bright Blue B23)

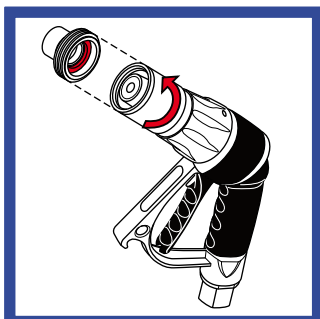


Check vehicle fill point

- Must be clean
- Rubber seal shall be in place
- Rubber seal and thread shall be in good condition

SQUARE &
INSTRUCTION
SYMBOL & TEXT
GASKET

— BLUE
— BLACK
— RED
(Signal Red R13)

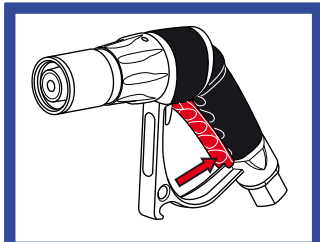


Connect nozzle

- Align connector to avoid cross-threading
- Turn connector clockwise until tight

SQUARE &
INSTRUCTION
SYMBOL & TEXT
GASKET

— BLUE
— BLACK
— RED



Squeeze lever fully and hold. If leak occurs:

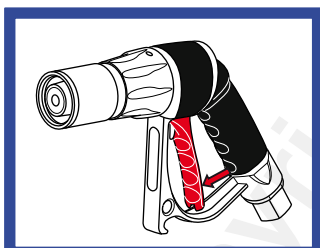
- Release lever
- Seek assistance immediately

WARNING! DO NOT JAM LEVER
OPEN WITH OBJECT

SQUARE &
INSTRUCTION
SYMBOL & TEXT
LEVER &
ARROW

— BLUE
— BLACK
— RED

WARNING & TEXT — RED



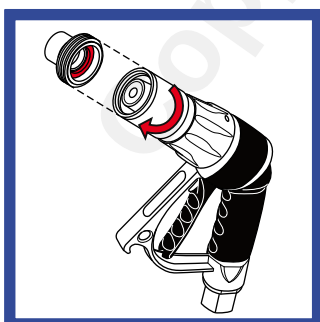
When filled release lever

- Expect small discharge of LP Gas

WARNING! RELEASE THE LEVER LATCH,
IF FITTED, BEFORE DISCONNECTING
THE NOZZLE

SQUARE &
INSTRUCTION
SYMBOL & TEXT
LEVER &
ARROW

— BLUE
— BLACK
— RED



Disconnect nozzle and stow

SQUARE &
INSTRUCTION
ARROW

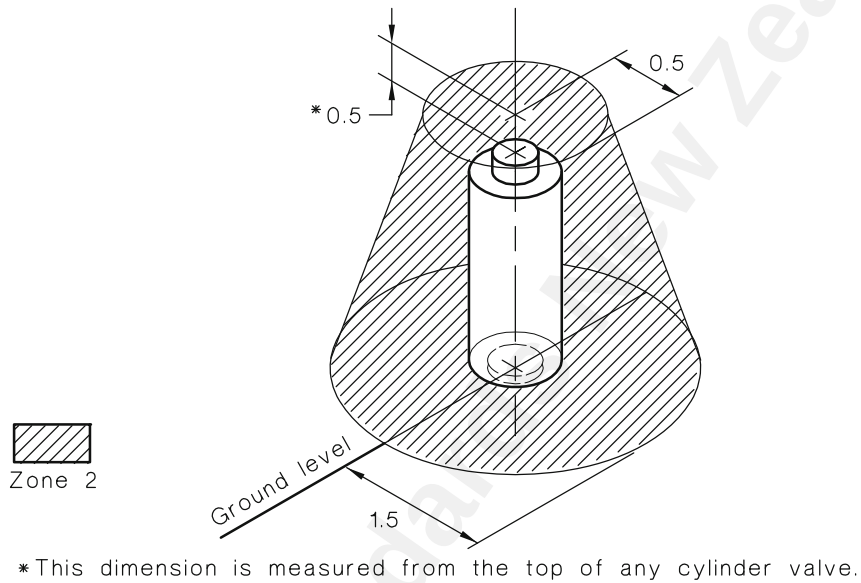
— BLUE
— RED

FIGURE D4 FILLING INSTRUCTIONS

APPENDIX E HAZARDOUS AREAS

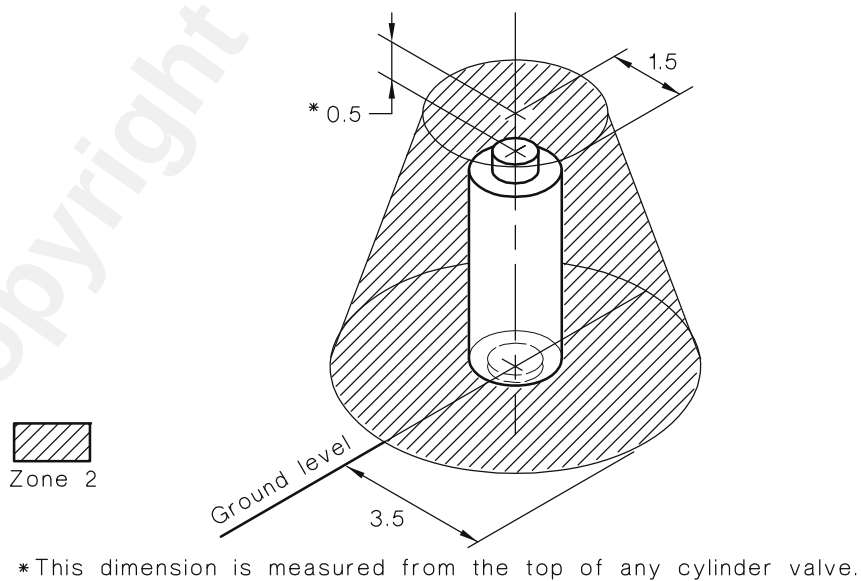
(Normative)

The hazardous areas illustrated in Figures E1 to E5 are taken from AS/NZS 60079.10.1. The examples provided are not complete and other situations and requirements that may be relevant to LP Gas installations are also provided in AS/NZS 60079.10.1.



DIMENSIONS IN METRES

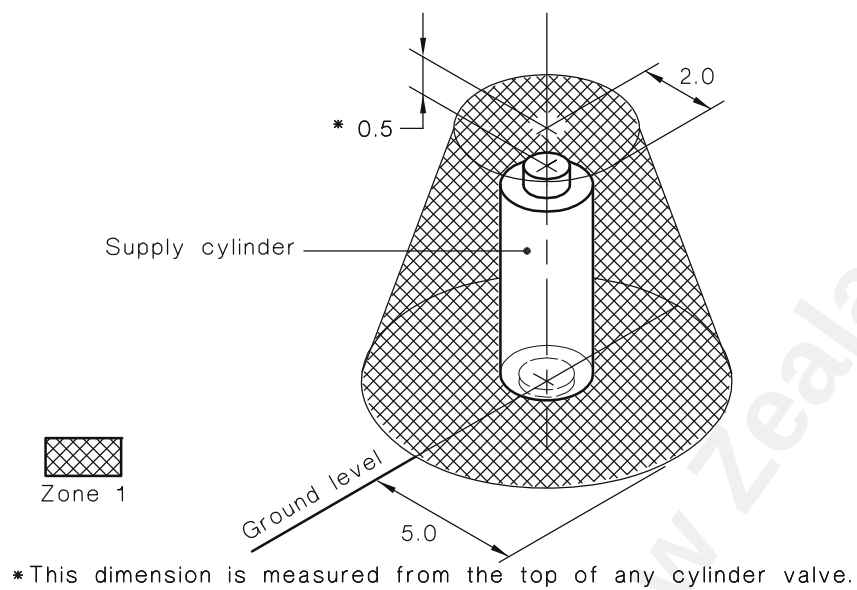
FIGURE E1 EXCHANGE CYLINDER



DIMENSIONS IN METRES

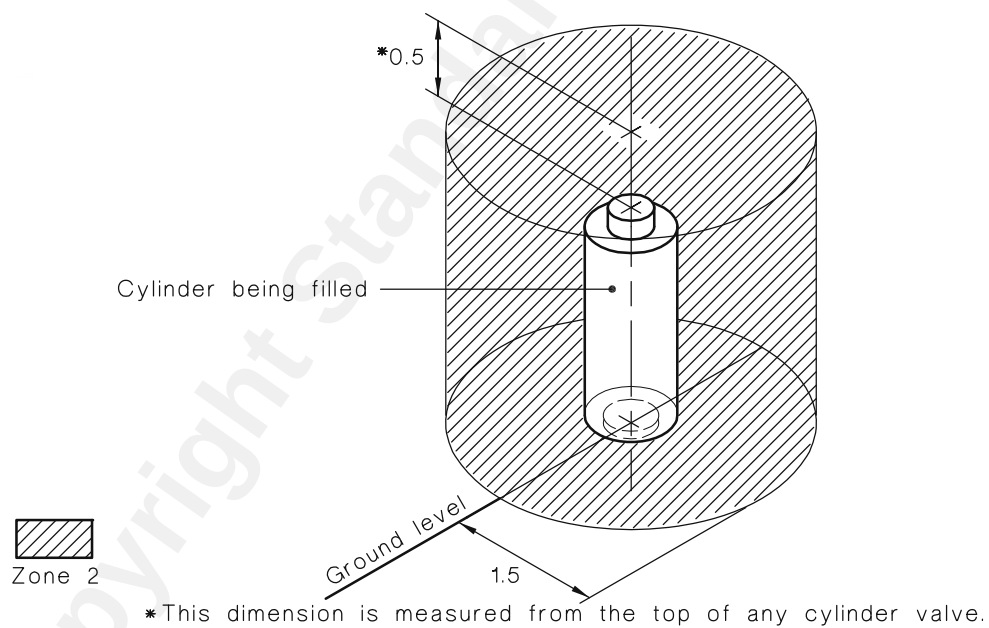
FIGURE E2 IN SITU FILL CYLINDER

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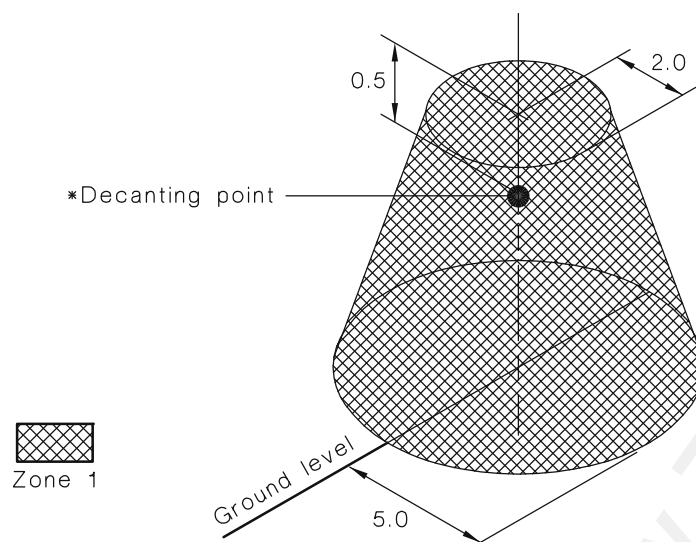
DIMENSIONS IN METRES

FIGURE E3 FILLING WITH GAS BLEEDING, IN ADEQUATELY VENTILATED LOCATION



DIMENSIONS IN METRES

FIGURE E4 FILLING WITHOUT GAS BLEEDING, IN ADEQUATELY VENTILATED LOCATION



- * Decanting point: the point where the decanting hose is connected to the supply.
It may be—
(a) a cylinder valve; or
(b) another supply point, e.g. a tank.

DIMENSIONS IN METRES

FIGURE E5 FILLING FROM CYLINDER OR TANK SUPPLY, IN ADEQUATELY VENTILATED LOCATION

APPENDIX F

EXCHANGE FACILITIES FOR PORTABLE CYLINDERS—AUSTRALIA

(Normative)

F1 INTRODUCTION

This Appendix provides requirements and recommendations for the location of cages in which portable exchange LP Gas cylinders are kept. These facilities provide customers with an exchange system for LP Gas cylinders for barbecues and other uses, in place of refilling the customer's own cylinder by decanting. Such facilities are usually located at service stations and hardware shops. A customer exchanges an empty cylinder for a filled cylinder of LP Gas of equivalent size.

This Appendix compiles the requirements already given elsewhere in this Standard and provides further guidance for this specific application.

F2 RESTRICTIONS

The following restrictions apply:

- (a) The maximum capacity of any individual cylinder shall be limited to 25 L.
- (b) The cage shall be sturdy and stable, and shall allow free air movement through it.
- (c) The cage shall be located away from, or be protected from, significant trafficable areas.
- (d) The maximum aggregate capacity of cylinders kept per cage or group of cages shall not exceed 2500 L.
- (e) Where cages are grouped together, their aggregate capacity shall not exceed 2500 L. There shall be a minimum distance of 3 m between the groups of cages.
- (f) The separation distances given in Table 4.1 shall apply.

F3 LOCATION OF CAGES

Before placing a cylinder cage at a site, an assessment shall be made on the need to provide additional security measures to prevent vandalism on the LP Gas cylinder cage.

If a vandalism risk is present, additional physical security should be provided, such as a site perimeter fence.

NOTE: As a guide, cylinder cages should only be located in—

- (a) high visibility areas, e.g. there is passing traffic after hours and on weekends;
- (b) areas with low incidence of vandalism; or
- (c) areas cleared of combustibles (e.g. plastic rubbish bins, empty boxes, etc.).

Cages shall be located—

- (a) outdoors;
- (b) at least 1 m from any opening;
- (c) at least 1 m from the hose reach of an LP Gas decanting cylinder;
- (d) at least 1.5 m horizontally or 0.5 m vertically from any ignition source;
- (e) at least 1.5 m from any pit, drain, basement, public place or dispenser for any type of fuel;
- (f) at least 5 m from any tank containing LP Gas;

- (g) at least 3 m from any above-ground tank containing dangerous goods other than LP Gas;
- (h) at least 0.5 m away from any accumulation of combustible materials;
- (i) at least 2 m from any structure limiting egress past any cage; and
- (j) clear on at least two sides from any wall, solid display or other item that could restrict air flow.

NOTE: The location of such cages as shown in Figure F1.

F4 SIGNS AND LABELS

Every individual cylinder cage shall be provided with signs and notices that are clearly visible and readily distinguishable from any advertising signs attached to the cage.

The following signs and notices shall be prominently displayed on the front of the cylinder cage:

- (a) A dangerous goods class label.

NOTE: This label should be 250 mm square.

- (b) A warning notice, reading 'FLAMMABLE GAS—NO SMOKING, NO FLAME'.

NOTE: Symbols may be used.

F5 PRIVATE TRANSPORT OF PORTABLE LP GAS CYLINDERS

Where cylinders of LP Gas are transported for private domestic use, Appendix O should be consulted.

F6 TRANSPORT OF PORTABLE LP GAS CYLINDERS FOR TRADE USE

Where cylinders of LP Gas are transported for trade use, Appendix P should be consulted.

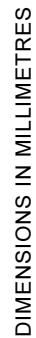


FIGURE F1 LOCATION OF CAGES FOR EXCHANGE CYLINDERS

APPENDIX G

EXCHANGE FACILITIES FOR PORTABLE CYLINDERS—NEW ZEALAND

(Normative)

G1 INTRODUCTION

Portable exchange LP Gas cylinders shall be kept in a secure cage or cages. This Appendix provides requirements and recommendations for the location of such cages. These facilities provide customers with an exchange system for LP Gas cylinders for barbeques and other uses, in place of refilling the customer's own cylinder. Such facilities are usually located at, but not limited to, service stations and hardware shops. A customer exchanges an empty cylinder for a filled cylinder of LP Gas of equivalent size.

This Appendix compiles the requirements already given elsewhere in this Standard and provides further guidance for this specific application.

G2 LOCATION OF SITE

The following factors shall be considered when deciding the location of an exchange facility:

- (a) Before placing a cylinder cage at a site, an assessment shall be made on the need to provide security measures to prevent vandalism on the LP Gas cylinder cage.

NOTE: If a vandalism risk is present, additional physical security measures should be provided, such as a site perimeter fence.

- (b) The cage shall not be located near significant trafficable areas, unless suitably protected by the use of bollards or guardrails or similar.

NOTE: As a guide, cylinder cages should only be located in—

- (a) high visibility areas e.g. there is passing traffic after hours and on weekends;
- (b) areas with low incidence of vandalism; or
- (c) areas cleared of combustibles (e.g. plastic rubbish bins, empty boxes, etc.).

G3 RESTRICTIONS

The following restrictions shall apply:

- (a) Cages shall be located outdoors.
- (b) The cage shall be sturdy and stable, and shall allow free air movement through it.
- (c) The maximum capacity of any individual cylinder shall be limited to 10 kg.
- (d) The maximum aggregate capacity of cylinders in a cage or single group of cages shall not exceed 1250 kg.
- (e) There shall be a minimum distance of 3 m between each aggregate quantity.

The positioning of the cage will depend on the quantity of LP Gas present; the following requirements apply:

- (i) Up to 100 kg of LP Gas may be located within 1 m of a building, provided there are no openings into the building below the top of the cylinders and within 1 m either side of the cylinders.

- (ii) Between 100 kg and up to 300 kg of LP Gas may be located within 2 m of a building, provided the wall of the building behind and 2 m either side of the cylinders is constructed of fire resisting materials and the building does not have any openings below the top of the cylinders and within 2 m either side of the cylinders.
- (iii) Between 300 kg and up to 1000 kg of LP Gas may be located within 2 m of a building provided the wall of the building (or a wall between the cylinders and the building) directly behind the cylinders is vapour tight and has a minimum fire resistance rating of 60/60/60 minutes extending at least 2 m either side of the cylinders and any building within 2 m of the cylinder does not have any opening located below the top of the cylinder and within 2 m either side of the cylinder.
- (iv) If the quantity is 1000 kg or more, the cylinder shall be separated as follows:

Quantity, kg	HILU, m	LILU, m
1000	2.5	2
1500	2.75	2
2000	3	2

NOTE: Toughened laminated glass of a thickness of 10–12 mm or greater that complies with one of the following standards is acceptable as fire resisting material:

- (a) Toughened Safety Glass Lic. No. 2446 T.F.A.
- (b) Tempafloat AS/NZS 2208:96 Lic. No. 2518 T.F.A.
- (c) Tempafloat AS/NZS 2208:96 Lic. No. 2465 T.F.A.
- (d) Metroglass tech AS/NZS 2208 Grade 1.
- (e) Pilkington AS/NZS 22011 LS Grade Laminated Safety Glass.

G4 MANAGEMENT OF CYLINDER CAGE

Cages shall be located as follows:

- (a) At least 1 m from the hose reach of an automotive dispenser.
- (b) At least 1.5 m horizontally and 0.5 m vertically from any ignition source.
- (c) At least 1.5 m from drains, pits, basements, public places or fuel dispensers. Where a drain or gully trap is unavoidable, the opening shall be securely covered or fitted with a water seal.
- (d) At least 5 m from any tank containing LP Gas.
- (e) At least 3 m from aboveground tanks containing hazardous substances other than LP Gas.
- (f) At least 2 m from structures limiting access past the cages.
- (g) Clear on at least 2 sides from walls, solid structures or obstacles that may restrict airflow.
- (h) So that the area within 1.5 m of the cylinder cage shall be kept clear of weeds, rubbish and other combustible materials.
- (i) The cylinder cage shall be periodically inspected and any defects, such as damage to the structure, shelves, padlocks or signage which shall be remedied.

G5 STORAGE OF CYLINDERS WITHIN THE CAGE

The following requirements apply:

- (a) Cylinders shall be stored or displayed with their valves uppermost. This ensures that if there is a leak, vapour and not liquid will be released.
- (b) The outlet valve of every cylinder shall be kept closed.
- (c) Only LP Gas cylinders shall be stored within the cylinder cage.
- (d) The cylinder cage shall be kept locked and under supervision.
- (e) There shall be at least one fire extinguisher with a minimum rating of 30B located within 30 metres of the cylinder cage. A 2 kg dry chemical extinguisher is the minimum requirement.
- (f) Cylinders in the cage shall be inspected regularly to ensure they are not damaged or leaking.

G6 SAFETY CONSIDERATIONS

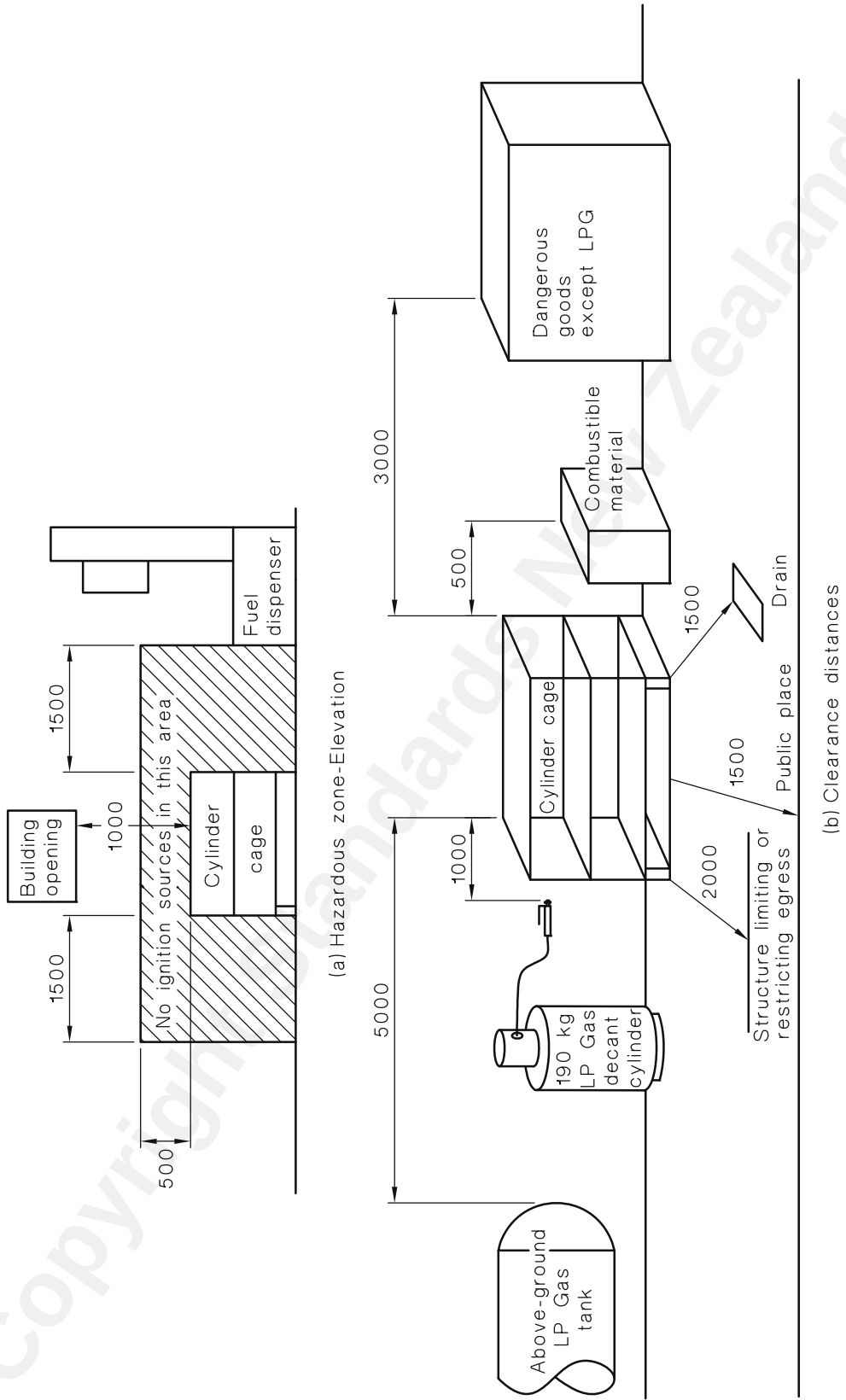
The following requirements apply:

- (a) Smoking or other sources of ignition shall not be permitted within 1.5 m of the cylinder cage.
NOTE: A vehicle is not regarded as being an ignition source while it is entering or leaving a hazardous area zone surrounding the cage.
- (b) No electrical apparatus shall be installed either in the cylinder cage or within 1.5 metres of the cylinder cage.
- (c) No flammable liquids, corrosive or oxidising materials shall be stored within 3 m of the cylinder cage unless they are separated by a 240/240/240 fire resistance rated structure.
- (d) The cylinder cage shall not hinder or endanger the means of escape from the premises or adjoining premises.

G7 SIGNS AND LABELS

Every individual cylinder cage where the aggregate quantity of LP Gas present is 250 kg or more shall be provided with signage. A sign that includes the following items shall meet the New Zealand legislative requirements:

- (a) State that hazardous substances are present.
- (b) State the hazardous property of the substance.
- (c) Describe the general type of hazard of the substance.
- (d) The precautions necessary to prevent unintended ignition.
- (e) Advise what action is to be taken in an emergency.



DIMENSIONS IN MILLIMETRES

FIGURE G1 LOCATION OF CAGES FOR EXCHANGE CYLINDERS

APPENDIX H

DISCHARGE CAPACITY OF SAFETY VALVES

(Normative in Australia only)

H1 INTRODUCTION

This Appendix is derived from the requirements of AS 1210 and provides equations for some common applications. For applications that are not covered below, the requirements of AS 1210 shall apply. In case of conflict, AS 1210 shall take precedence.

H2 TANKS

The aggregate discharge capacity in cubic metres per minute of air at 120% of the set pressure for safety valves to be used on uninsulated tanks shall be calculated from the following equations:

- (a) For above-ground tanks: $q_v = GA^{0.82}$... H1
- (b) For underground tanks: $q_v = \frac{GA^{0.82}}{100} \times 30$... H2

where

q_v = required flow capacity of the valves, in cubic metres per minute of air at 15°C and 101.5 kPa (absolute)

G = factor for the particular gas (see below)

A = total outside surface area of tank, in square metres

Where the surface area is not stamped on the nameplate or where the marking is not legible, the area shall be calculated from one of the following equations:

- (i) Cylindrical vessel with hemispherical heads:
Area = overall length $\times$ outside diameter $\times$ 3.1416 ... H3
- (ii) Cylindrical vessel with semi-ellipsoidal heads:
Area = (overall length + 0.3 outside dia.) $\times$ outside dia. $\times$ 3.1416 ... H4
- (iii) Spherical vessel:
Area = (outside diameter)² $\times$ 3.1416 ... H5

The value of G for tanks for the following gases, for the design pressure (given in parentheses), is as follows:

- (A) Propane 10.66 (1.75 MPa design pressure).
- (B) Butane 8.98 (0.70 MPa design pressure).
- (C) Propylene 10.13 (1.80 MPa design pressure).

WARNING: WHERE THE GAS TO BE STORED IN A TANK IS OF A LOWER VAPOUR PRESSURE THAN THAT FOR WHICH THE TANK, AND HENCE THE SAFETY VALVE PRESSURE SETTING, WAS ORIGINALLY DESIGNED, THESE EQUATIONS WILL NOT BE STRICTLY ACCURATE AND IT IS NECESSARY TO RECHECK THE CAPACITY OF THE VALVE BY MEANS OF THE BASIC EQUATIONS GIVEN IN AS 1210.

H3 VAPORIZERS

The same procedure shall apply as for tanks, i.e. as in Paragraph H1, except that the value of A is obtained by adding together the areas of those surfaces of the shell and the heat exchanger that are in direct contact with LP Gas.

APPENDIX I

LEAKAGE TESTING OF PIPEWORK

(Normative)

I1 CONNECTIONS BETWEEN TANK OR CYLINDER AND FIRST STAGE REGULATOR

Testing of the connections between storage tanks or cylinders and the first-stage regulator shall be made at LP Gas storage tank or cylinder pressure, using a non-corrosive soap and water, detergent solution or leakage detection fluid to detect leaks at all joints.

CAUTION: THE AMMONIA PRESENT IN SOME SOAPS AND DETERGENTS CAN REACT WITH BRASS FITTINGS AND CAUSE SUCH FITTINGS TO CRACK AFTER A SHORT PERIOD OF TIME. CAUTION SHOULD THEREFORE BE EXERCISED WHEN USING SOAP SOLUTIONS ON BRASS FITTINGS, AND ALL CONNECTIONS SHOULD BE RINSED THOROUGHLY WITH FRESH WATER AS SOON AS POSSIBLE AFTER THE APPLICATION OF THE SOAP SOLUTION.

I2 ALL OTHER PIPEWORK

I2.1 General

All other pipework not provided for in Paragraph I1 shall be tested in accordance with Paragraph I2.2 or Paragraph I2.3.

I2.2 Low pressure pipework

Pipework operating at pressures not exceeding 7 kPa shall be tested with air or nitrogen at a pressure of 7.0 kPa, or twice the operating pressure, whichever is the greater. A manometer, or a pressure gauge graduated in kilopascals and having a scale range indicating at least 20% above the test pressure, shall be used.

There shall be no loss in pressure after a period of 5 min plus 5 min for each 30 L (0.03 kL) of pipe capacity. Alternatively, bubble leak detectors may be used.

I2.3 High pressure pipework

Pipework operating at pressures exceeding 7 kPa shall be tested as follows:

- (a) All prefabricated pipework shall be tested on assembly in the fabricator's shop either hydrostatically at $1.5p$ or pneumatically at $1.1p$, where p is 1.75 MPa for propane or 0.825 MPa for butane. Where a pipeline is to be used for vapour only and a safety valve is fitted on that line, the test pressure may be taken as the setting of the safety valve. In such cases, the regulator and safety valve settings shall be sealed.
- (b) After erection and assembly in the field, the complete piping installation shall be either pneumatically tested using air or nitrogen at 0.7 MPa or hydrostatically tested at $1.5p$ [values of p as in Item (a)].
- (c) For pneumatic testing, a preliminary test at not greater than 25% of the test pressure shall be performed and any defects rectified. The pressure shall then be gradually increased to the test pressure.
- (d) All pneumatic testing shall be carried out in conjunction with a test for leaks at all joints and connections, using a non-corrosive soap and water, detergent solution or leakage detection fluid.
- (e) The test pressure shall not exceed the test pressure of any component in the installation.

- (f) For testing as under Item (b), the pipework following temperature stabilization shall sustain a holding test for a period of 1 h without a drop in pressure.
- (g) At the conclusion of a satisfactory test, the LP Gas pressure shall be gradually raised in the system by introducing vapour. To avoid cold spots, liquid LP Gas shall not be introduced until full operating pressure is reached. During this operation, all joints and connections shall be inspected and tested for leaks using a non-corrosive soap and water, detergent solution or leakage detection fluid.

CAUTION: THE AMMONIA PRESENT IN SOME SOAPS AND DETERGENTS CAN REACT WITH BRASS FITTINGS AND CAUSE SUCH FITTINGS TO CRACK AFTER A SHORT PERIOD OF TIME. CAUTION SHOULD THEREFORE BE EXERCISED WHEN USING SOAP SOLUTIONS ON BRASS FITTINGS, AND ALL CONNECTIONS SHOULD BE RINSED THOROUGHLY WITH FRESH WATER AS SOON AS POSSIBLE AFTER THE APPLICATION OF THE SOAP SOLUTION.

Pipework operating at pressures exceeding 200 kPa shall also be tested in accordance with AS 4041 or equivalent Standard, in addition to the requirements given above.

WARNING: HAMMER TESTING OF JOINTS AND PIPEWORK SHALL NOT BE CARRIED OUT DURING ANY TESTING.

Where tanks are purged with an inert gas, care shall be taken that only vapour is used, to ensure that the steel is not chilled.

APPENDIX J DECANTING PROCEDURE

(Informative)

WARNING: IF LP GAS IN LIQUID FORM COMES IN CONTACT WITH THE BARE SKIN OR EYES, IMMEDIATELY WASH THE AFFECTED AREAS WITH WATER FOR AT LEAST 10 MINUTES AND SEE A DOCTOR.

The following procedure should be followed when filling cylinders by decanting:

- (a) Inspect the customer's cylinder for suitable test date, corrosion and dents, paying particular attention to the underside of the cylinder. If significant corrosion is present, do not fill the cylinder and refer the customer to a registered cylinder test station.
NOTE: Operators should ensure that it is less than 10 years since last test date. Where the original test date is exceeded, operators should check for any subsequent test date stamping.
- (b) Unlock the decant storage vessel cover/valve. Ensure that the dead-man's valve on the filling hose assembly is closed and the assembly is tightly connected to the decant storage vessel liquid outlet valve.
- (c) Position customer's cylinder on a non-insulating surface (e.g. not inside or on top of plastic crates). Then connect filling hose assembly.
- (d) Place one bare hand on the decant storage vessel and the other on the customer's decant cylinder to discharge any static energy.
- (e) Put on gloves and open the decant storage vessel valve slowly to pressurize the decant filling hose.
- (f) Open the dead-man's handle fully, then slowly open the customer's cylinder valve while observing for leaks. If a leak is observed immediately cease decant operations and close all valves.
NOTE: By opening the deadman's handle first, then slowly opening the decant cylinder, this will avoid actuation of the excess flow valve which is located in the upstream end of the POL connection of the decant hose.
- (g) Check that the bleeder valve on the customer's cylinder is directed away from the operator and towards a well-ventilated area before opening the bleeder valve no more than 1 full turn. This will reduce the pressure in the customer's cylinder to be filled.
- (h) Remain in attendance during the entire process of filling. Do not use any device (e.g. clamps, rubber bands) to hold the dead-man's valve open. Observe the hose and all connections during filling for possible leaks. If a leak is observed immediately cease decant operations and close all valves.
- (i) Allow filling to continue until a white cloud/mist emerges from the customer's cylinder bleed valve, which indicates that the cylinder is full (the standard fill level has been reached).
- (j) Immediately release the dead-man's valve to stop filling.
- (k) On observing the white mist has reduced to a vapour only release, close the customer's cylinder bleed valve screw.
- (l) Remove gloves and touch both the decant storage vessel and the customer's decant cylinder with bare hands (to discharge any static electricity from the body of the person decanting). Replace gloves and continue.

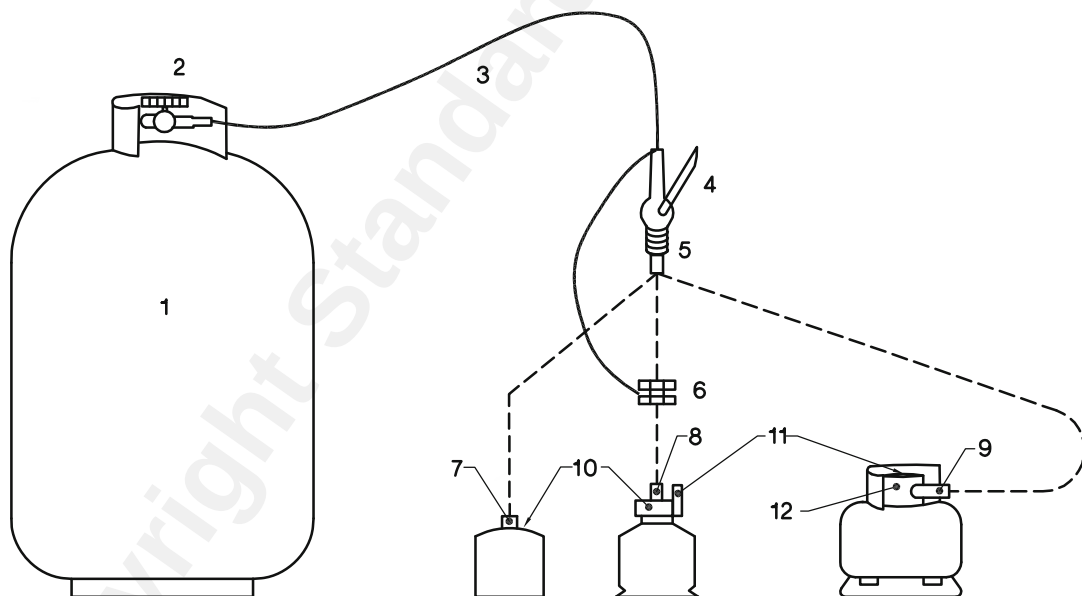
- (m) Shut off the customer's decant cylinder main valve and carefully disconnect the filling hose assembly from the customer's cylinder. When breaking the connection, a small amount of vapour or liquid trapped between the two valves will escape.
- (n) Confirm the customer's cylinder is leak-free by applying soapy water with a small paint brush or spray bottle to the following:
 - (i) Outlet of the main valve.
 - (ii) Thread where the main valve is attached to the customer's cylinder.
 - (iii) The pressure relief valve.
 - (iv) Bleed valve opening.

NOTE: For cylinders that leak after filling, the cylinders should be placed in an open area to avoid any accumulation of flammable vapour and away from ignition sources. Seek specialist advice.

- (o) If no further immediate decanting is to be carried out, Close the main valve on the decant storage vessel and safely depressurize the decant hose assembly.
- (p) Lock the cover/valve of the decant vessel.

NOTES:

- 1 In some states, regulations require the cylinder outlet to be plugged.
- 2 Figure J1 illustrates decanting equipment and its terminology.



LEGEND:

- 1 = Decant vessel
- 2 = Decant vessel main valve
- 3 = Decanting hose
- 4 = Spring-loaded 'dead-man's' nozzle valve
- 5 = Nozzle
- 6 = Adaptor for external threads

- 7 = Fitting—Internally threaded outlet.
The plunger valve inside the cylinder opens or closes as the nozzle is connected or disconnected
- 8 = Fitting—Externally threaded outlet
- 9 = Standard fitting—POL outlet
- 10 = Bleed valve—opened with a spanner
- 11 = Cylinder main valve—controlled by hand
- 12 = Bleed valve—adjusted with an Allen key or screwdriver

FIGURE J1 DECANTING EQUIPMENT

APPENDIX K

EMERGENCY PLANNING

(Informative)

K1 EMERGENCY PLANS

K1.1 General

A site storing LP Gas may be required by its state regulatory authority to have an emergency plan. Regulations will determine the minimum quantity of storage that requires such a plan, and dictate minimum requirements for the contents of the plan. Emergency plans seek to minimize the effect of an incident at a site and requires the timely application of defined procedures by people with adequate training and resources.

K1.2 Premises emergency plan

In support of available regulatory guidance, an emergency plan should include the following information as a minimum:

- (a) Emergency services contact details—‘000’ for Australia and ‘111’ for New Zealand.
- (b) Name and street address of the site.
- (c) The nature of operations on the site.
- (d) Detailed maps of the site and neighbouring land, to the extent of potential disruption in an emergency. In particular, these should identify any other dangerous goods storages and any infrastructure that could be affected.
- (e) A site plan showing the location of all LP Gas, and any other placard quantity of dangerous goods, with maximum quantities shown. The site plan should also show—
 - (i) the location of fire services (e.g. pumps, hydrant mains, hydrants, deluge systems, hose reels, extinguishers);
 - (ii) other critical isolation valves (e.g. dangerous goods pipes, stormwater, and containment systems including bunds);
 - (iii) vehicular entry points and access within the site;
 - (iv) the location of isolation points for site services (e.g. electricity, gas, water);
 - (v) staging points for emergency services; and
 - (vi) emergency assembly points.
- (f) After-hours contact details for relevant site personnel.
- (g) The means of notification of an emergency to neighbouring sites.
- (h) An evacuation plan for the site.
- (i) Details of actions to be taken in response to identified major incidents (e.g. fire threatening the site, fire involving LP Gas, leak without fire).
- (j) Identification of a person to act as a contact for emergency services.

Emergency plans should be prepared in consultation with local emergency services, and should be reviewed and updated as necessary.

K1.3 Initial advice to the emergency services

The role, responsibilities and duties of the person/s nominated to advise the emergency services of an emergency should be identified. The advice would usually be given by dialling '000' and asking for the relevant emergency service agency (typically Fire). The information provided should include the following details, where available:

- (a) Name and location of the site (suburb, street, nearest cross street to relevant site entry).
- (b) Number of injured persons or casualties and the nature of injuries.
- (c) The nature of the emergency including a brief description.
- (d) Hazards involved (including details of substances, including UN Numbers, names of substances, quantities involved).
- (e) Telephone contact number (for any return messages).
- (f) Name of person making the call.
- (g) Any other useful information (e.g. wind speed and wind direction, etc.).

K2 DOCUMENTS PROVIDING GUIDELINES FOR THE PREPARATION OF EMERGENCY PLANS

- 1 NSW Department of Planning and Infrastructure. *Hazardous Industry Planning Advisory Paper No. 1: Emergency Planning*. Sydney 2011.
- 2 Fire and Rescue NSW. *Policy No. 1 Guidelines for Emergency Plans at sites having Dangerous Goods, Explosives and Major Hazard Facilities*. Sydney 2006.
- 3 Safe Work Australia. *Guidance Material: Major hazard facilities—Emergency Plans*. Canberra 2012.
- 4 Queensland Department of Emergency Services (Chemical Hazards and Emergency Management Unit). *Emergency Planning: Guidelines for Hazardous Industry*. Brisbane 1998.
- 5 AS 3745, *Planning for emergencies in facilities*.
- 6 Safe Work Australia. *Fact sheets: Emergency plans fact sheet*. Canberra 2012.

APPENDIX L

MOTOR VEHICLE REFUELLING

(Informative)

Any training in the refuelling of vehicles should incorporate at least the following procedures and precautions as a minimum:

- (a) Do not begin to refuel the vehicle unless—
 - (i) there is no smoking or flame within 6 m of the vehicle;
 - (ii) the vehicle ignition is switched off;
 - (iii) there are no obvious leaks or faults in the LP Gas equipment of the vehicle and the gas dispensing system; and
 - (iv) the vehicle fill connection is in good condition.
- (b) Do not leave the vehicle unattended whilst filling.
- (c) Before connecting the filler nozzle, check whether the vehicle has an automatic fill limiter valve (AFL). Such vehicles carry a marking 'AFL FITTED: Bleeding during filling not required'. The procedure in Item (e) is not required for such vehicles.
- (d) Connect the filler nozzle.
- (e) If an AFL is not fitted, use the fixed liquid level gauge to prevent overfilling, by the following procedure:
 - (i) Wear protective gloves when operating the fixed liquid level gauge screw. Protective gloves must be available on the site for use during an emergency.
 - (ii) The fixed liquid level gauge screw should be opened as soon as the tank is about half full.
 - (iii) When liquid emerges from the fixed level gauge, immediately shut the filler nozzle valve.
 - (iv) Ensure that the fixed liquid gauge is securely closed at the completion of refuelling.
- (f) When using a filler nozzle that incorporates a nozzle bleed valve, open the valve to empty the nozzle before disconnecting.
- (g) Securely replace the protective cap on the vehicle-filling connection at the end of the filling operation.
- (h) If a continuous gas escape is apparent when the filler nozzle is disconnected, the non-return valve has malfunctioned, probably because of dirt under the seat. Reconnect nozzle and operate it briefly to clear the seat. If this proves unsuccessful, seal the vehicle fill connection with a metal cap. Advise the driver to have the fault rectified immediately.
- (i) At the end of the day's work, before the site or service station is closed down, check all dispensing equipment to ensure that the tank valves are closed, hoses are properly stowed, and electrical equipment is switched off.

APPENDIX M
FIRE EXPOSURE PROTECTION
(Normative)

M1 INTRODUCTION

This Appendix provides a procedure for determining whether any individual tank containing LP Gas needs protection from radiant heat that could be emitted from a nearby fire. The procedure is not a comprehensive treatment for all possibilities, and the following limitations need to be considered before applying it:

- (a) This procedure applies only to installations of more than 50 kL aggregate capacity.
- (b) The procedure has been simplified in that it deals only with the effects of fire in a conventional building adjoining the tank, so that the conditions described below can reasonably be assumed to apply.
- (c) The procedure deals with the simplest situation: a single structure in relation to a single tank. Where a building presents two or more faces to a tank, or there are likely to be multiple simultaneous sources of radiant heat, individual calculations are required.
- (d) The procedure cannot apply in situations where the source of radiant heat is another LP Gas tank, flammable liquids storage or the like. In such situations a site specific calculation is necessary based on the source of radiant heat, e.g. jet or pool fire hazards from adjacent tanks and activities.

Several protection methods are available as outlined in Clause 13.5.2.

M2 ASSUMPTIONS

The assessment given in Paragraph M3 is based on the following assumptions:

- (a) The temperature of the tank shell should not be allowed to exceed 300°C after 45 min of exposure to the heat radiation. Although this heat gain may result in the safety valve lifting, it is not sufficient to weaken the tank shell.
- (b) The temperature of the tank shell will not exceed this temperature if the heat radiation at the shell's surface does not exceed 10 kW/m². This estimate is based on a worst case, i.e. the tank being effectively empty of LP Gas liquid and the tank's surface having deteriorated to the point where it cannot reflect heat.
- (c) The temperature of the building fire will be 1000°C, so the emission will be 150 kW/m².

NOTES:

- 1 This is based on an average building fire, according to the National Building Technology Centre Technical Record 517 and confirmed from other international sources.
- 2 This temperature is only valid for fires in buildings. It should not be extrapolated to other types of fire load, as the nature of the burning material or its storage arrangement could greatly increase the fire temperature. The duration of the fire also needs to be considered. These cases are outside the scope of this Appendix and need to be considered in consultation with a fire protection specialist.

M3 ASSESSMENT PROCEDURE

The possibility of a nearby fire shall be estimated, and then the likely radiation from such a fire shall be calculated. Taking into account the distance from the building to the tank, whether the radiation to which the tank could be exposed is within acceptable limits shall be determined. If the projected radiation exposure is greater than the acceptable limit, measures need to be taken to protect the tank.

M4 CALCULATION OF FIRE EXPOSURE PROTECTION

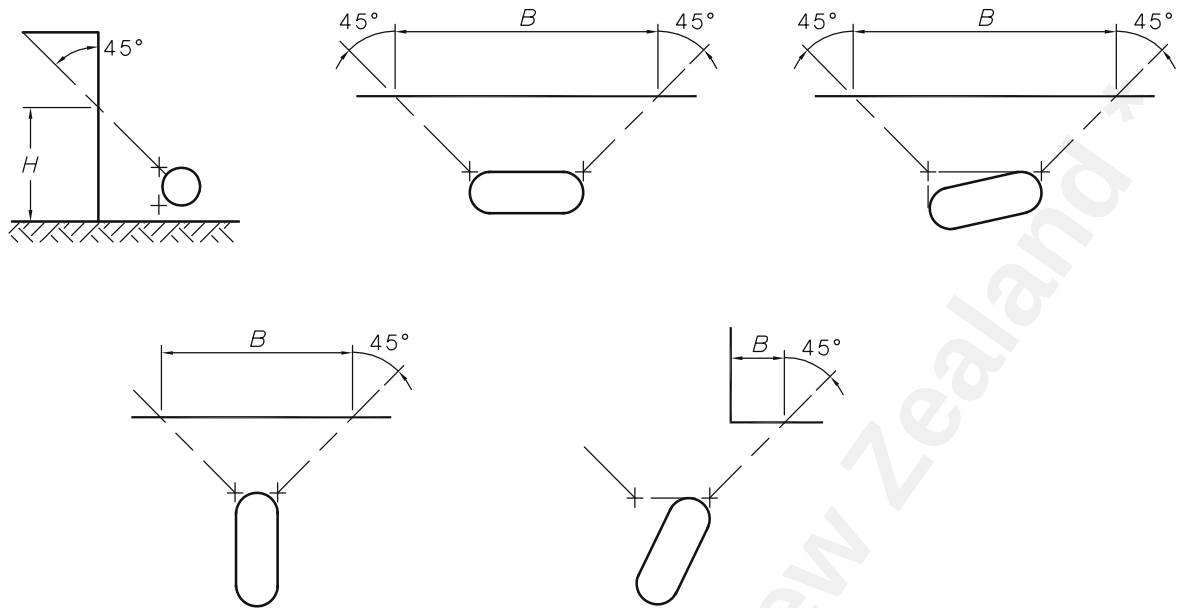
Each tank's need for protection shall be determined from radiant heat as follows:

- (a) From Table 6.1, note the separation distance from the tank to a protected place. Note any structure, storage or stack that could be a source of fire visible from the tank under consideration. Ignore any such fire source that is more than 3 times the protected-place separation distance from the tank.
- (b) Outline the surface area as shown in Figure M1(a) and determine the extent of the area (H high, B across) within which a fire would be significant to the tank.
- (c) Identify the areas as shown in Figure M1(b) and determine what portions of that area $H \times B$ will be visible fire area. This is the net fire area (A) in square metres. Use the following guidelines to estimate the net fire area (A):
 - (i) Where the whole of the area in Item (b) is combustible (e.g. a timber wall) or is of material likely to collapse in an internal fire (e.g. fibrous cement, glass, sheet metal), then A is the whole of the area $H \times B$ measured as in Item (b).
 - (ii) Where the building face is not combustible but has openings (e.g. brick wall with windows), A is the total of those openings, plus 25% for possible external fire plumes unless fire windows, fire doors or fire shutters are fitted.
 - (iii) Where a building is divided into fire-isolated compartments in accordance with the BCA/NZBC, the calculation may be based on the fire being limited to the compartment nearest the tank.
 - (iv) Parked cars may be ignored in the calculation, but any trucks that are a regularly on-site need to be included.
 - (v) Stacks of combustible solids, tanks holding flammable liquids, and other special hazards require independent assessment.
- (d) Calculate the distance D from the following equation (or read from Figure M2):

$$D = 2.2\sqrt{A} \quad \dots \text{M1}$$

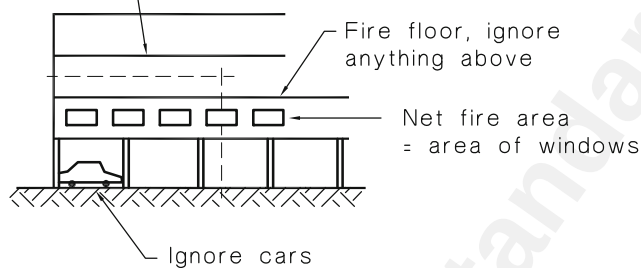
M5 EVALUATION

Where the calculated safe distance D is greater than the actual separation planned, a method of tank protection is required (see Clause 13.5.2).

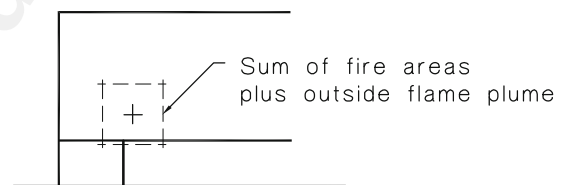


(a) Outline the total surface area of the suspected hazard within 45° zone of view

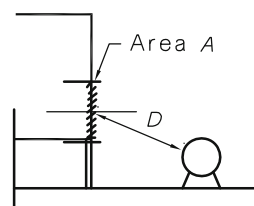
Area
 $H \times B$
as above



(b) Pick out areas at which a visible fire is possible

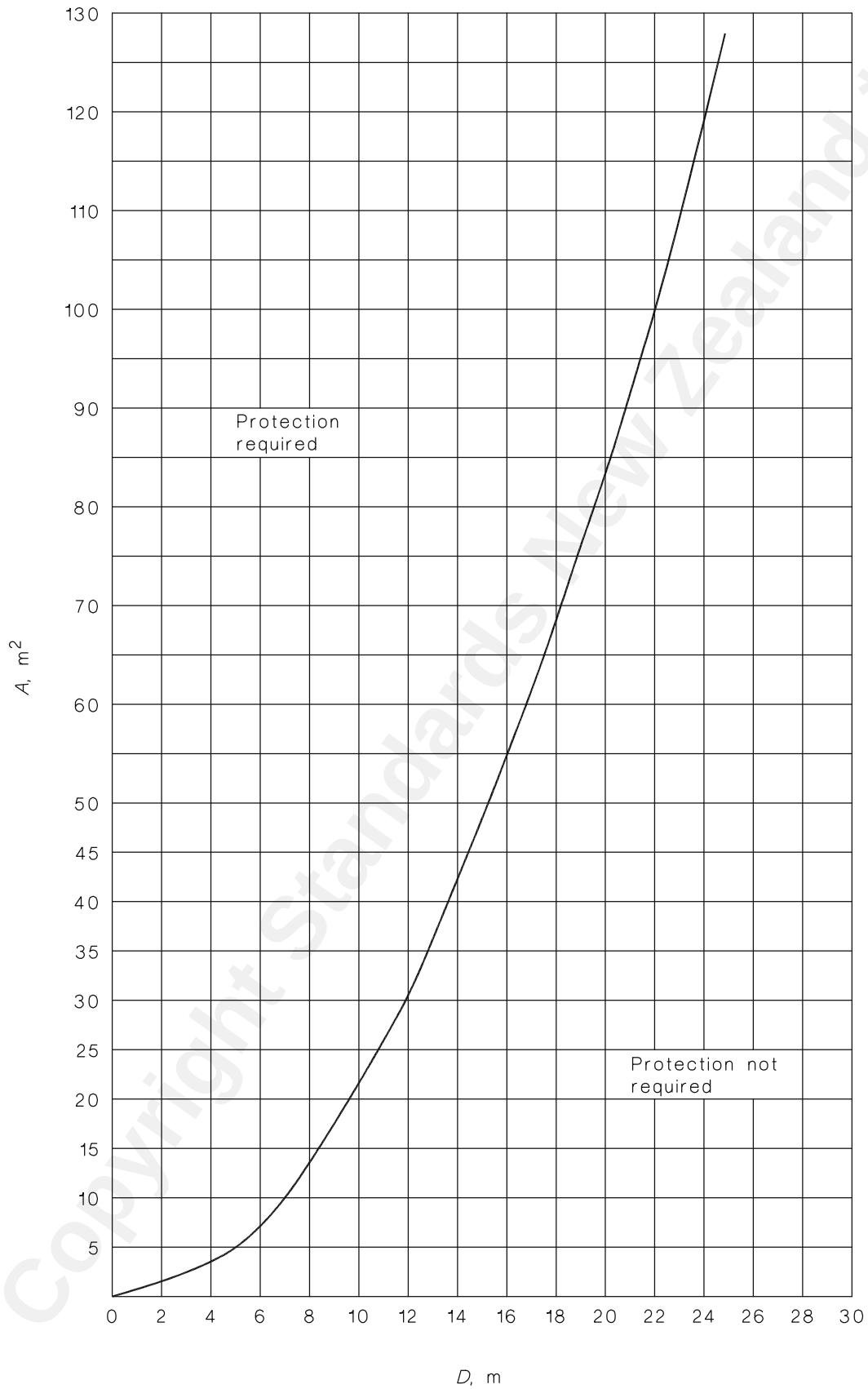


(c) Merge the visible fire areas, add 25% if an outside plume is possible. This is A for equation



(d) Check D

FIGURE M1 STEPS IN FINDING DISTANCE FROM A POTENTIAL FIRE

FIGURE M2 VALUES OF D

APPENDIX N

FIRE SAFETY

(Informative)

Safety is the fundamental objective of this Standard and is the most important single issue when storing and using LP Gas.

A fire will not occur if any one of the three essential elements, i.e. fuel, air and an ignition source, is missing. It is sometimes possible to eliminate air, for example by the use of underground and mounded tanks, but ignition sources and fuel escapes are usually more amenable to control.

Water systems alone cannot ensure total fire safety. The principal means of protection is the engineering of the installation. The components, their arrangement, the site layout and the operating and maintenance procedures all form part of the primary fire safety provisions, the purpose of which is to ensure that the secondary provisions, i.e. the water systems, never need to be used in a real emergency.

Risk of ignition can be minimized by separating potential leaks from ignition sources, controlling access, controlling on-site procedures and activities and good housekeeping by cleaning combustibles from the site. Obviously these measures can only be aimed at coping with the predictable, such as normal operational releases and minor mishaps. To try to cater for gross escapes bordering on the catastrophic would result in separation distances that are out of the question in practical terms, so it is necessary to reduce the probability of any major releases to a level that can reasonably be considered to be negligible.

The prevention of gas escapes, i.e. containment, is considered to be the single most important aspect of this Standard. If gas remains contained, there can be no fire risk. If an escape can be terminated quickly and preferably automatically, the risk of ignition and the consequences are minimized. Thus engineering the containment and valving provisions are considered vital to fire safety.

Loss of containment can result from accidental impact, corrosion, breakdown of seals and gaskets because of long-term deterioration, or rapid breakdown because of fire.

Physical damage arises mainly from vehicles, most commonly either colliding with the installation or driving off with a hose still connected. The solutions are impact protection, the choice of safe location and a variety of measures to either prevent drive-away or make provision against the consequences.

Equipment deterioration is a management matter. Operating and maintenance procedures should be set up in the first place, should be implemented and should not be allowed to lapse.

Fire threats fall roughly into two classes, i.e. a nearby fire radiating heat to a tank or a fire around the tank and impinging directly on it. A tank can tolerate a certain amount of heat influx, but the level is not high. Unintentional heating can lead to a high risk of gas discharge, either through a safety valve or through the failure of some feature of the installation.

An accidental on-site fire is dealt with in the main by preventive measures and procedures. Flammable liquid spills can be prevented from becoming a hazard to the LP Gas installation by means of bunding, kerbing, grading and other spillage control measures. Rubbish and combustible materials should not accumulate on a well-kept site, but, if a minor fire should start, the extinguishers and hose reels specified should cope.

Nearby high radiation fires are an important consideration. There is obviously no point in providing elaborate cooling systems if there is nothing nearby to burn, yet there will be cases where there is a real risk. A survey of each site is necessary. The radiation level from an average building fire is in the order of 150 kW/m^2 , and the tolerable heat flux at the tank surface is 10 kW/m^2 , hence the limiting distances to potential hazards can be calculated and a decision on the need for heat protection can be made.

When a fire has developed on the LP Gas system itself, there is a great potential for escalation of the incident. A leak, particularly of liquid, can throw a flame a considerable distance. If such a flame impinges on a tank, the heat flux is almost always considerably in excess of the tolerable level. It is particularly serious if the flame impingement is on the vapour space of the tank, and this does not necessarily mean the top, as it must be borne in mind that a tank that is virtually empty is entirely vapour space.

Escalation of a flame impingement on a tank may cause a Boiling Liquid Expanding Vapour Explosion (BLEVE) and large fireball with potentially catastrophic consequences for several hundred metres in all directions.

Therefore the engineering of the installation constitutes the most significant and most effective element in fire safety considerations. Fire safety is achieved principally by mechanical means, i.e. a system of valves which control all outflow of LP Gas, whether liquid or vapour, and can shut down the system, preferably automatically, should an incident occur.

A site-specific hazard analysis and safety assessment such as a fire safety study may be required to demonstrate fire safety for certain LP Gas installations. In particular, consideration of fire safety requirements to protect against escalation to catastrophic events and offsite threats is required when major hazard facility legislation applies to the LP Gas facility.

This Standard has been written from the point of view that, where gas remains contained there can be no risk. If an escape occurs it must be terminated quickly and preferably automatically, so that the risk of ignition and the consequences of it are minimized. The resulting fire cannot be fought by conventional methods, using hoses or extinguishers. The only effective way to fight a gas fire is to turn off the gas.

Attention has been paid to determining possible causes of gas escape and the capabilities and shortcomings of particular types of equipment. Key engineering aspects of the installation are as follows:

- (a) Any impingement of flame on an unwetted (vapour space) tank surface is critical. The worst possible event, a tank rupture, should not be allowed to occur. To achieve this, it should be inherent in the design of an installation that losses of containment are prevented where possible or otherwise controlled or safely directed. The engineering of the containment provisions should virtually eliminate the risk of a gas fire within the installation.
- (b) An escape of liquid is more serious than an escape of vapour, except that any impingement of flame on an unwetted (vapour space) tank surface is critical.
- (c) Openings into a tank should not be more numerous, nor larger, than they need to be.
- (d) For the requirement of shut-off system, see Clause 5.3.4.
- (e) Careful thought should be given in the design stage to ensure that any filling or withdrawal connection, shear point, screwed or flanged connection, or other feature to which flame could flash in the event of fire is located and directed to avoid flame impingement. Pipework should be designed to minimize the number of flanges and joints.
- (f) Generally the equipment should have adequate short-term tolerance for radiated heat sufficient to allow time to set up and bring into operation protective cooling measures.

- (g) Water sprays and sprinkler systems are not a substitute for protective valving. More work on the valve system design is preferable to more work on the water system. However, the use of cooling water may help by slowing down or preventing escalation, depending on the scale of the incident.
- (h) Training, operating, and maintenance procedures need to be set up in the first place, implemented, and not subsequently allowed to lapse.
- (i) A certain basic level of firefighting equipment is required for all but the most minor installations. Water sprays are not necessarily mandatory from the point of view of tank size. They are one of several alternative forms of incidental heat protection, and the need is determined from a survey of the actual site conditions.

The requirements of the Standard regarding the handling of a fire emergency involving LP Gas storages are based on the following elements:

- (i) Rapid evaluation of the nature of the fire is imperative.
- (ii) If it is an adjacent fire in some other structure or material, then the problem is whether the heat radiation to the tank is sufficient to require remedial action.
- (iii) If gas is escaping the priority tasks are to prevent escalation, to stabilize, then to terminate. The twin needs are to shut off the gas flow and, in the meantime, to cool any areas that may need it.
- (iv) If stability can be achieved, there is nothing wrong with letting the gas burn if it is doing no harm, even to the extent of burning off all the stored gas if this is the safest thing to do.
- (v) If the situation appears to be escalating, evacuation needs to be considered. The required distance for evacuation will vary with the size of the tank.
- (vi) Spray systems can protect against incident radiation, but should not be relied upon to cope with a concentrated flame impingement.

The automotive filling installation requirements of this Standard incorporate location requirements based on risk assessment. These requirements may be used as indicative guidance for designers, developers, planners and authorities dealing with location and land-use planning aspects of LP Gas service stations. The application of local planning arrangements and advice on separation distance must take precedence over this indicative guidance.

These requirements are based on hazard analysis, quantified risk assessment and event trees for a generalized LP Gas service station and its operation.

A significant program of fire testing, conducted to demonstrate actual release rates and flame lengths prevailing under various conditions involving loss of containment, was performed by industry. The resultant experimental data, together with additional failure data, risk reduction measures and safety management system, have been incorporated into event trees to provide an in-depth assessment on which the requirements are based.

The assessment specifically considered the technical, operational and safety requirements associated with a single 8000 L above-ground storage tank, its associated equipment and activities. With respect to a residential area, the assessment provided a cumulative risk of a fatality at 7.5 m of $1 \times E-6$ per annum, being a borderline result that was particularly sensitive to assumptions of input, ignition and delivery frequency, and a distance of 10 m was considered representative of the QRA outcome. A conservative figure of 15 m has been used in this Standard.

Uncertainty of applicable local failure-frequency data was found to be the primary limiting factor in the generic analysis and as a result any risk based separation distances should have regard to available site specific failure rate data and a new state of knowledge concerning LP Gas hazard and risk at LP Gas service stations.

APPENDIX O

TRANSPORT OF PORTABLE LP GAS CYLINDERS

(Informative)

Where cylinders of LP Gas are transported for private, domestic use, the following conditions, recommendations and instructions apply:

- (a) Cylinders need to be in good condition prior to being transported. Each cylinder should be checked for leaks, either by checking for odour, or applying a non-corrosive soap and water, detergent solution or leakage detection fluid around the cylinder valve and checking for bubbles.

CAUTION: THE AMMONIA PRESENT IN SOME SOAPS AND DETERGENTS CAN REACT WITH BRASS FITTINGS AND CAUSE SUCH FITTINGS TO CRACK AFTER A SHORT PERIOD OF TIME. CAUTION SHOULD THEREFORE BE EXERCISED WHEN USING SOAP SOLUTIONS ON BRASS FITTINGS, AND ALL CONNECTIONS SHOULD BE RINSED THOROUGHLY WITH FRESH WATER AS SOON AS POSSIBLE AFTER THE APPLICATION OF THE SOAP SOLUTION.

- (b) All control valves should be safeguarded against damage by means of a valve protection ring or protective caps. If an outlet valve plug is available, it should be used.

NOTE: LP Gas is flammable and can cause drowsiness or asphyxiation if released in a confined space.

- (c) The cylinder should be disconnected from any regulator, hose or appliance.
- (d) The total quantity of LP Gas transported inside any enclosed vehicle compartment should not exceed 33 L water capacity (i.e. one nominal 13 kg cylinder).
- (e) The total quantity of LP Gas transported by the vehicle should not exceed 65 L water capacity.
- (f) Cylinders should be secured in an upright position during transport.

NOTE: This is to ensure that any relief device is in direct contact with the vapour space of the cylinder.

- (g) Where a cylinder is to be transported in the passenger compartment of a vehicle, it should be in the rear of the compartment. The cylinder should be restrained in order to prevent movement or physical damage to the cylinder or the vehicle's occupants.

NOTE: Unrestrained cylinders can cause serious injury if the vehicle suddenly brakes or is involved in an accident.

- (h) Where a cylinder is to be transported in the boot of a vehicle, it needs to be secured so that it cannot fall over during transport.
- (i) Cylinders should not be left unattended inside the vehicle, particularly during hot weather.

NOTE: A moderate increase in temperature can cause the gas pressure inside the cylinder to rise and can lead to an abrupt discharge of LP Gas.

- (j) LP Gas is flammable; do not smoke or have any naked flame present while handling or transporting the cylinder.
- (k) Discharge any personal static build-up by touching a grounded (earthed) metal object (e.g. the body of the car) with your bare hands immediately before handling the LP Gas cylinders.
- (l) Once transported to their destination, the cylinders should be unloaded without delay.

APPENDIX P

TRANSPORT OF PORTABLE LP GAS CYLINDERS FOR TRADE USE

(Informative)

P1 GENERAL

Where cylinders of LP Gas are transported for trade use, the following conditions, recommendations and instructions apply.

NOTE: For quantities greater than 60 L (in Australia) and 125 L (in New Zealand) additional requirements are specified in the ADG Code (in Australia) and New Zealand Land Transport Rules (in New Zealand).

- (a) Cylinders should be in good condition prior to being transported. Each cylinder should be checked for leaks, either by checking for odour, or applying a non-corrosive soap and water, detergent solution or leakage detection fluid around the cylinder valve and checking for bubbles.

CAUTION: THE AMMONIA PRESENT IN SOME SOAPS AND DETERGENTS CAN REACT WITH BRASS FITTINGS AND CAUSE SUCH FITTINGS TO CRACK AFTER A SHORT PERIOD OF TIME. CAUTION SHOULD THEREFORE BE EXERCISED WHEN USING SOAP SOLUTIONS ON BRASS FITTINGS, AND ALL CONNECTIONS SHOULD BE RINSED THOROUGHLY WITH FRESH WATER AS SOON AS POSSIBLE AFTER THE APPLICATION OF THE SOAP SOLUTION.

- (b) All control valves should be safeguarded against damage by means of a valve protection ring or protective caps. If an outlet valve plug is available, it should be used.

NOTE: LP Gas is flammable and can cause drowsiness or asphyxiation if released in a confined space.

- (c) The cylinder should be disconnected from any regulator, hose or appliance.
- (d) Cylinders should be secured in an upright position during transport.
NOTE: This is to ensure that any relief device is in direct contact with the vapour space of the cylinder.
- (e) Cylinders should not be capable of being struck by loose objects.
- (f) At least one 10B dry-powder fire extinguisher is fitted in the driver's cabin.
- (g) LP Gas is flammable; do not smoke or have any naked flame present while handling or transporting the cylinder.
- (h) Discharge any personal static build-up by touching a grounded (earthed) metal object (e.g. the body of the car) with your bare hands immediately before handling the LP Gas cylinders.
- (i) Once transported to their destination, the cylinders should be unloaded without delay.
- (j) Cylinders should not be used while still in a closed vehicle.

P2 OPEN-TYPE VEHICLES

Open-type vehicles have cargo areas open to the air with unrestricted ventilation. The cargo areas of some open-type vehicles with sides (e.g. utility vehicles) can potentially retain escaped gas.

This is due to escaped gas flowing to and accumulating in low, enclosed and poorly ventilated areas. If you are not sure if your cargo area may allow gas to accumulate, assume it can until a competent person determines otherwise. The cargo area should be assessed by a competent person for its potential to retain gas, and to identify the likely areas where gas could accumulate. Consider installing vents in these areas.

P3 CLOSED-TYPE VEHICLES

Closed-type vehicles are vehicles, such as vans, utilities and the like, that have cargo areas with restricted natural air movement and ventilation. This includes vehicles where gas cylinders are stored under canopies, inside service bodies or toolboxes, and vehicles where cylinders are covered by tarps or tonneau covers.

If a closed-type vehicle cannot be avoided, it should be fitted with a separate gas storage cabinet that is—

- (a) vapour-tight from the rest of the vehicle;
- (b) secured to the vehicle of sufficient size to store all cylinders carried in the vehicle, including empty cylinders;
- (c) designed and installed to ensure it is vented to outside the vehicle;
- (d) with one or more permanent vents or drains located in the bottom of the cabinet (either floor, door or side) and with an internal diameter of at least 25 mm that cannot be blocked when the cylinders are in the cabinet;
- (e) fitted with a method of securing cylinders in an upright and stable position (e.g. straps);
- (f) fitted with a visible flammable gas label on the outside of the cabinet door; and
- (g) be securely sealed whenever cylinders are stored in the cabinet.

P4 INFORMATION AND TRAINING

Ensure workers are informed and trained in the following procedures:

- (a) Checking and testing for leaking cylinders.
- (b) Securing cylinders and sealing gas cabinets.
- (c) Regularly checking cabinet seals and vents.
- (d) Dealing with damage to the gas cabinet, gas leaks or explosion.

In Australia, transportation and loading of LP Gas cylinders are required to comply with local regulations for dangerous goods transport and the ADG Code. Listing of the Competent Authorities for dangerous goods transport can be found in the ADG Code. Copy of the ADG Code is available for download at the National Transport Commission website: www.ntc.gov.au

APPENDIX Q

IMPACT PROTECTION

(Informative)

Q1 GENERAL

This Appendix provides guidance on an assessment method to determine the impact protection requirements of LP Gas containers due to exposure to collision by vehicular traffic.

Q2 VULNERABILITY

Vulnerability of LP Gas installations is assessed using a standard risk analysis (RMA) or hazard operability process (HAZOP) that includes factors, such as—

- (a) proximity to, and management of any traffic flow; and
- (b) the characteristics of that traffic flow.

Q3 RANGE OF INSTALLATIONS

For small containers stored in cages which are not fixed and with the valves closed, adequate protection is provided by that cage if 'shunted' by an encroaching vehicle. Larger storage tanks, fixed to a structure, must be protected from impact as movement may damage pipe-work and fittings.

Q4 PROTECTION OF LP GAS INSTALLATIONS

Protection systems are designed to absorb a proportion of the energy generated by the momentum of the vehicle in motion. Vehicles are designed in a similar manner, with 'crumple zones' built into the vehicle's structure.

It is the momentum (mass of the vehicle and the speed of the vehicle) and the probability of impact that determines the required LP Gas installation protection.

Options for such protection include the following:

- (a) Separation (no protection is required), for example—
 - (i) a 110 L cylinder attached to a house with an adjacent driveway to the garage(s);
 - (ii) a tank installed on a farm away from traffic; and
 - (iii) a tank at the rear of a restaurant where there is either no vehicular traffic or very low speed vehicles on infrequent occasions.
- (b) Visual—Painted lines and plastic cones, such as in a storage area in which a pallet truck is used, or a walk-way where pedestrians frequent—no vehicles or very low speed vehicles.
- (c) Low energy protection—Simple kerbing, purpose-built storage cage, bolt down bollards such as in a car-park, service station forecourt, factory apron or school yard—all vehicles at low speeds.
- (d) High energy protection—W-guard rail, Armco fencing, bollards in concrete for use in busy streets, chain wire fence in lane-ways and driveways—all vehicles at various speeds.

Q5 EXAMPLE OF IMPACT PROTECTION ASSESSMENT

Table Q1 provides details for the likely combinations of LP Gas installations and vehicle momentum. These combinations are given an assessment category number (A to H) and points allocated according to levels of risk. By determining the point value for each category, the ranking can be determined from the following site storage vehicle assessment formula:

$$\text{Ranking} = A + B + C + D + E + F + G + H$$

Table Q2 lists ranges of ranking values and provides examples of LP Gas installation protection that such a ranking would require. Figure Q1 illustrates these examples.

TABLE Q1
ASSESSMENT OF LP GAS INSTALLATION PROTECTION REQUIREMENTS FROM VEHICULAR TRAFFIC

Section	Measurement needed	Guide	Description	Points	Input column 0 = No 1 = Yes	Tally = Points × input
A Separation	Separation distance between the LP Gas installation and nearest traffic pathway, of any type	Worst	<3 m	100	0	0
			3 to 10 m	50	0	0
		Best	>10 m	0	0	0
Guidance note for assessor:	How close can vehicles get to the LP Gas installation while on the vehicle pathway?					
B Elevation	The elevation of the LP Gas installation relative to the nearest vehicle pathway	Worst	Pathway is above the LP Gas installation	100	0	0
			Pathway is on level plane with roadway	75	0	0
		Best	Pathway is lower than LP Gas installation	50	0	0
Guidance note for assessor:	An LP Gas installation is at a higher risk if an out-of-control vehicle can run down-hill and impact this.					
C Access	The quality of traffic on the nearest pathway	Worst	Unrestricted—General public in vehicles	900	0	0
			Controlled—Factory, industrial, commercial	500	0	0
			Access is controlled under constant supervision	0	0	0
		Best	Domestic—Driveways	0	0	0
Guidance note for assessor:	Risk increases with higher volumes of unsupervised drivers. Where traffic enters a site which has a visual and physical barrier such as a fenced yard with open access gates, this is adequate to achieve a level of control to minimize risk. Sites which are only subjected to services, such as, rubbish or regular deliveries have a significant reduction in the likelihood of an incident due to higher competency driver skills and low-level traffic volume.					

(continued)

TABLE Q1 (continued)

Section	Measurement needed	Guide	Description	Points	Input column 0 = No 1 = Yes	Tally = Points × input
D Direction	Direction of the flow traffic along the nearest traffic pathway, relative to the LP Gas installation or storage	Worst	Direct—Head on	100	0	0
			Indirect—Changes of direction required	30	0	0
			Parallel—No proximity	25	0	0
		Best	Restricted authorized service vehicles only	0	0	0
Guidance note for assessor:	Changes of direction require reduced speed and provide a risk control mechanism. LP Gas installations not in the general path of vehicles are at a much lower risk than those in the same direction as the traffic.					
E Position	Position of LP Gas storage or installation in relation to final traffic destination	Worst	Approach side	50	0	0
		Best	Departure side	25	0	0
Guidance note for assessor:	Vehicles leaving a site are more orderly and have a higher degree of control than those approaching their destination from a main road or street.					
F Taming	All traffic taming devices in the vehicle pathway	Worst	None	50	0	0
			Speed restrictions, signs	35	0	0
		Best	Speed humps, chicanes	25	0	0
G Pipework	External attached pipe work, valves and fittings within the LP Gas installation	Worst	External pipework attached/fixed	300	0	0
		Best	Non (free-standing installation or storage)	0	0	0
Guidance note for assessor:	Attached fixed pipework may increase the severity of leakage following accidental impact.					

(continued)

TABLE Q1 (continued)

Section	Measurement needed	Guide	Description	Points	Input column 0 = No 1 = Yes	Tally = Points × input
H Modifiers	NOTE: Kerbing 150 mm high is widely used to control traffic, and it can be combined with any of the above methods (see Note 1).		Reduction in ranking of 300 Points if kerbing is used in conjunction with other protection.	-300	0	0
	NOTE: Traffic management plans are recognized as providing a high level of on-site incident control resulting in reduced incidents and reduced consequences and severity.		Where unauthorized and approved traffic management plan is implemented.	-300	0	0
Guidance note for assessor:	Traffic taming and directional control devices provide a physical and visual separation both of which result in the control on traffic quality and behaviour.					

NOTES:

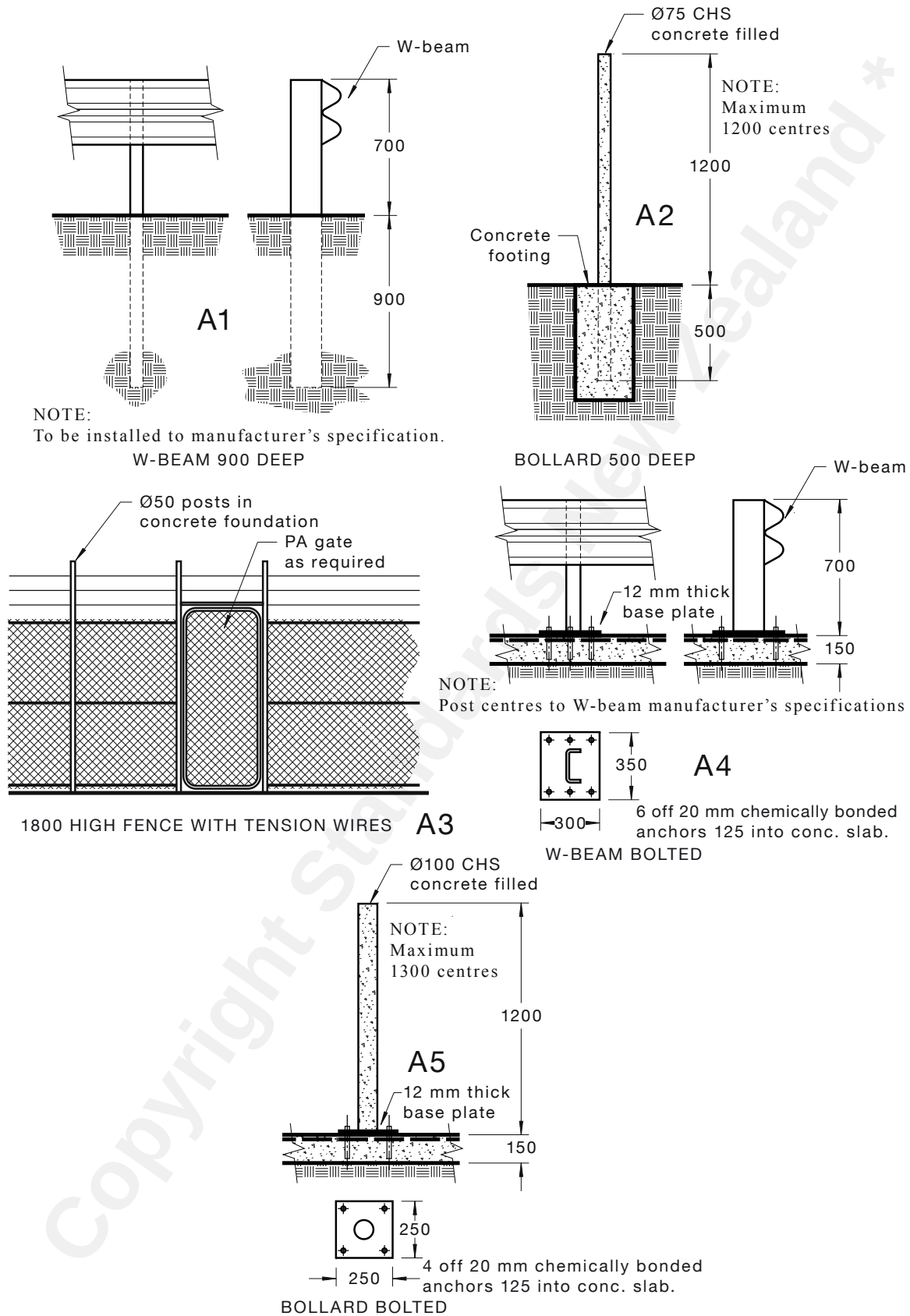
- One input per section except for Section H 'Modifiers' where none, either or both may apply
- If any kerbing, gutter, edge-strip or landscaping of 150 mm in height or greater, exists between vehicle passageway and the LP Gas installation, subtract 300 points from the ranking total. Where an authorized and approved site traffic management plan is implemented, subtract 300 points from the ranking total.
- The ranking must fall into the range specified to permit the use of the installation protection described for that range.
- All installation protection must be in accordance with the detail described in this Appendix, or the manufacturer's stated installation methods, whichever is the more stringent.
- Refer to Table Q3 for separation distances to LP Gas storage.

TABLE Q2
PROTECTION FOR ASSESSMENT RANKING

Ranking range	Detail drawing (see Figure Q1)	Description of required LP Gas installation protection
1600 to 1300 points	A1	W-Rail fitted to in-ground posts
1300 to 1000 points	A2	Bollards 75 mm OD, in ground, set in concrete
	A3	Chain wire fence, 50 mm posts in concrete, with strainer posts and brace wire
	A4	W-Rail with bolt down posts 125 mm into concrete
	A5	Bollards 100 mm OD bolt down 125 mm into concrete
	A6	Impact absorbing W-Rail with bolt down posts
1000 to 400 points	B1	W-Rail with bolt down posts 75 mm into concrete
	B2	Bollards 100 mm OD bolt down 125 mm into concrete
	C1	Self-contained (purpose built) movable frame suit single or multiple cylinders
	D1	Self-contained (purpose built) free-standing steel cage for exchange cylinders (max. cylinder capacity 30 litres water capacity)
700 to 400 points	C2	Kerbing 150 mm high
Score less than 400 points		No vehicle protection required

TABLE Q3
SEPARATION DISTANCES

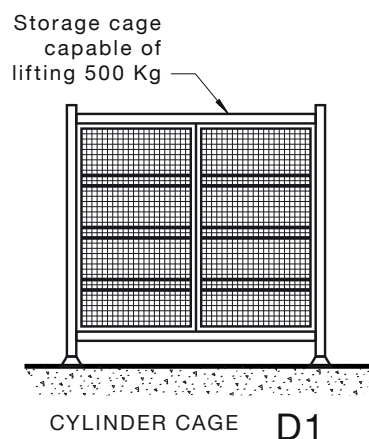
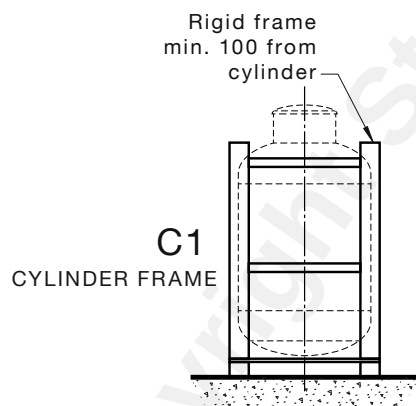
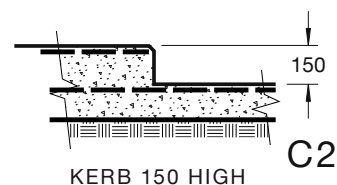
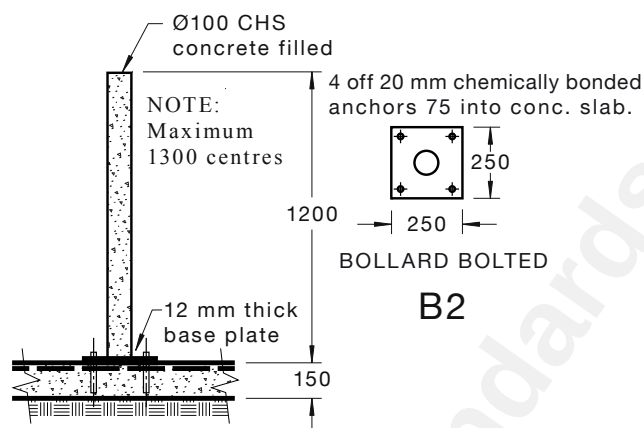
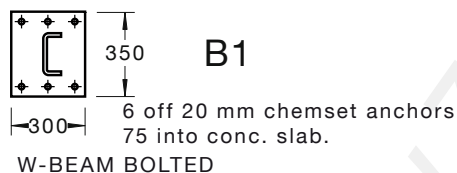
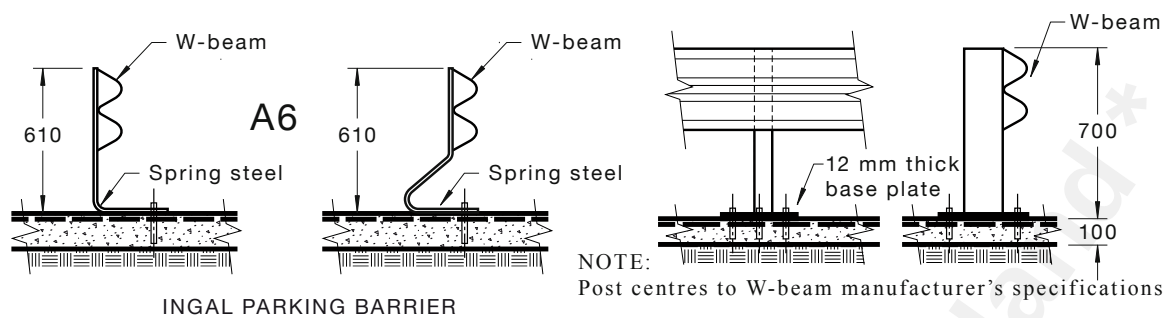
Detail drawing (see Figure Q1)	Any tank vehicle speed more than 15 km/h	Tank greater than or equal to 8 K Lt vehicle speed less than 15 km/h	Tank less than 8 K Lt vehicle speed less than 15 km/h	Cylinders greater than 30 Lt WC connected to service line vehicle speed less than 15 km/h	Cylinders greater than 30 Lt WC not connected to service line vehicle speed less than 15 km/h	Group of cylinders less than 30 Lt WC not connected to service line vehicle speed less than 15 km/h
A1	1500	1500	Vessel diameter	150	100	100
A2	1500	1500	Vessel diameter	150	100	100
A3	1500	1500	Vessel diameter	1000	100	100
A4	1500	1500	Vessel diameter	150	100	100
A5	1500	1500	Vessel diameter	150	100	100
A6	1500	1500	Vessel diameter	150	100	100
B1	Not suitable	Not suitable	Vessel diameter	150	100	100
B2	Not suitable	Not suitable	Vessel diameter	150	100	100
C1	Not suitable	Not suitable	Not suitable	150	100	Not suitable
C2	Not suitable	Not suitable	Not suitable	1000	1000	Not suitable
D1	Not suitable	Not suitable	Not suitable	Not suitable	Not suitable	Suitable



(a)

DIMENSIONS IN MILLIMETRES

FIGURE Q1 (in part) SITE STORAGE VEHICLE PROTECTION



(b)

DIMENSIONS IN MILLIMETRES

FIGURE Q1 (in part) SITE STORAGE VEHICLE PROTECTION

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